Week 3: Non-Parametric Models

Matthew Caldwell

COMP0088 Introduction to Machine Learning • UCL Computer Science • Autumn 2026

Admin

- Pulse surveys

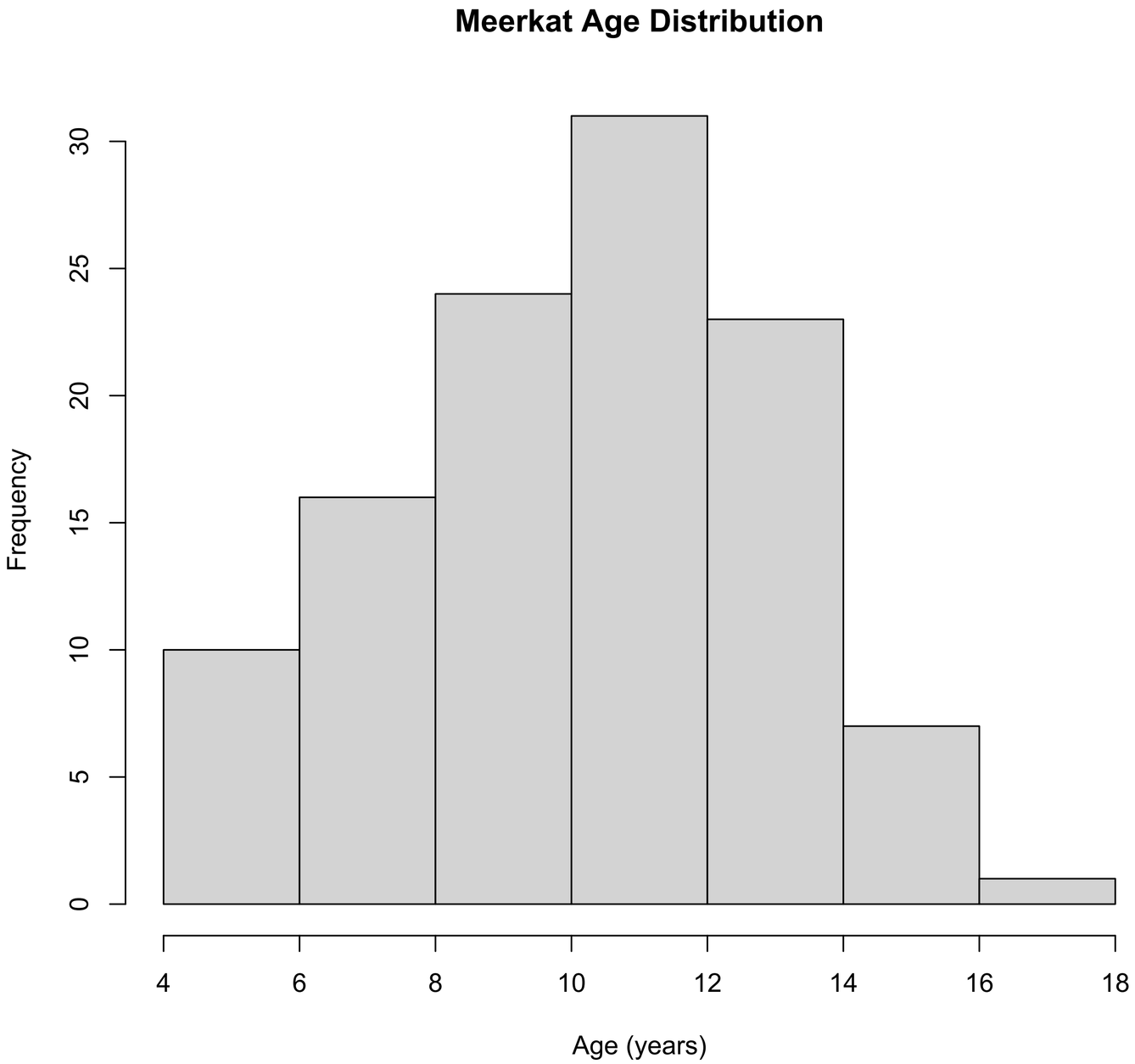
Week 3 Recap

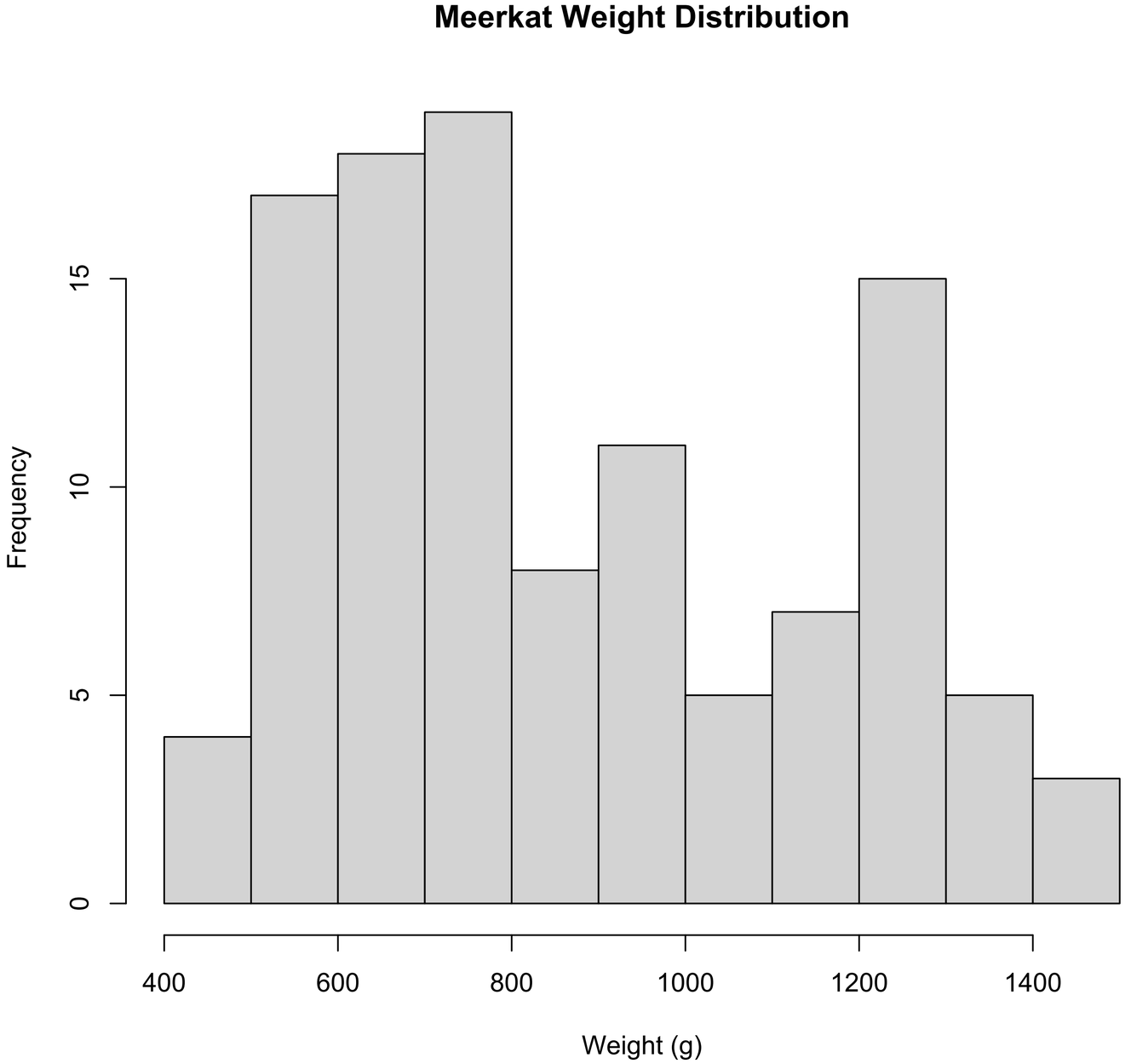
Crosseyed and Painless

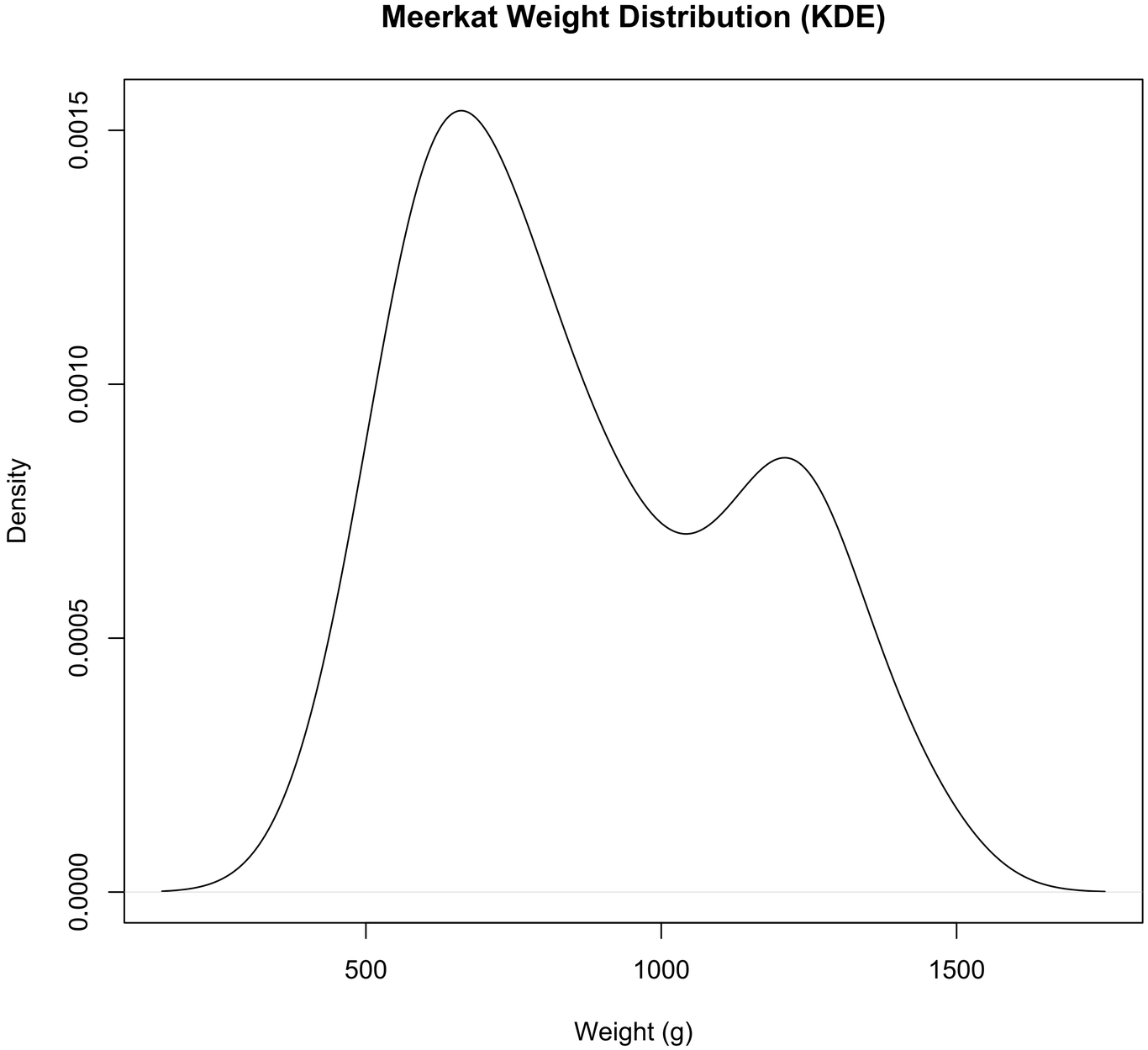


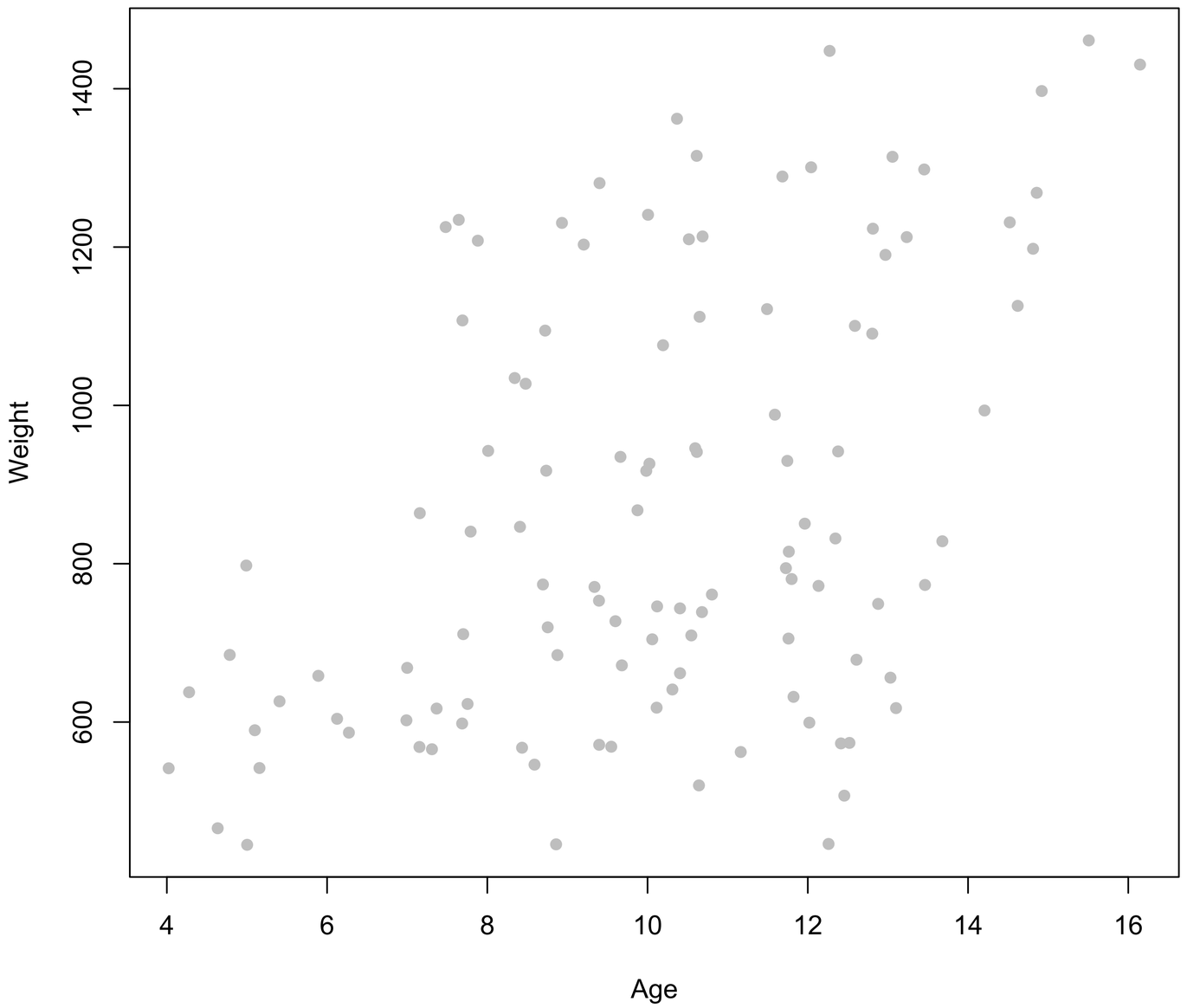
Meerkats

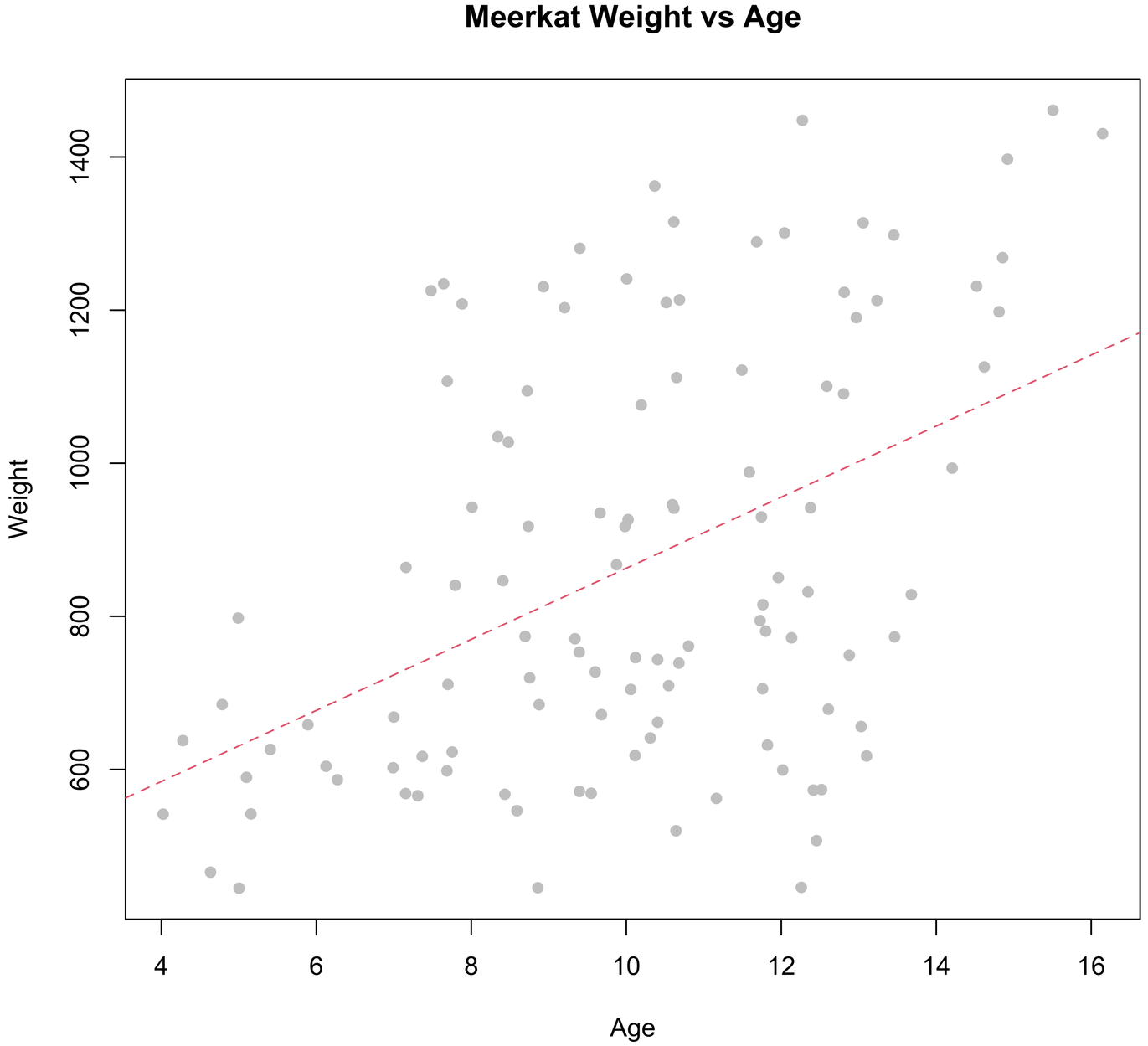
#	Age	Weight	Sex	Captive	...
1	11.8	815	1	0	
2	9.7	672	1	0	
3	8.9	446	1	0	
4	10.8	761	0	0	
5	8.3	1035	0	1	
6	11.7	930	1	1	
7	8.5	1027	0	1	
8	7.6	1234	0	1	
9	15.5	1461	0	1	
10	8.8	720	1	0	
11	12.8	1223	0	1	
12	7.7	711	1	0	
13	7.2	526			

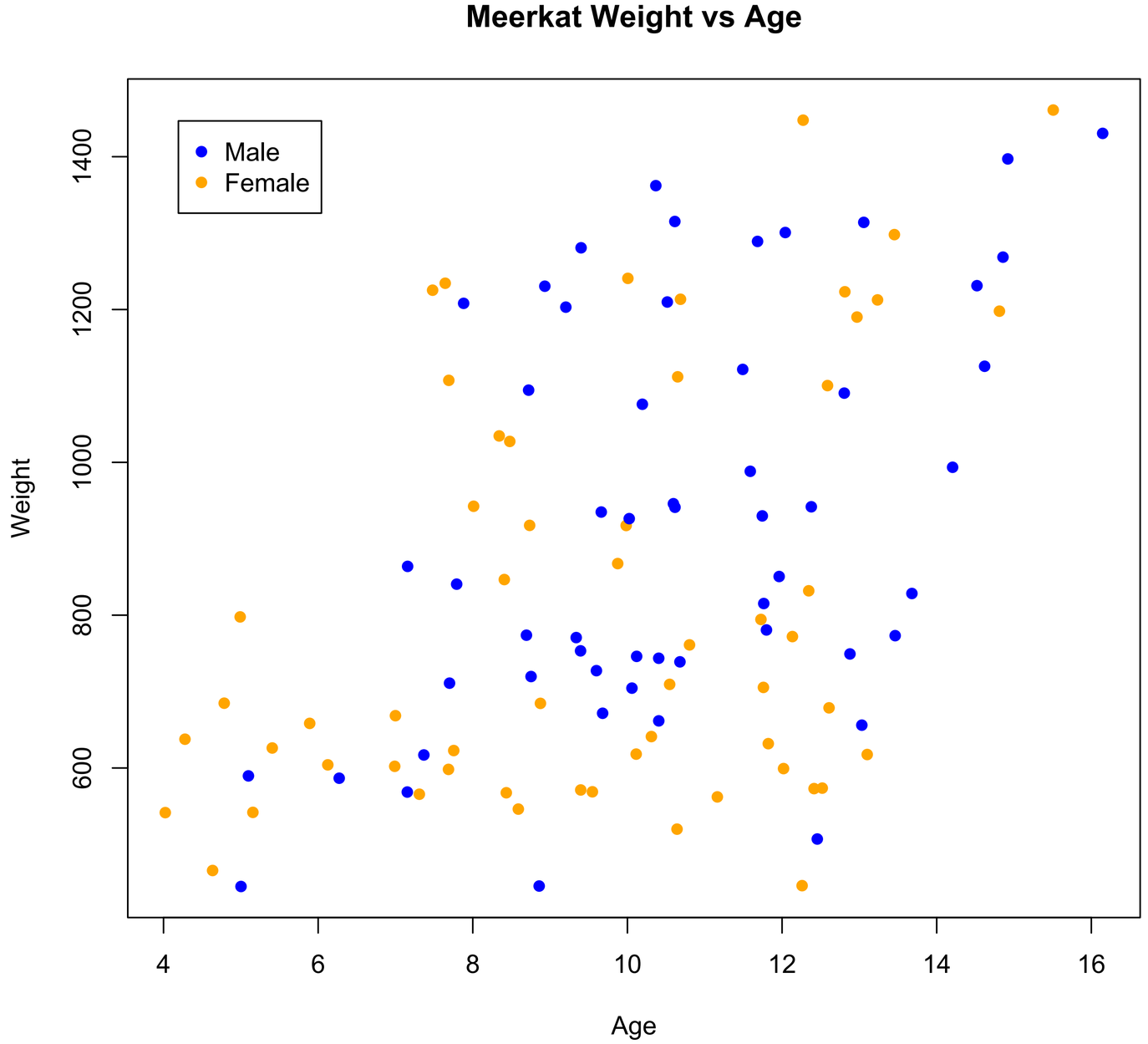

Meerkat Age Distribution
Frequency
0
5
10
15
20
25
30
4
6
8
10
12
14
16
18
Age (years)

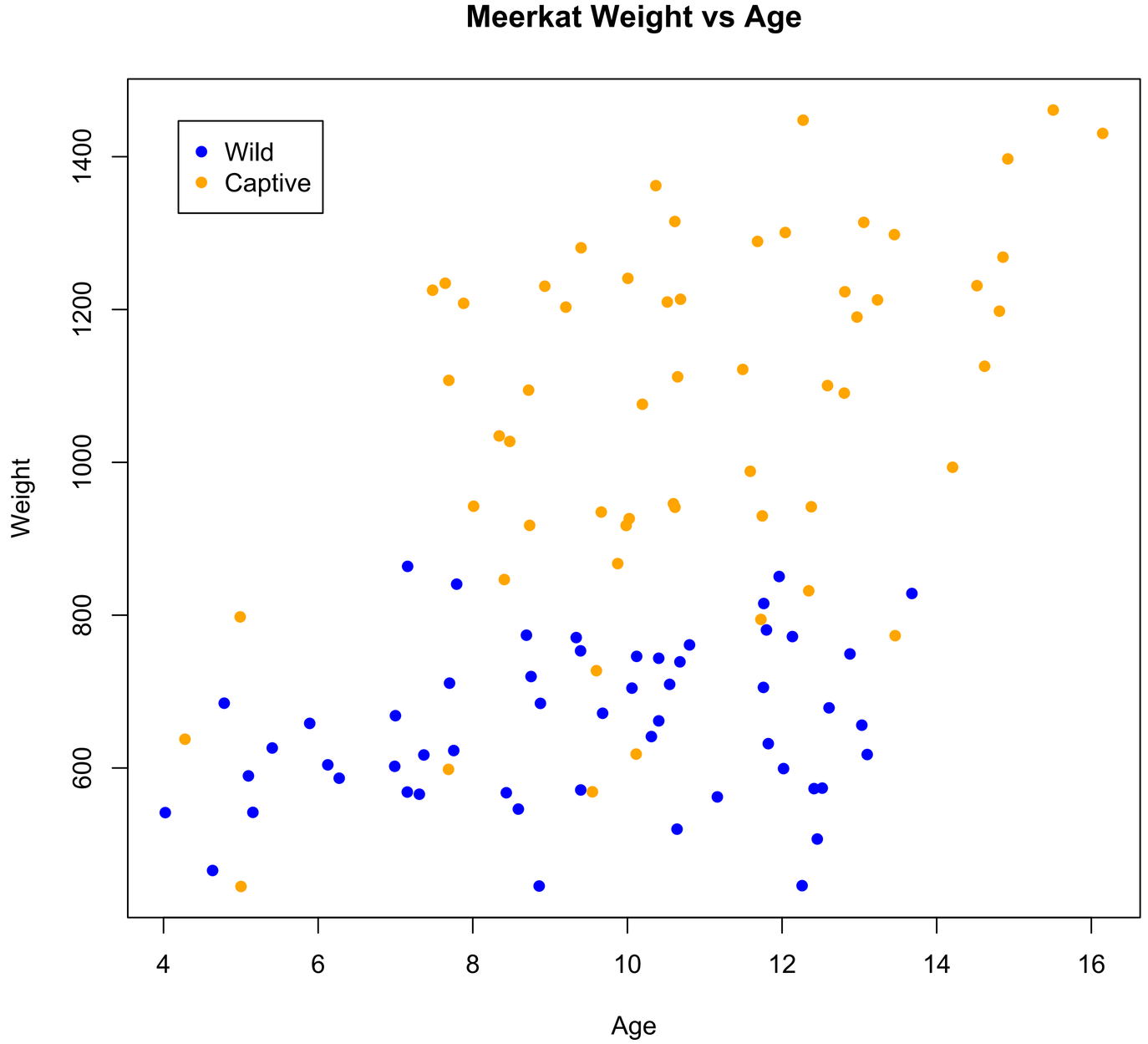

Meerkat Weight Distribution
Frequency
0
5
10
15
400
600
800
1000
1200
1400
Weight (g)

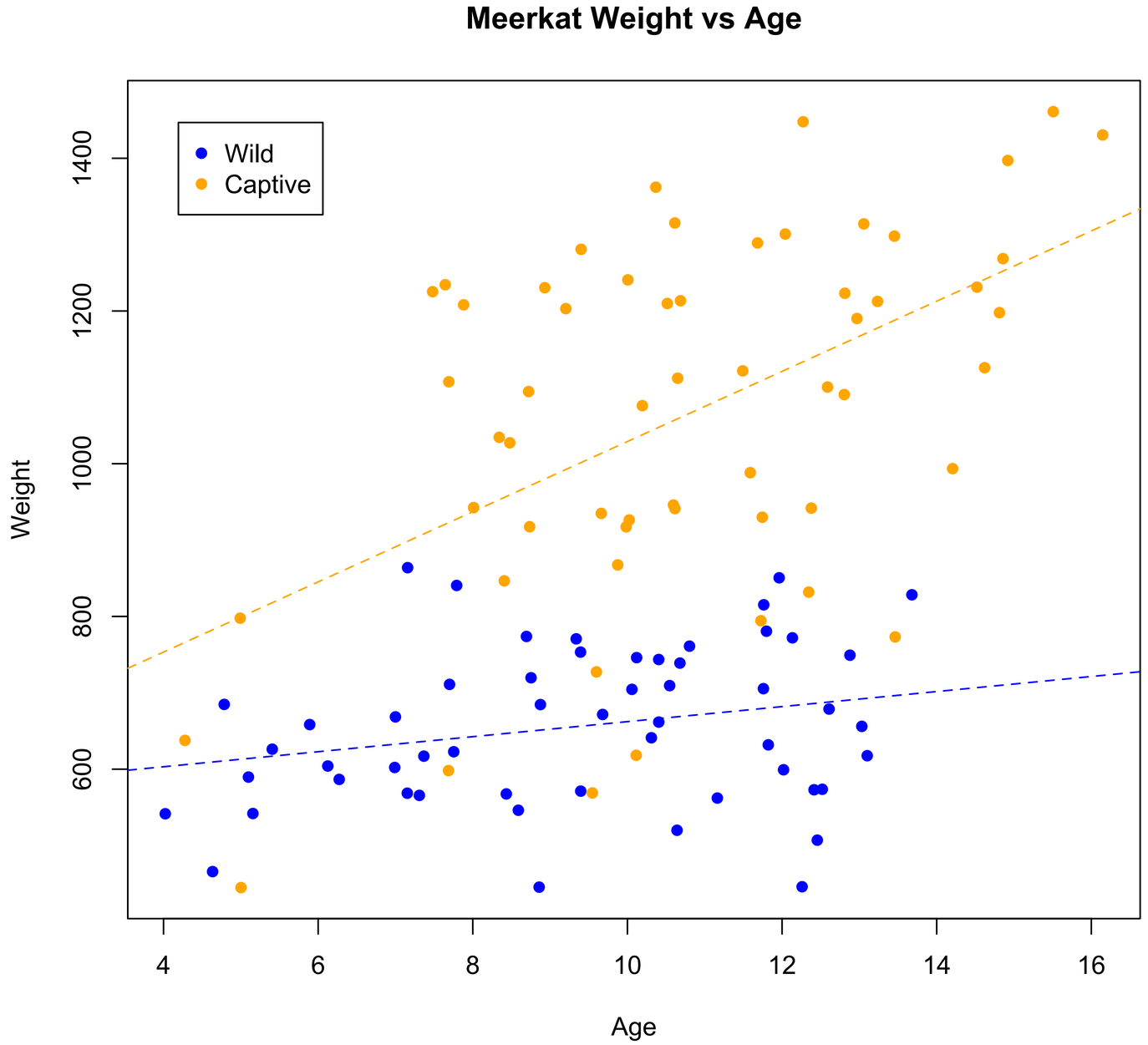

Meerkat Weight Distribution (KDE)
Density
0.0000
0.0005
0.0010
0.0015
500
1000
1500
Weight (g)

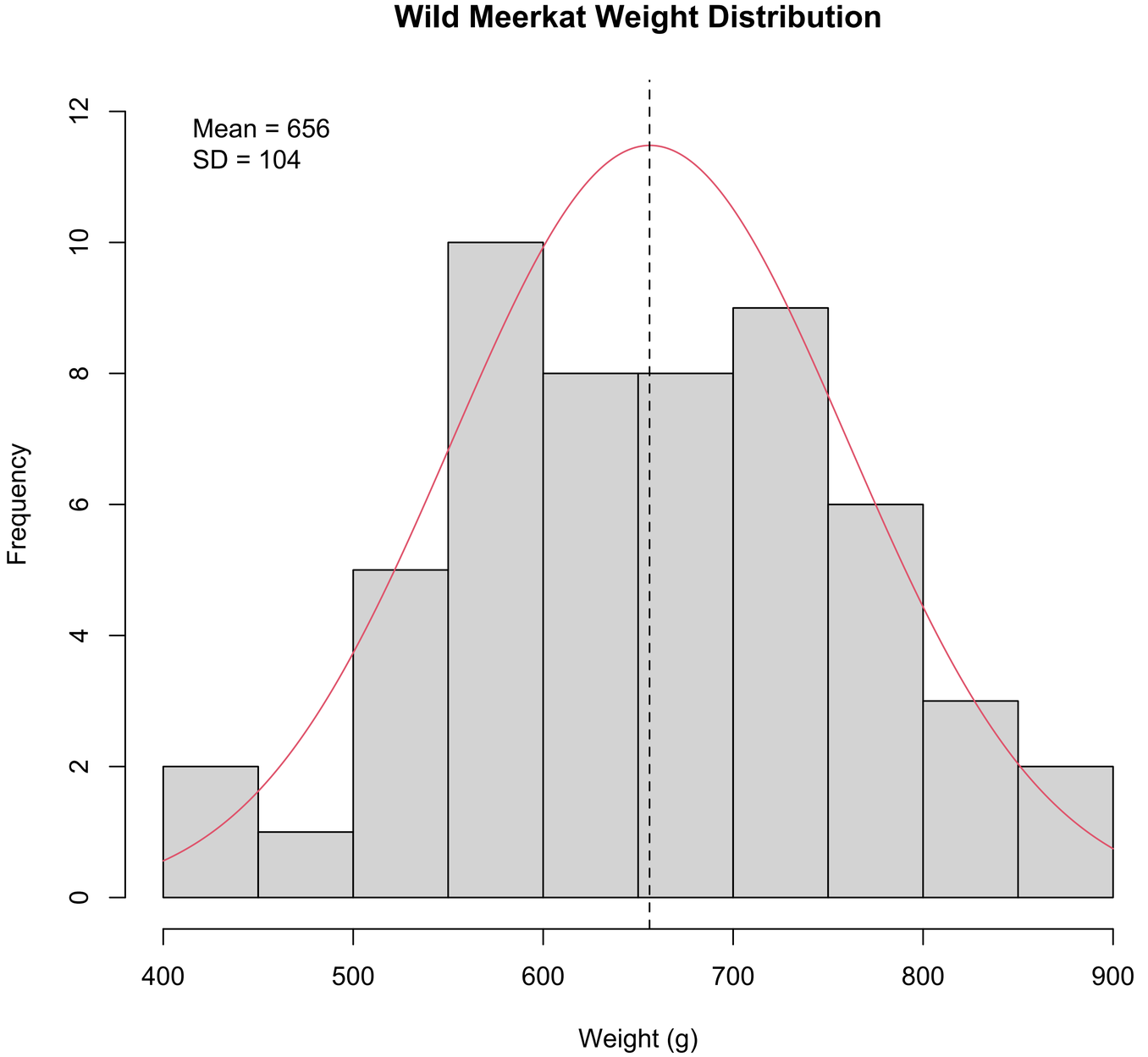

Meerkat Weight vs Age
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

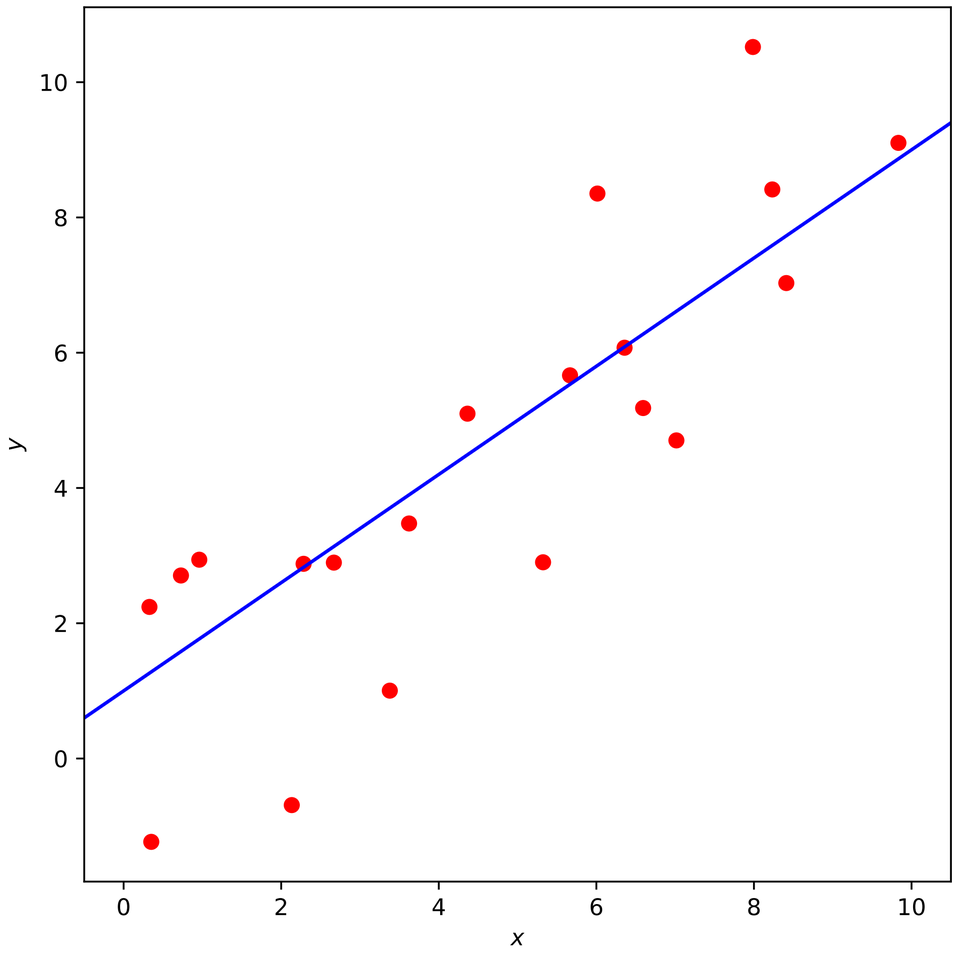

Meerkat Weight vs Age
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

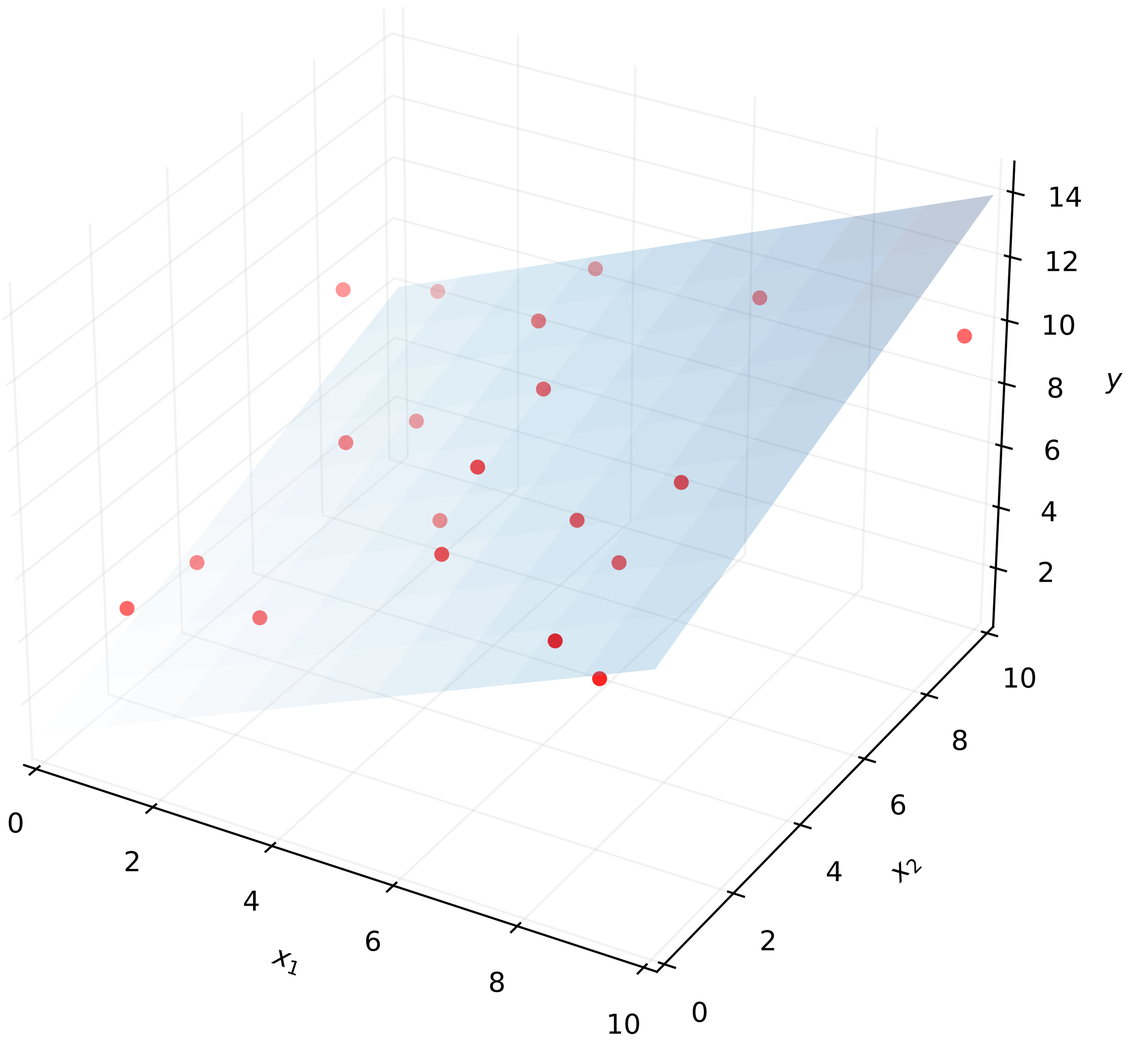

Meerkat Weight vs Age
Male
Female
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

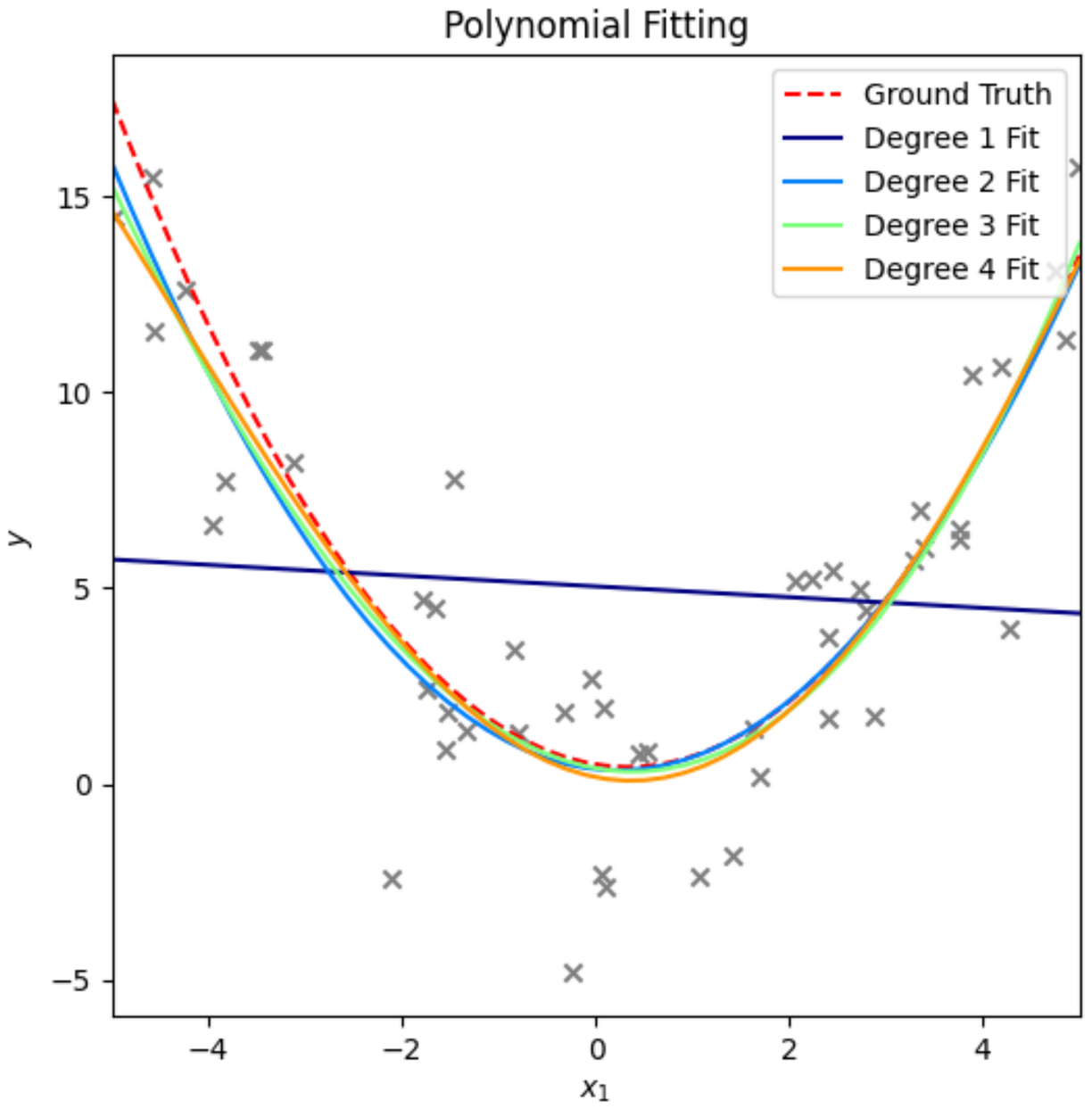

Meerkat Weight vs Age
Wild
Captive
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

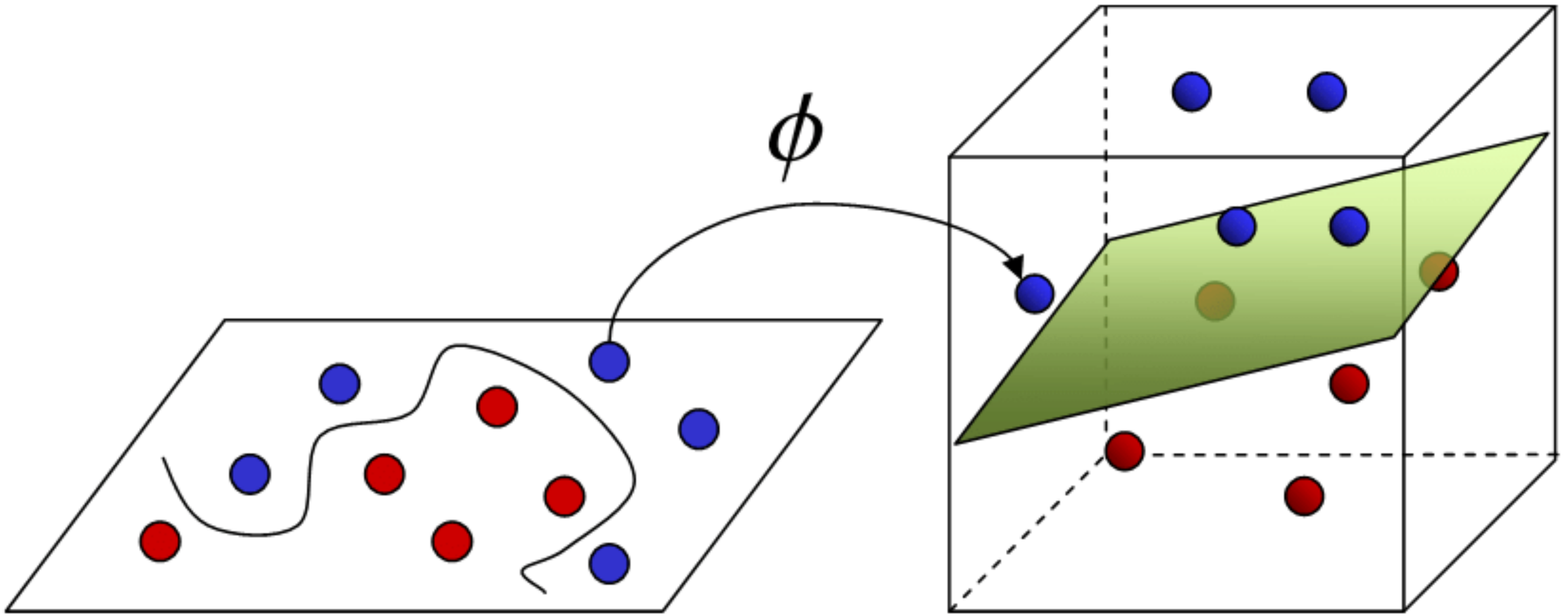

Meerkat Weight vs Age
Wild
Captive
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

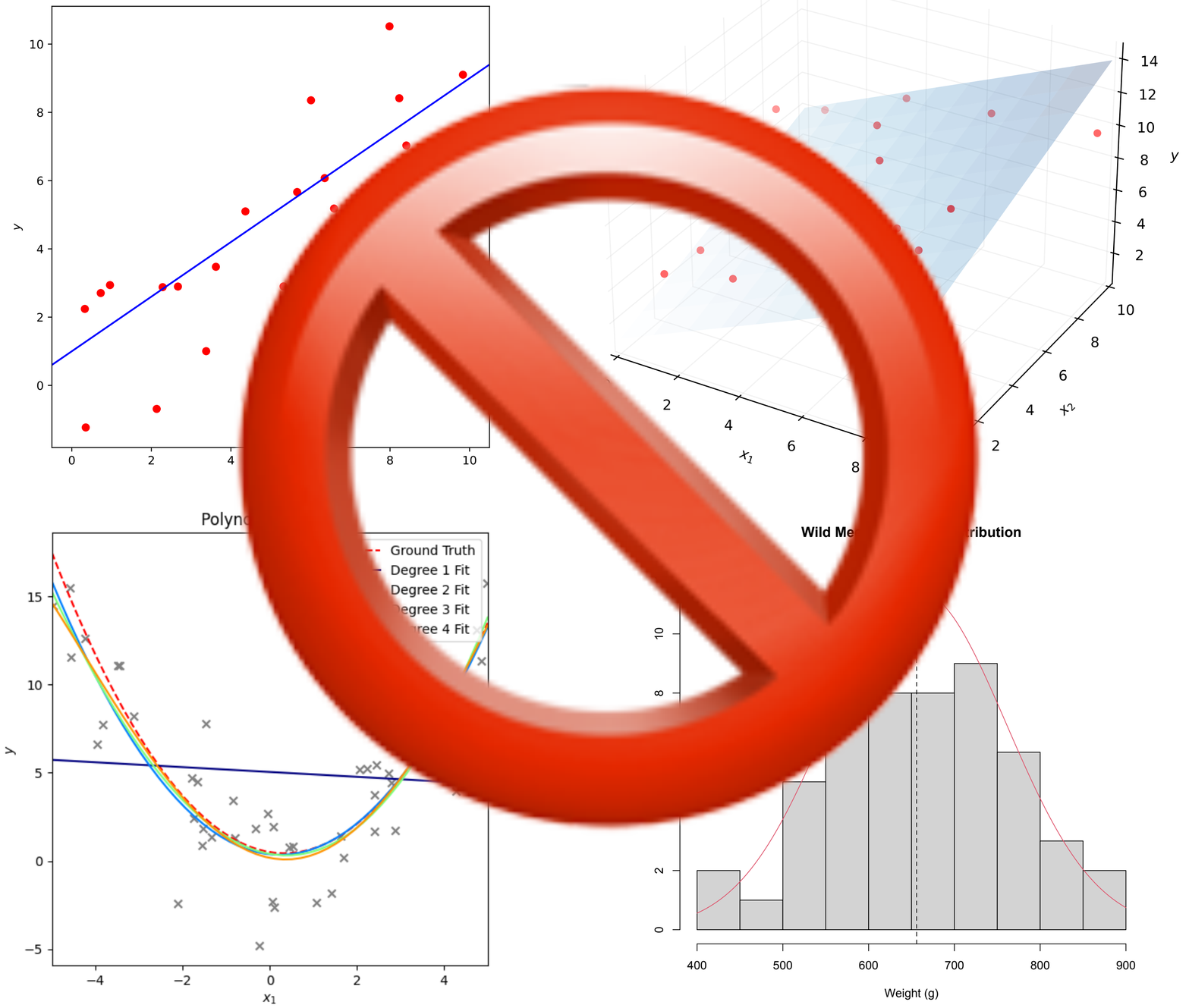

Wild Meerkat Weight Distribution
Mean = 656
SD = 104
Frequency
0
2
4
6
8
10
12
400
500
600
700
800
900
Weight (g)

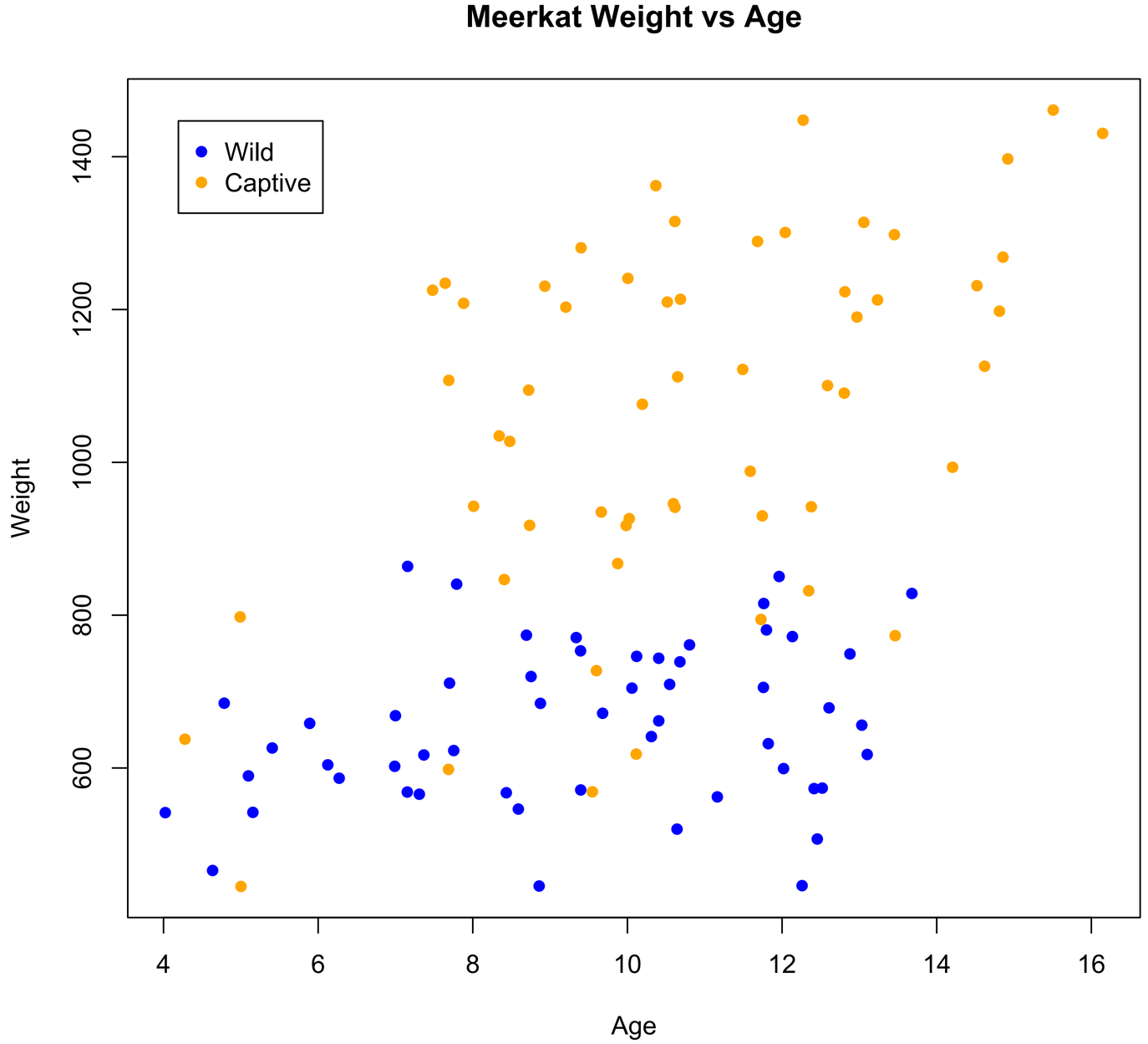

y
x
0
2
4
6
8
10


y
x_1
x_2
0
2
4
6
8
10
12
14

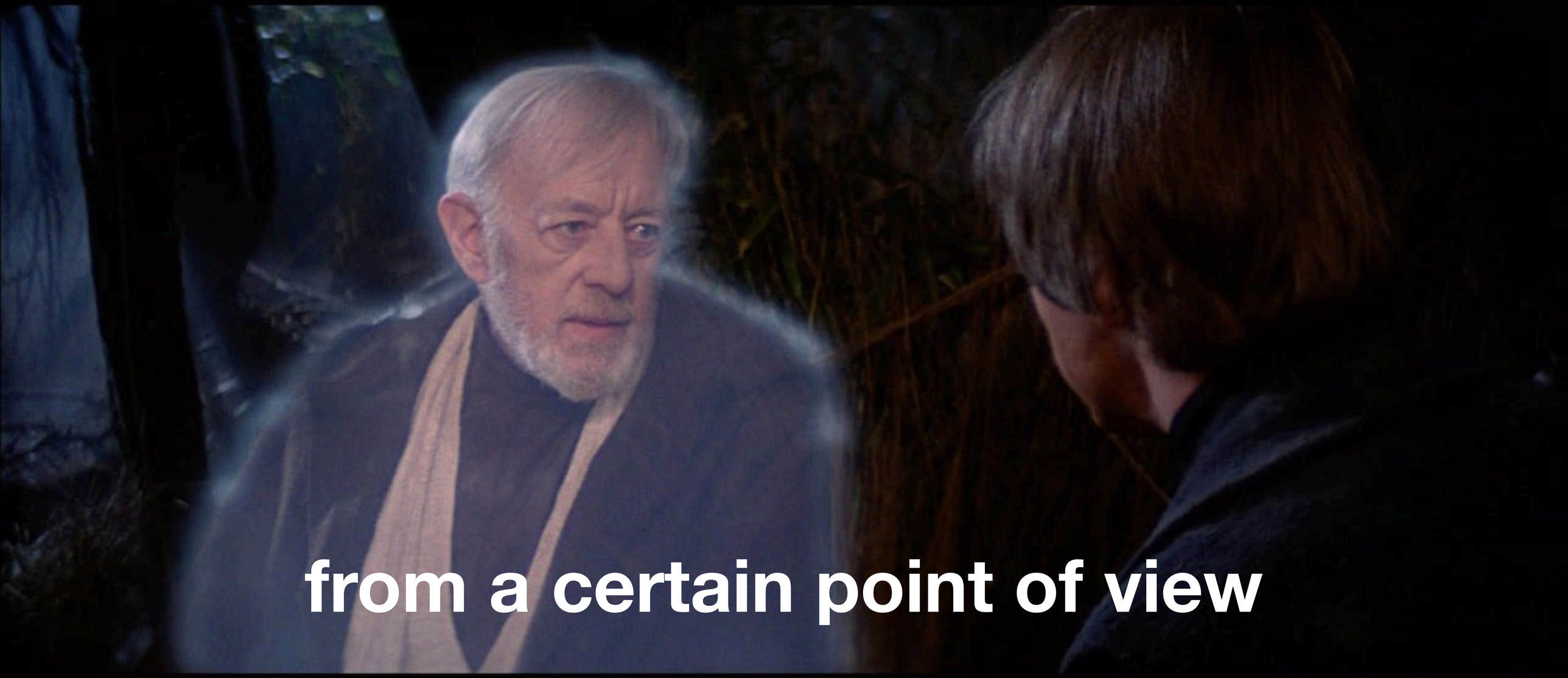

Polynomial Fitting
Ground Truth
Degree 1 Fit
Degree 2 Fit
Degree 3 Fit
Degree 4 Fit
15
10
5
0
−5
y
−4
−2
0
2
4
x_1


ϕ
a) Input Space
b) Feature Space

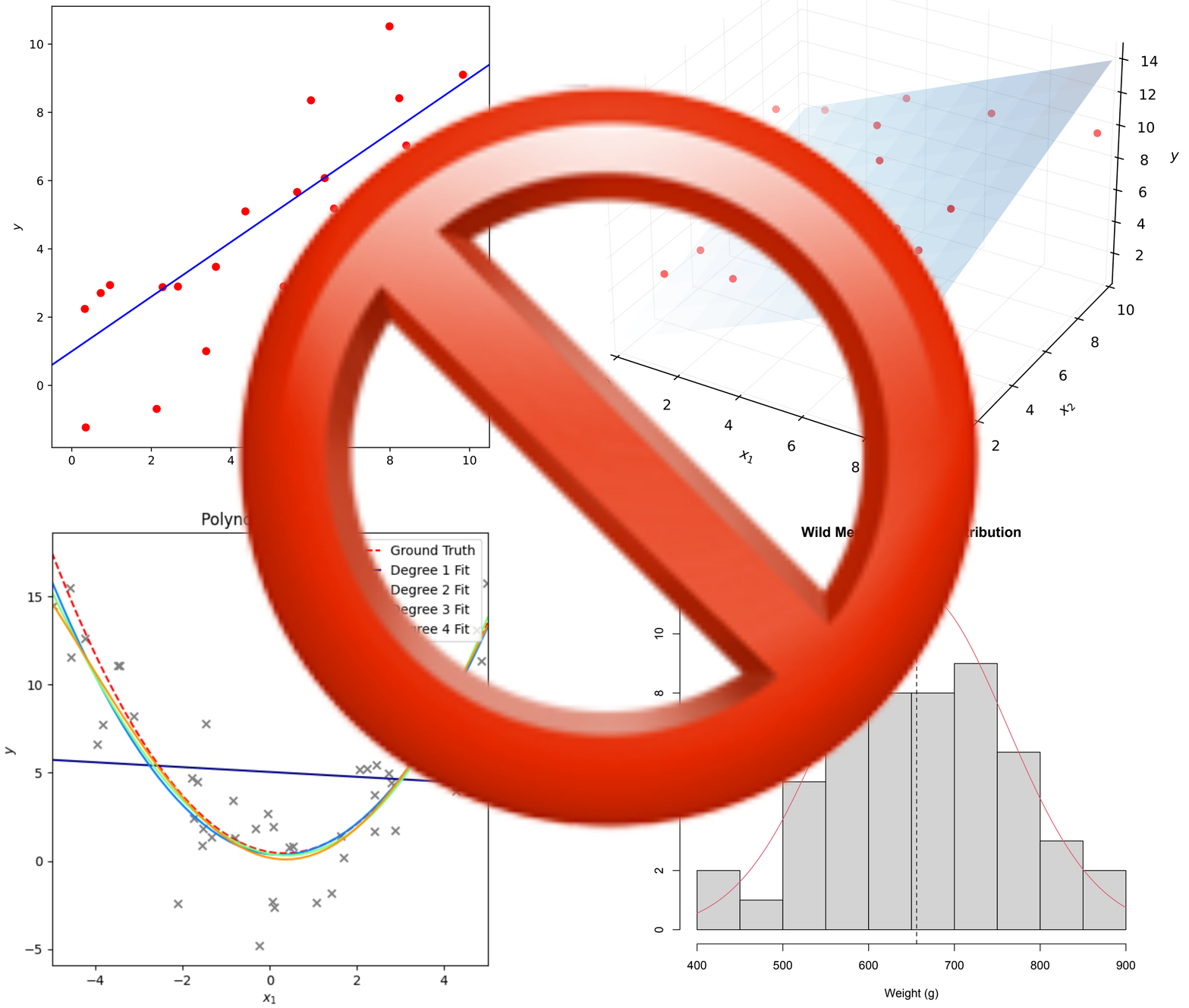

Ground Truth
Degree 1 Fit
Degree 2 Fit
Degree 3 Fit
Degree 4 Fit
Weight (g)

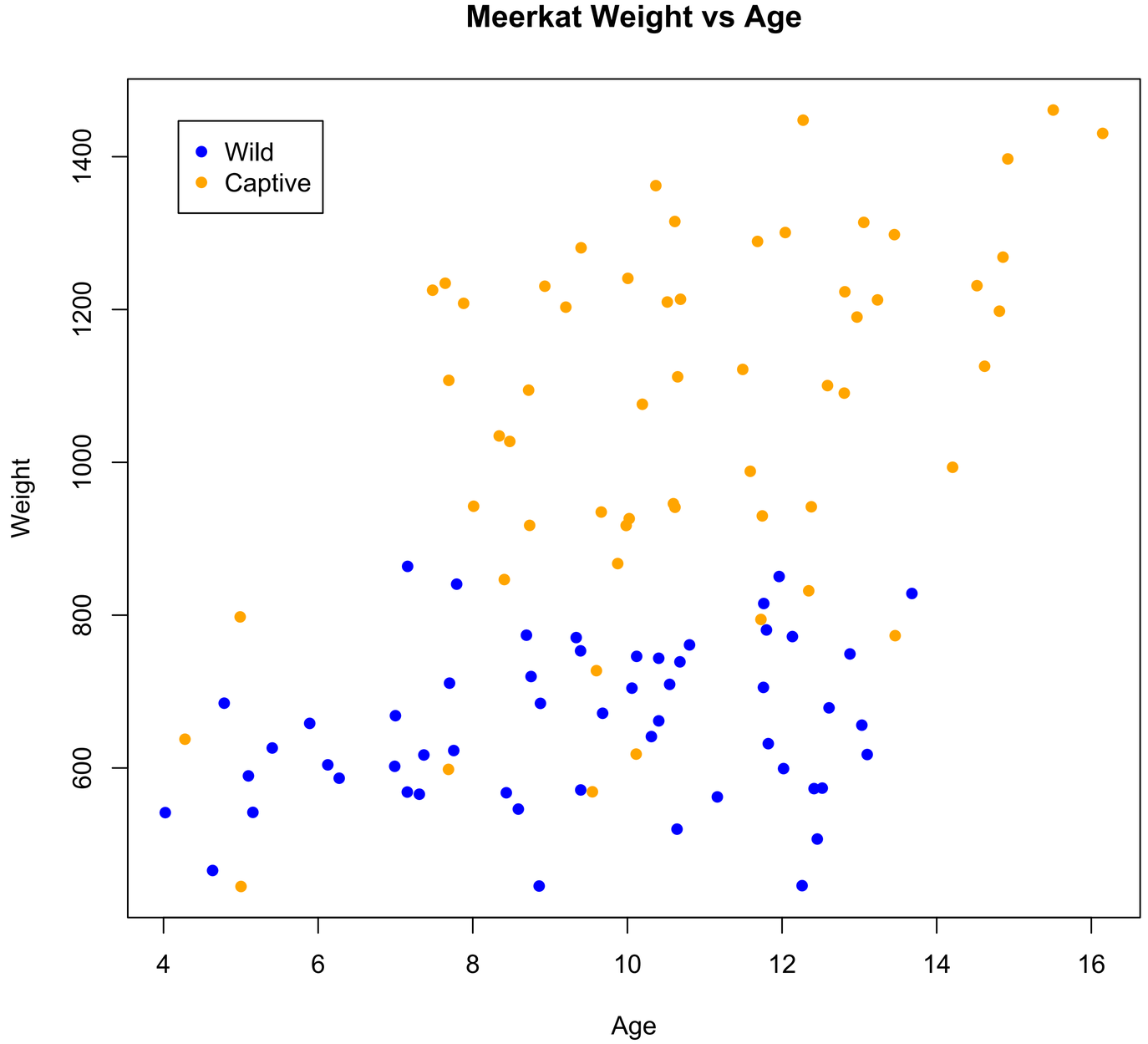

Meerkat Weight vs Age
Wild
Captive
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

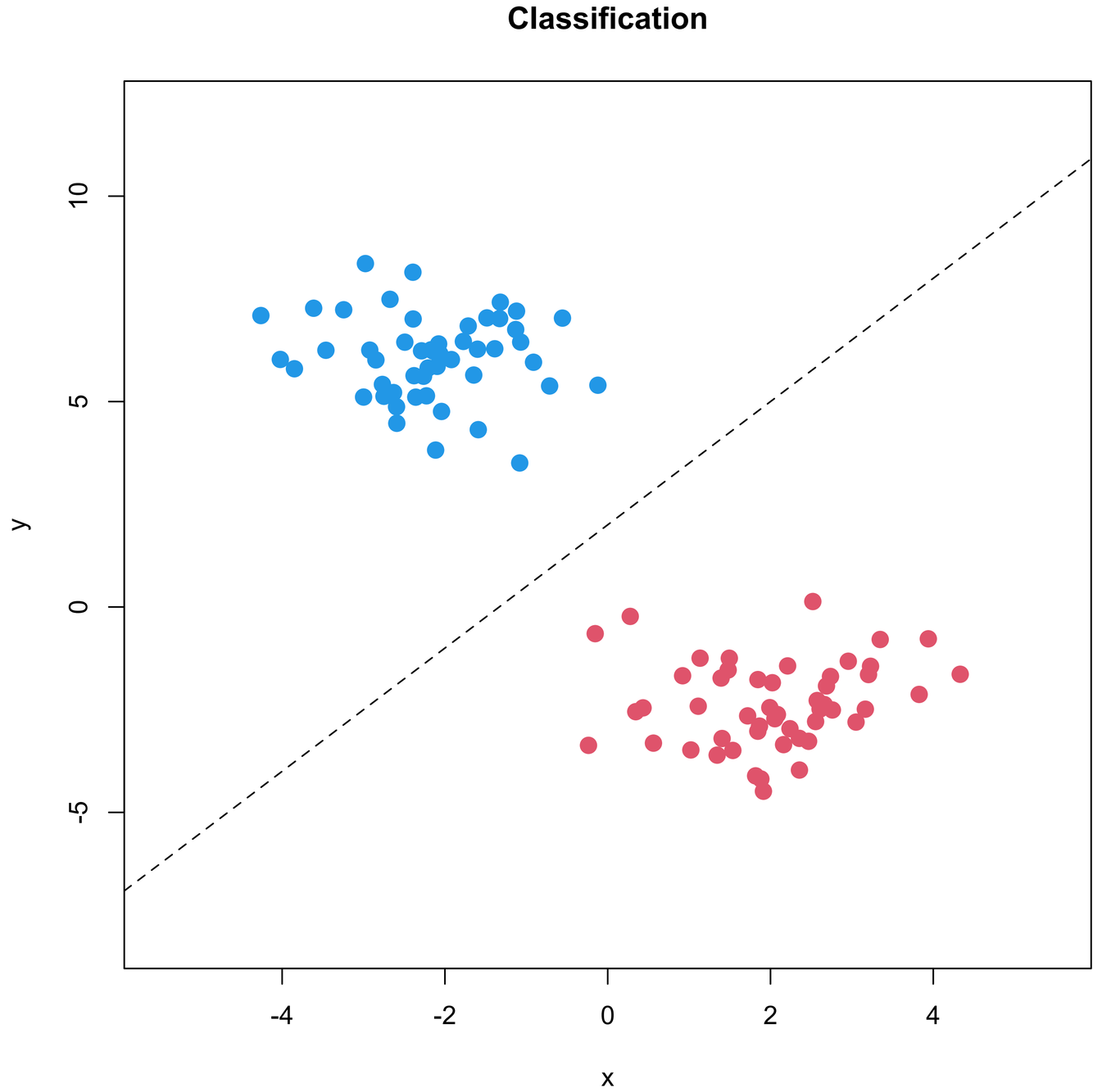


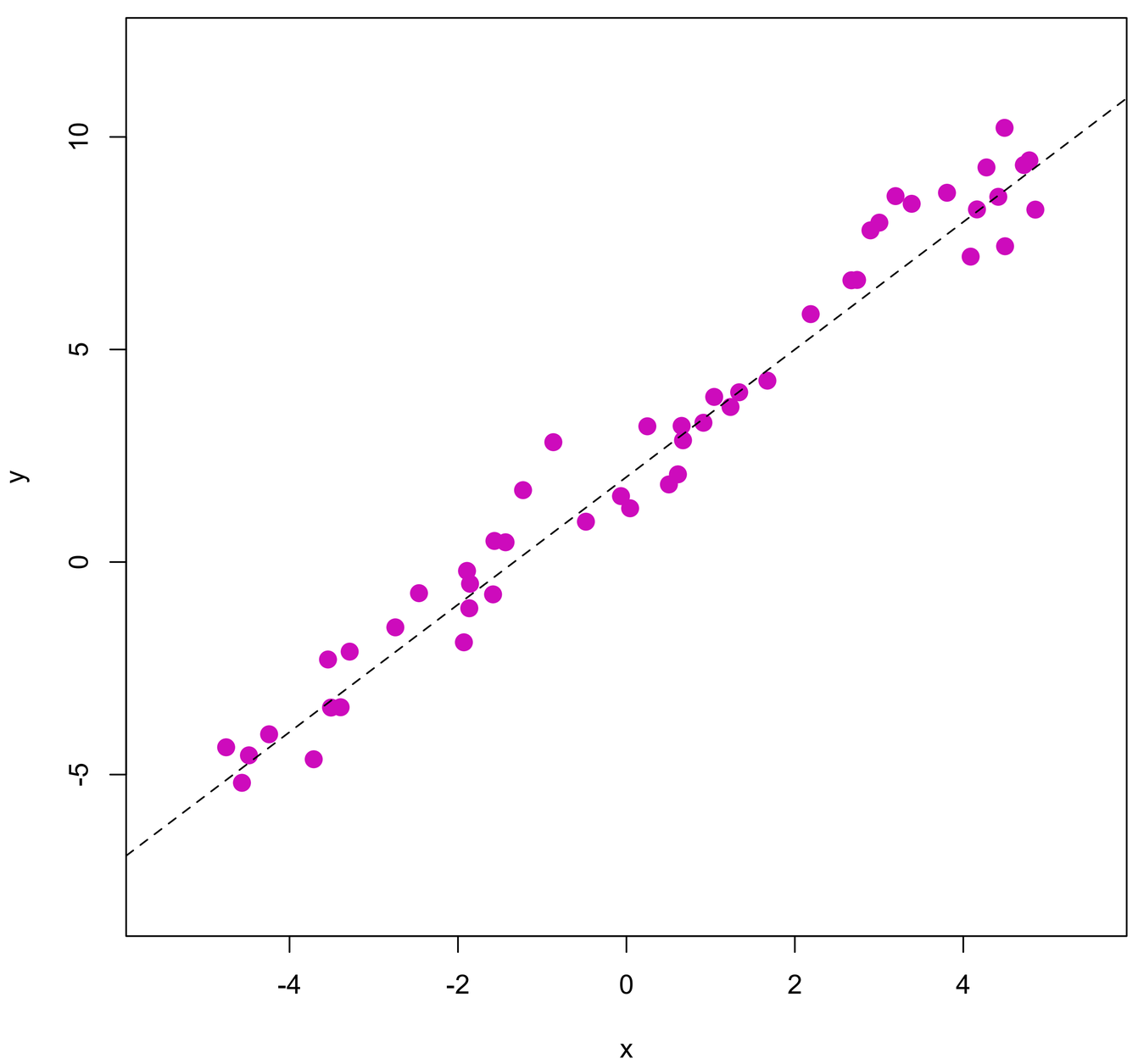

from a certain point of view

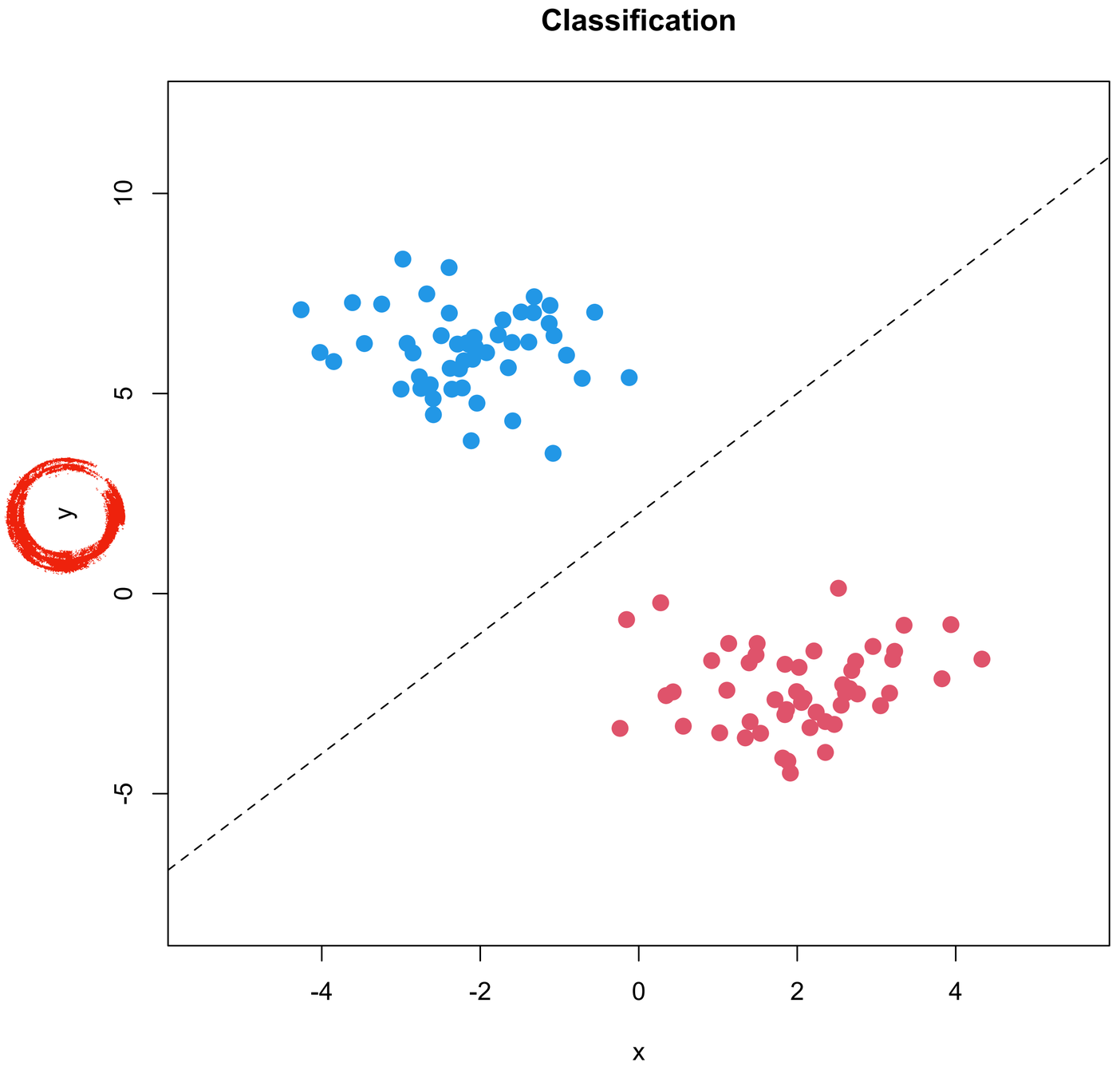
$$L(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}) = ??$$

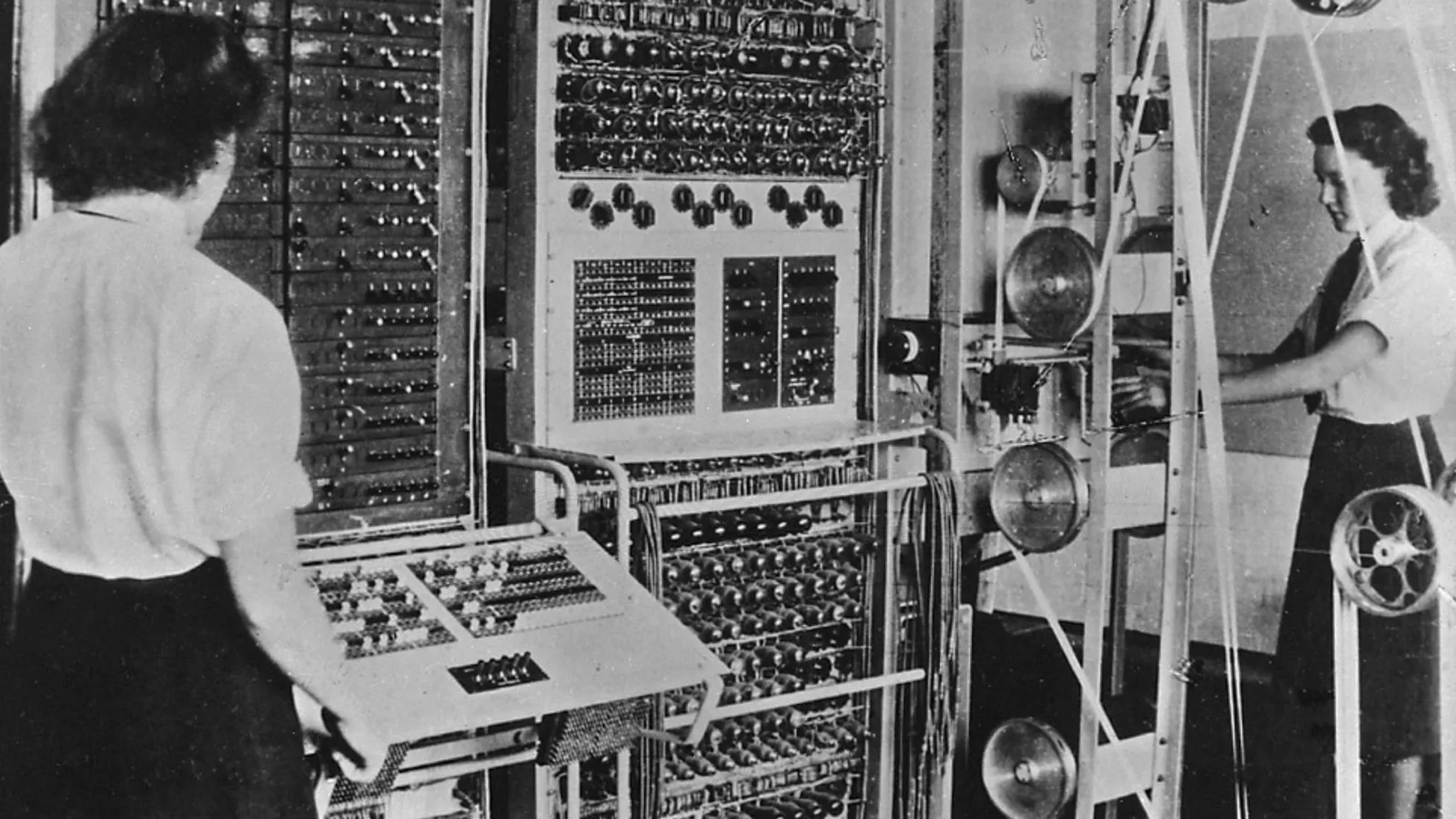

Ground Truth
Degree 1 Fit
Degree 2 Fit
Degree 3 Fit
Degree 4 Fit
Weight (g)


Meerkat Weight vs Age
Wild
Captive
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

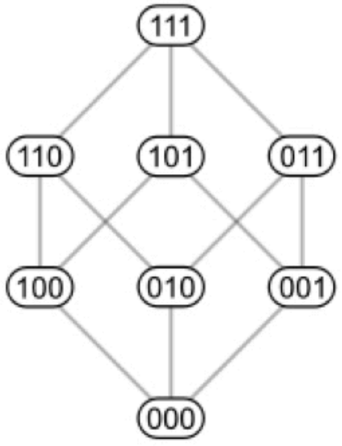

Classification
y
10
5
0
-5
-4
-2
0
2
4
x

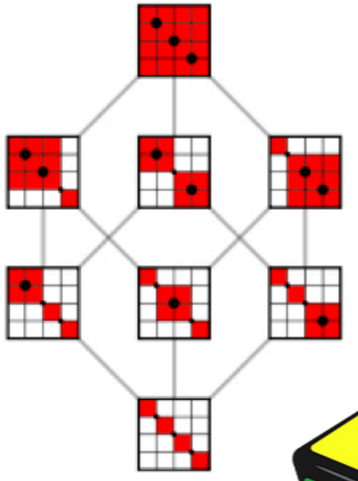

Regression
y
x
10
5
0
-5
-4
-2
0
2
4

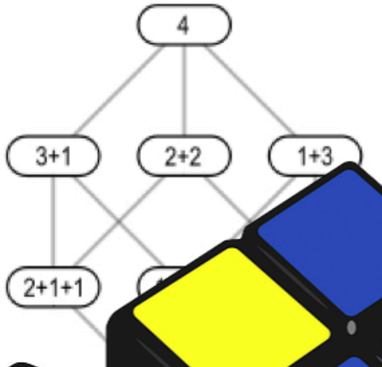

Classification
y
10
5
0
-5
-4
-2
0
2
4
x

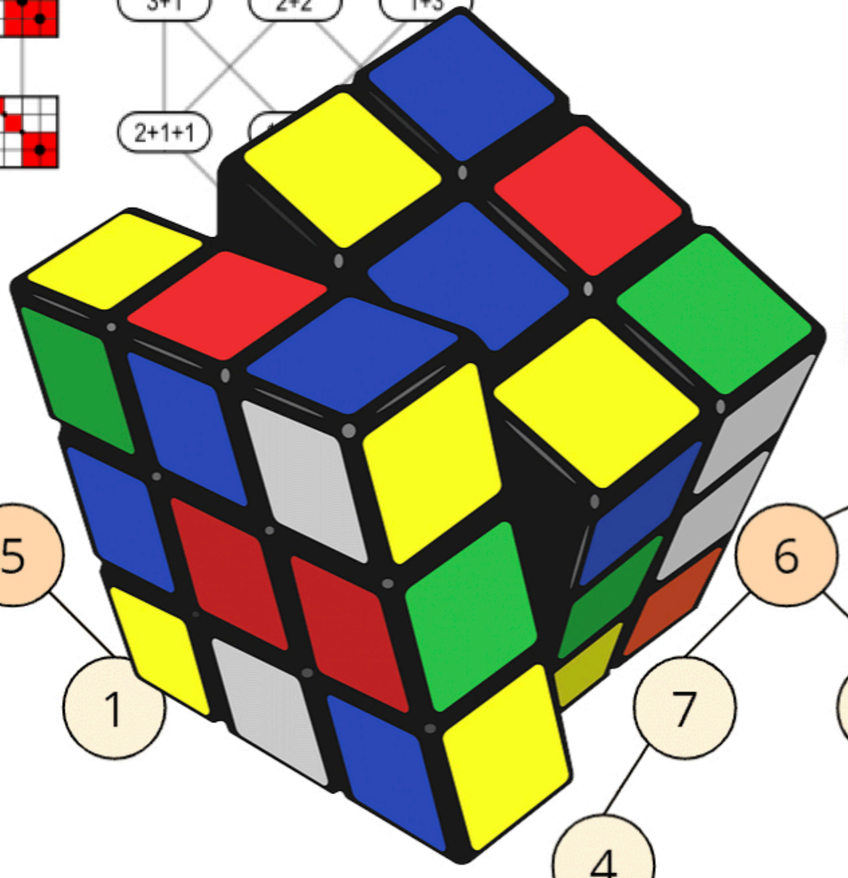


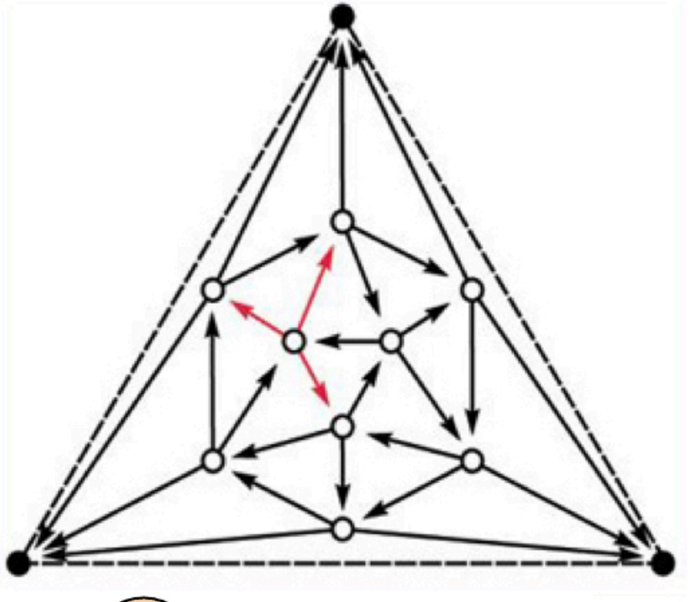


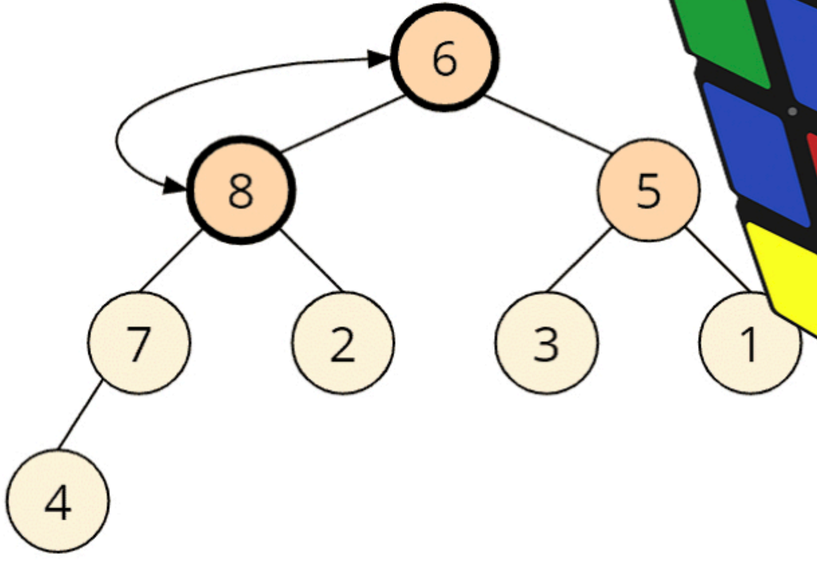

111
110
101
011
100
010
001
000

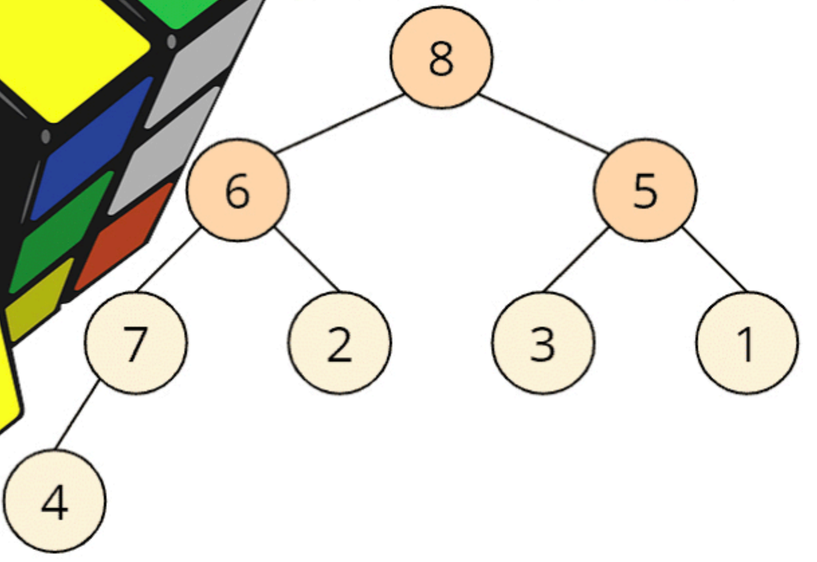



4
3+1
2+2
1+3
2+1+1



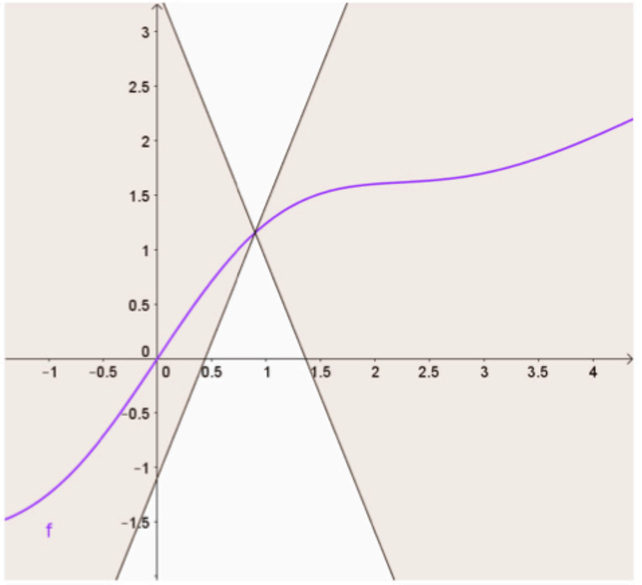


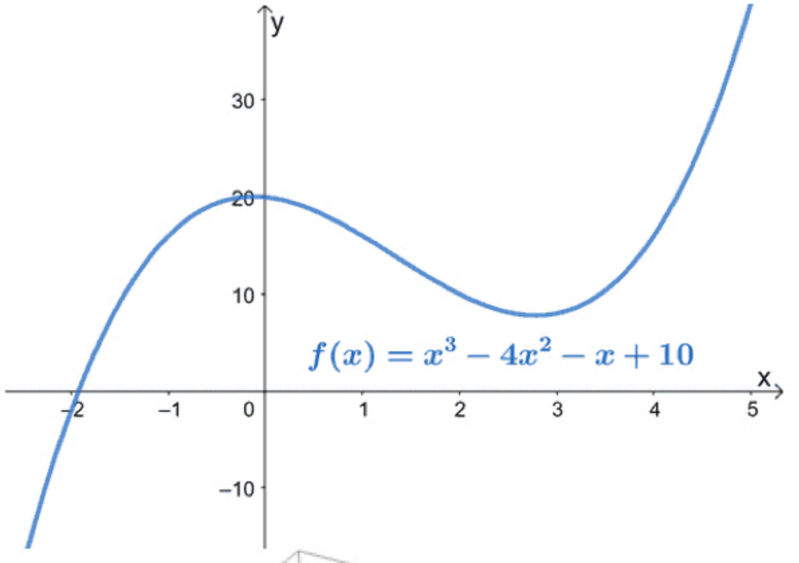

6
8
5
7
2
3
1
4

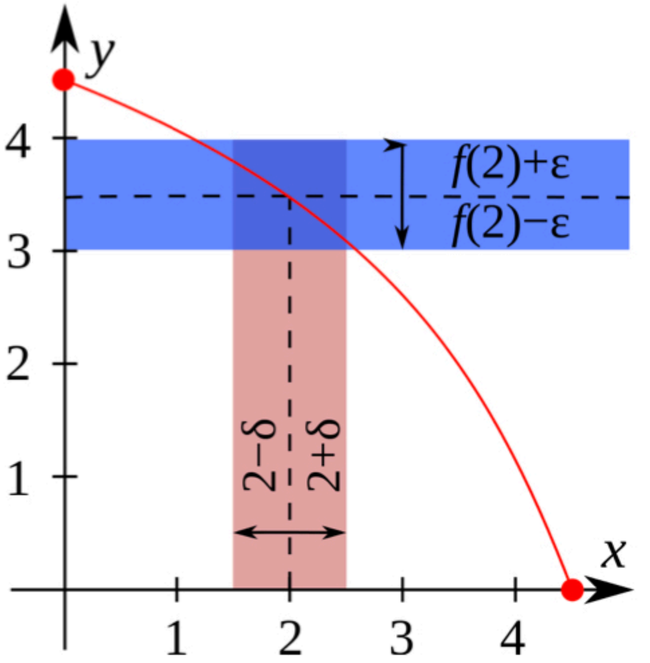

8
6
5
7
2
3
1
4

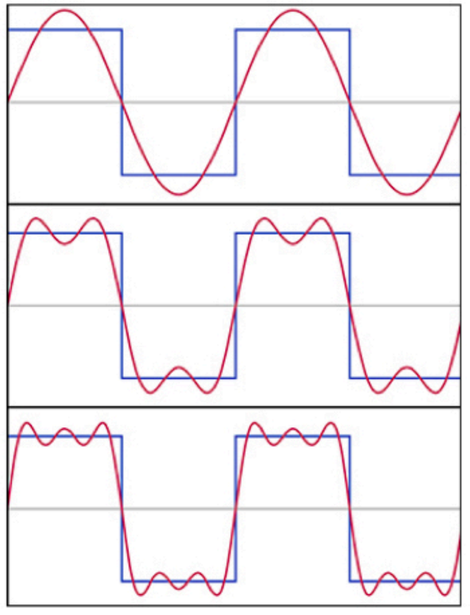

6
8
5
7
2
3
1
4
9

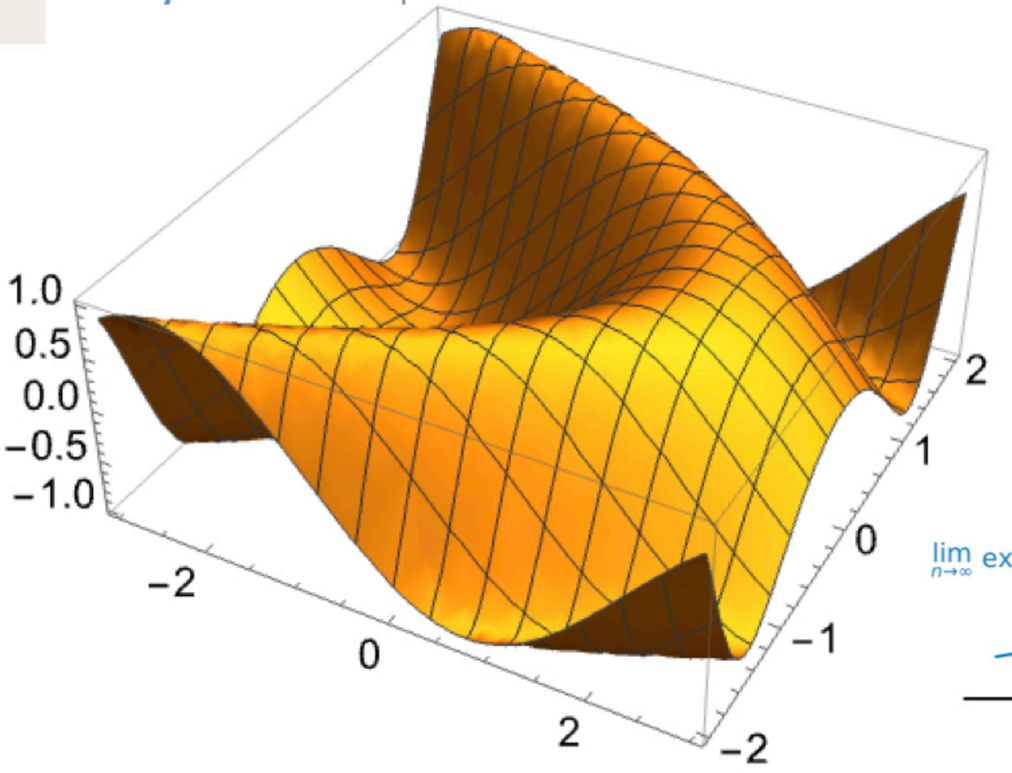

8
6
5
7
2
3
1
4
9

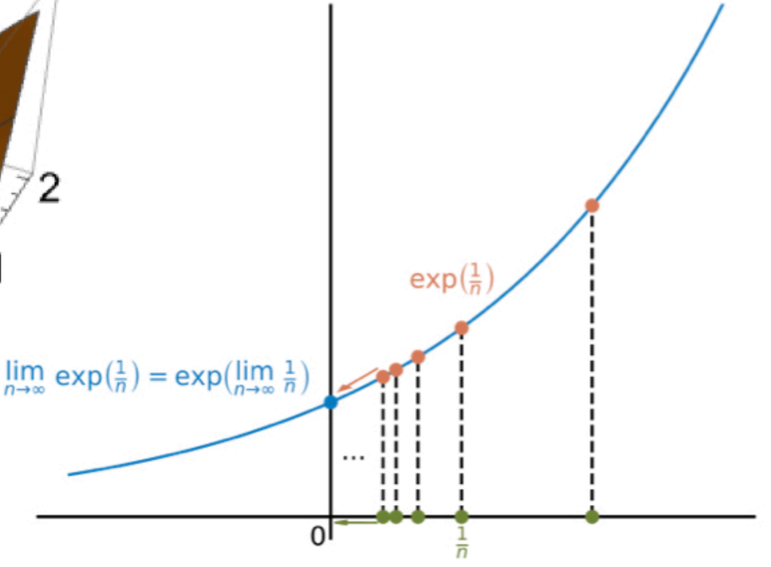

f


y
x
$f(x) = x^3 - 4x^2 - x + 10$

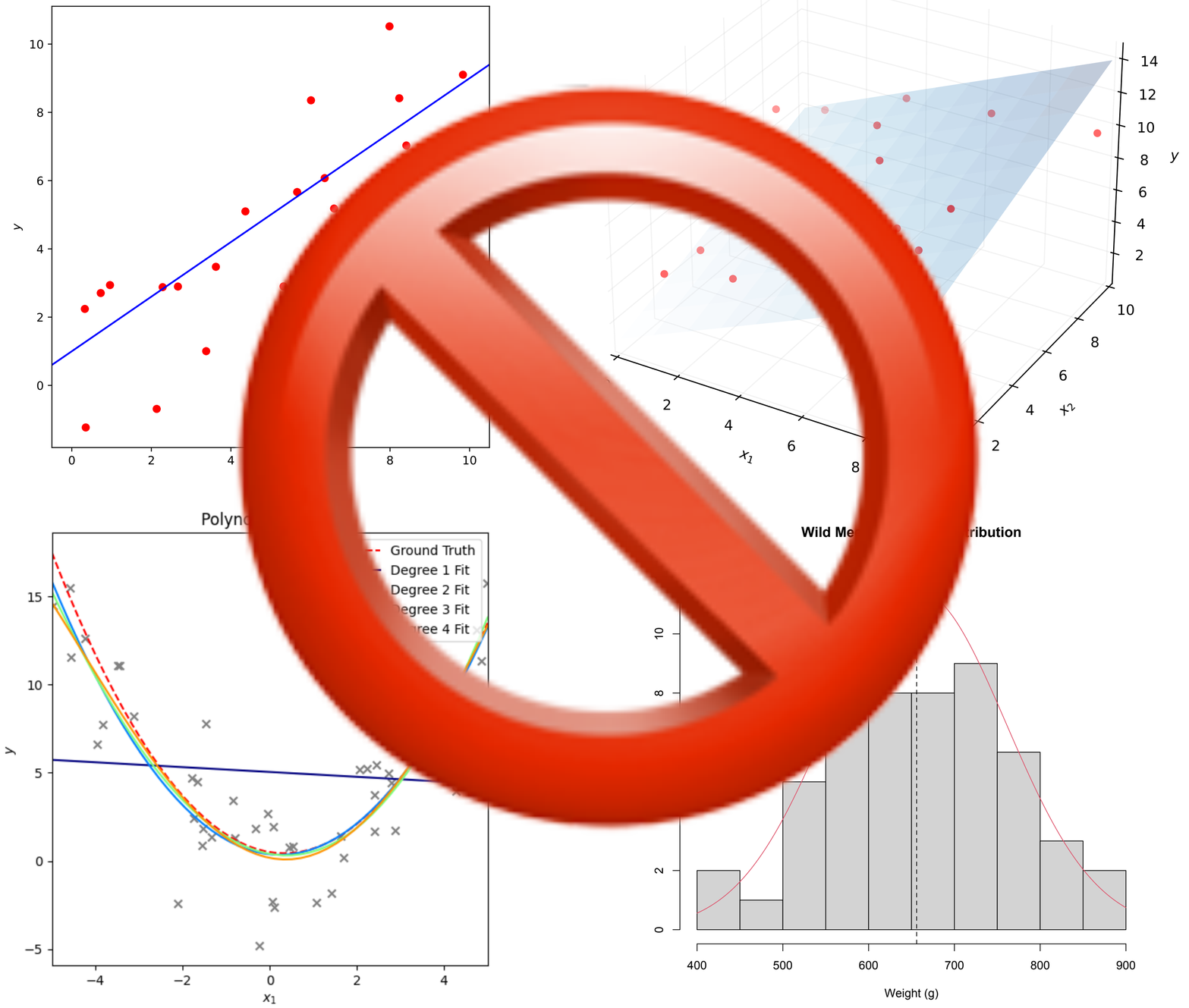

y
x
$f(2)+\varepsilon$
$f(2)-\varepsilon$
$2-\delta$
$2+\delta$



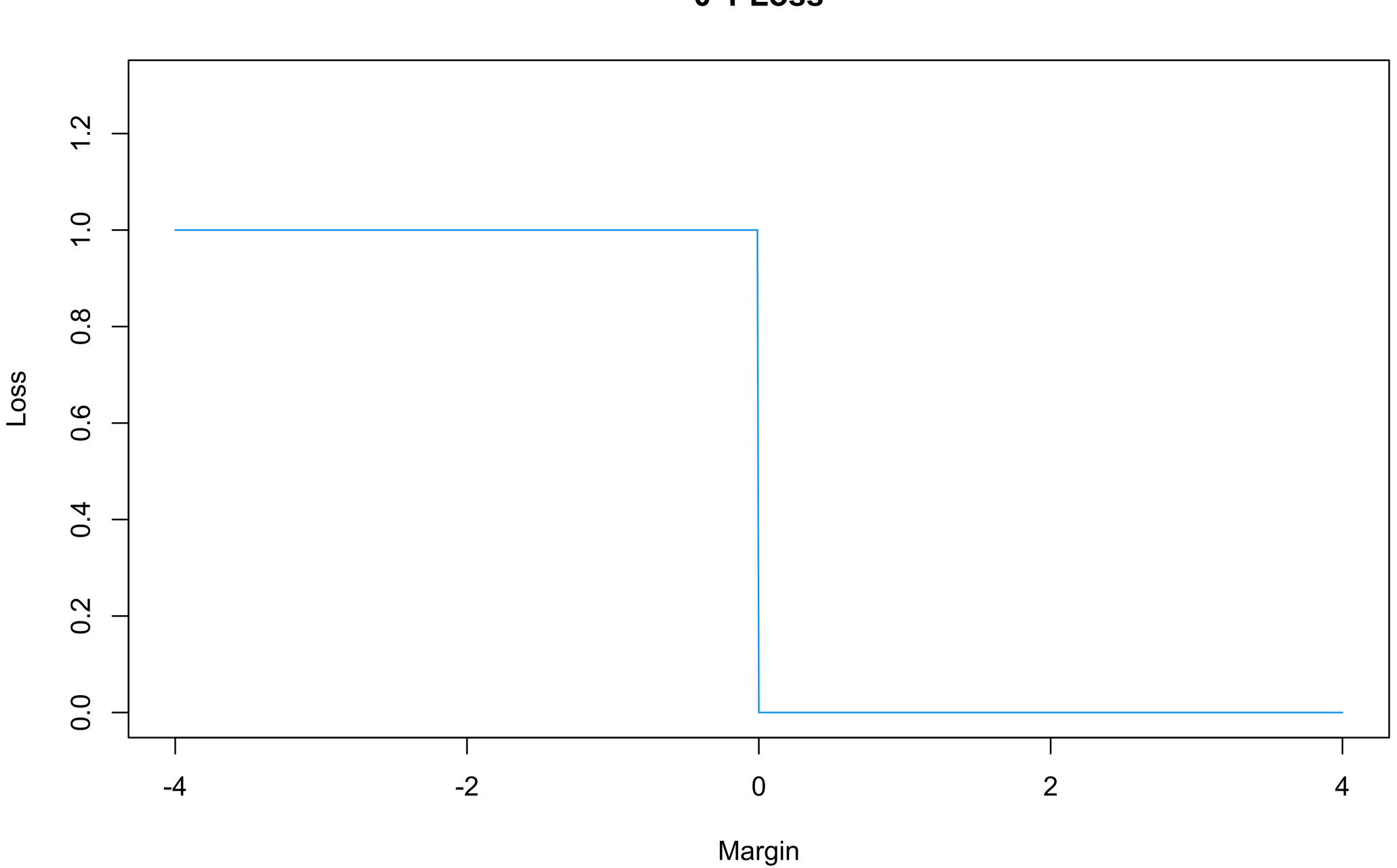


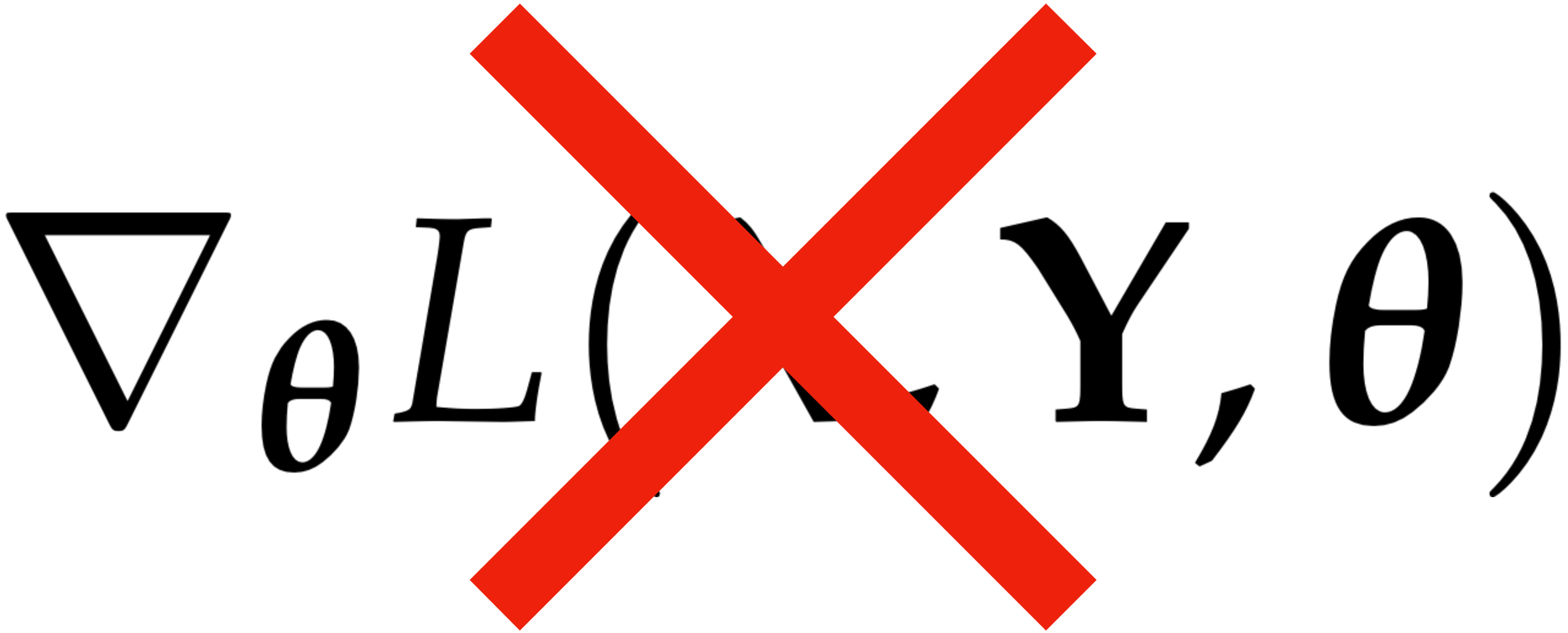

$\lim_{n\to\infty} \exp(\frac{1}{n}) = \exp(\lim_{n\to\infty} \frac{1}{n})$
$\exp(\frac{1}{n})$
$\frac{1}{n}$


$\frac{\partial y}{\partial x_1} = WTAF?$

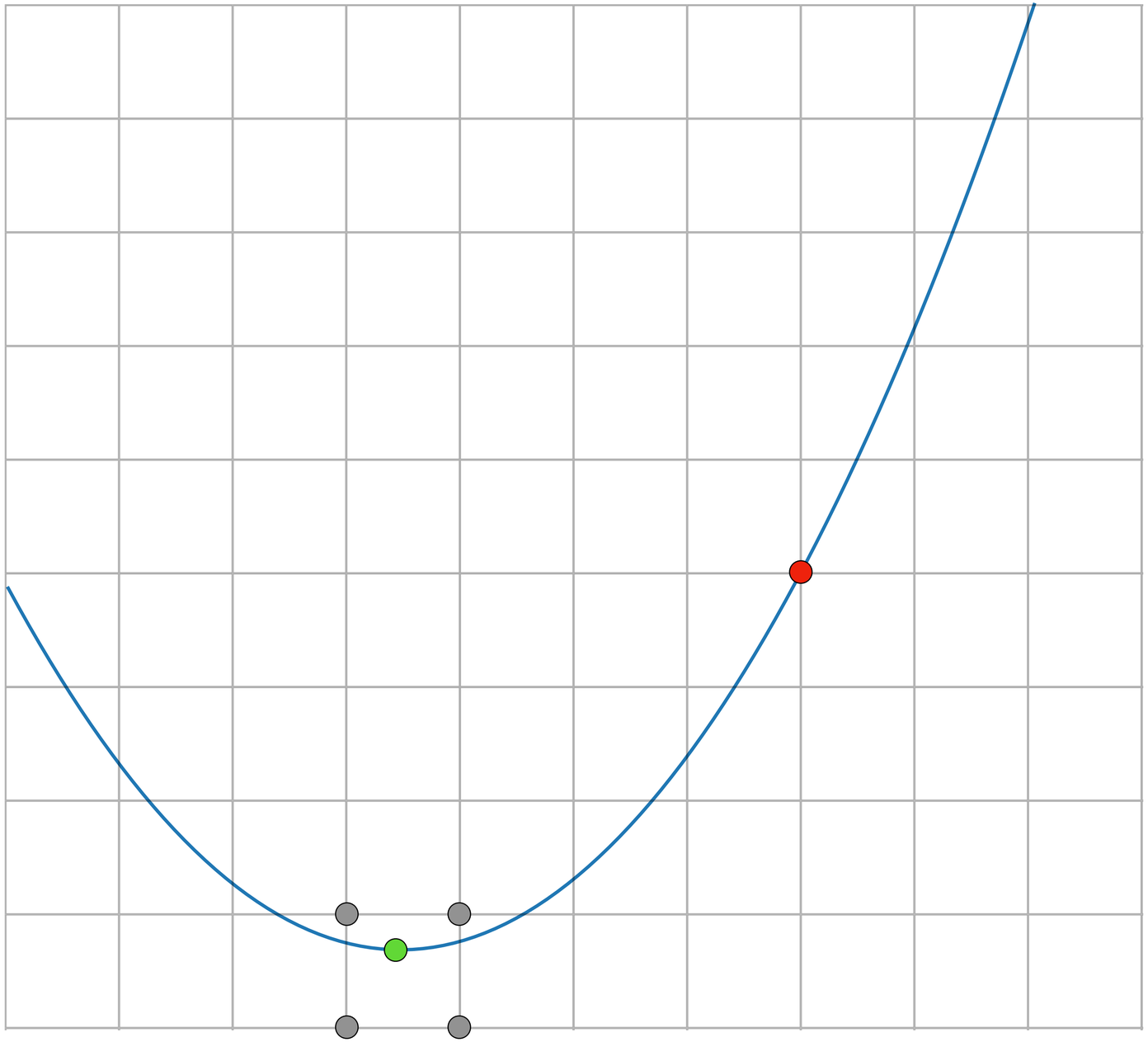

Ground Truth
Degree 1 Fit
Degree 2 Fit
Degree 3 Fit
Degree 4 Fit
Weight (g)

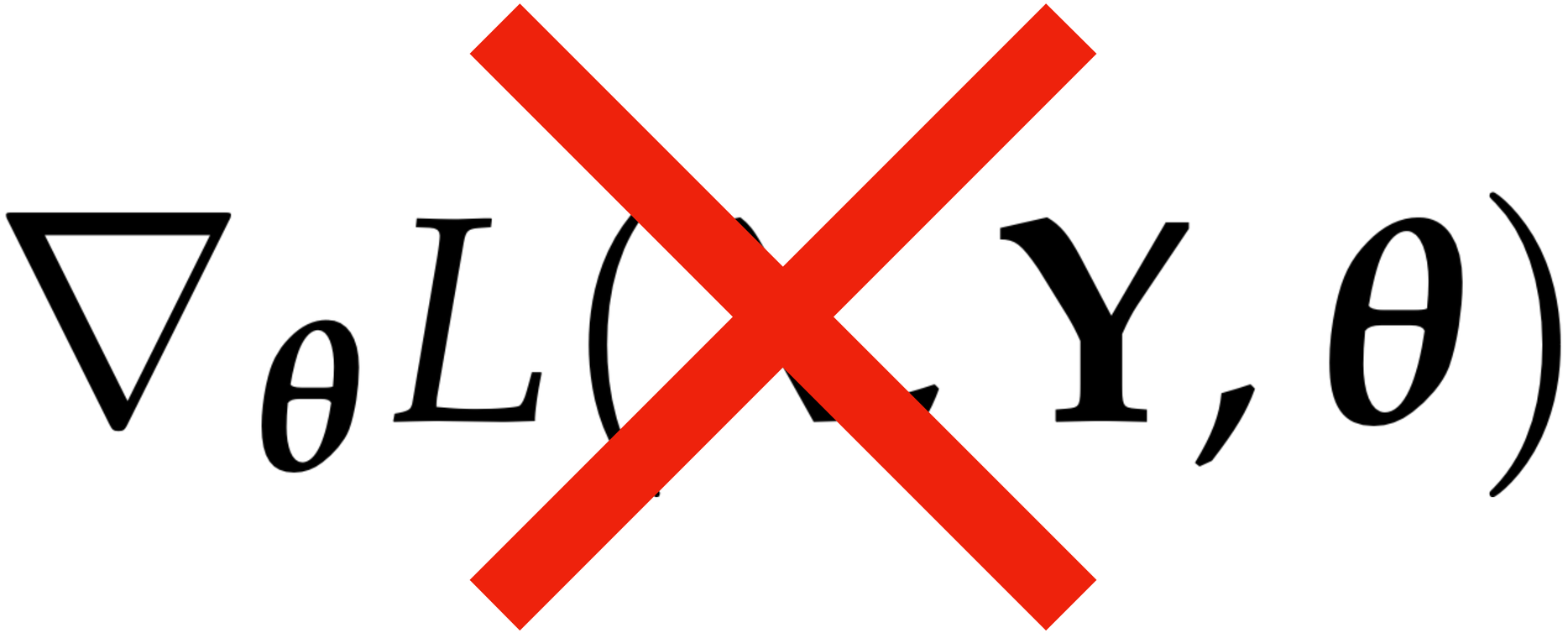
$$L(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}) = ??$$


0-1 Loss
Loss
0.0
0.2
0.4
0.6
0.8
1.0
1.2
-4
-2
0
2
4
Margin



constraint relaxation

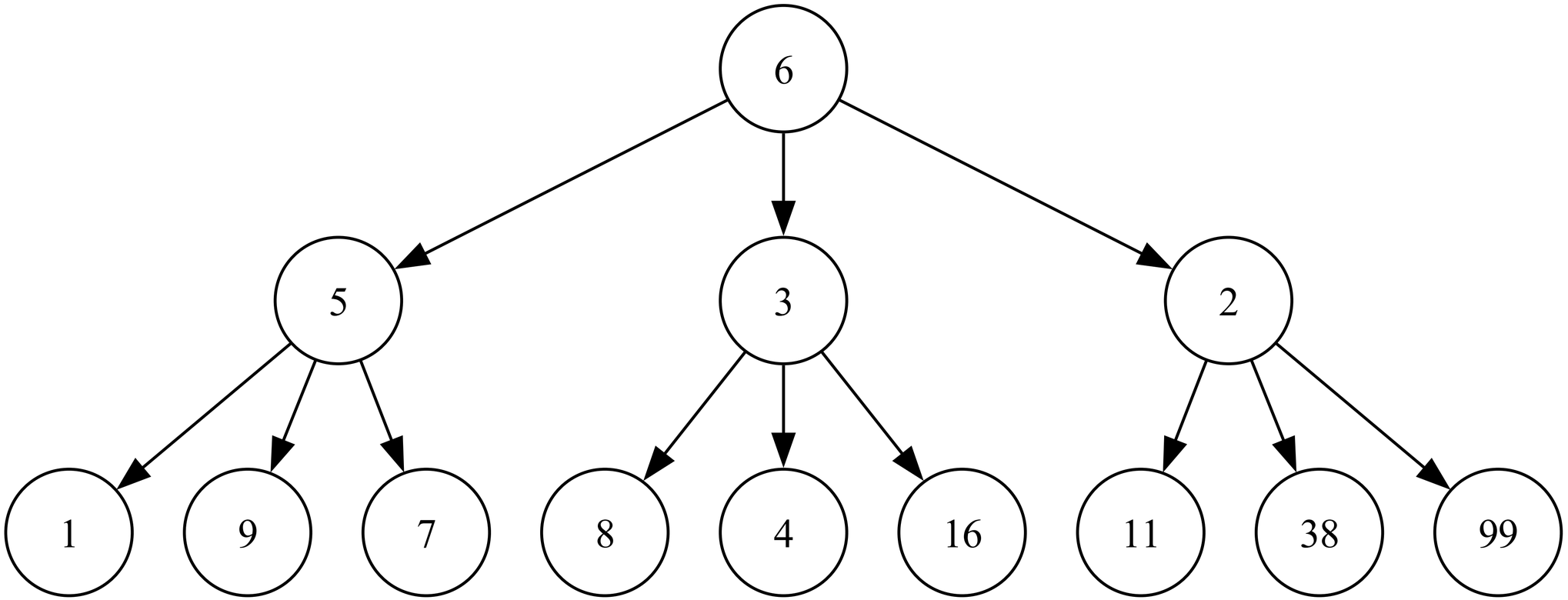


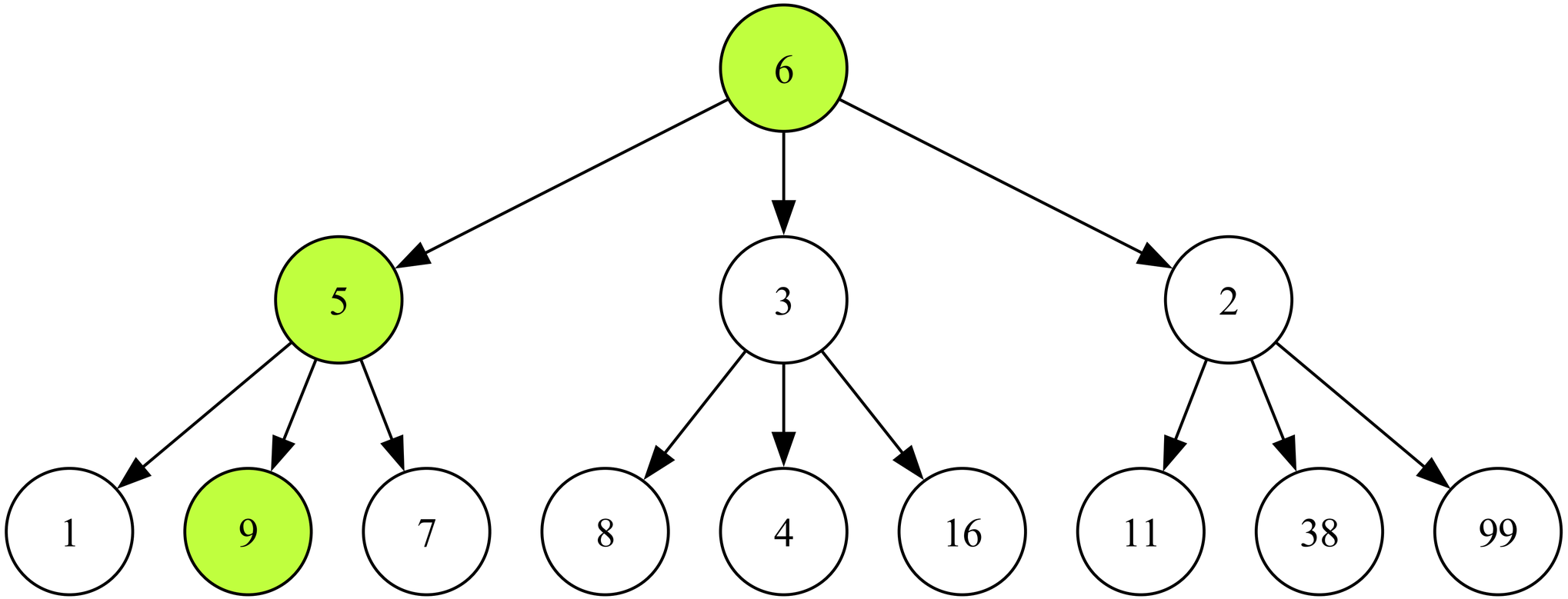


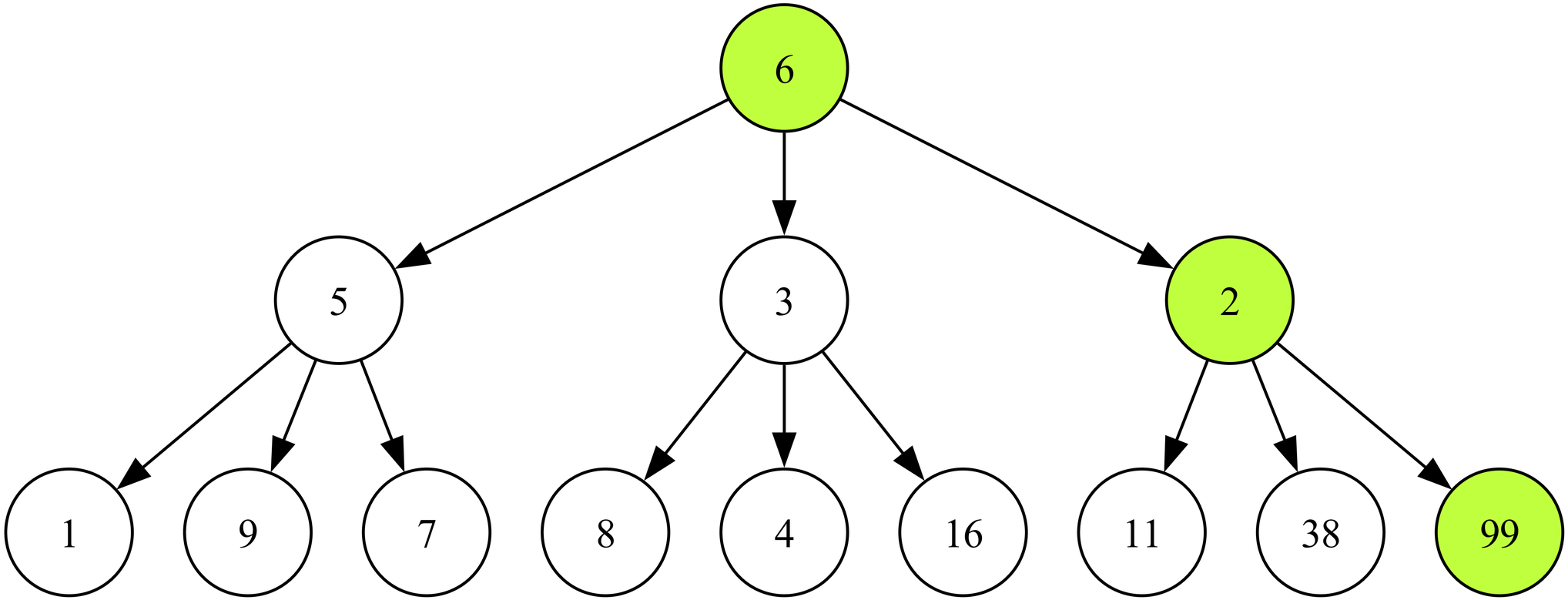


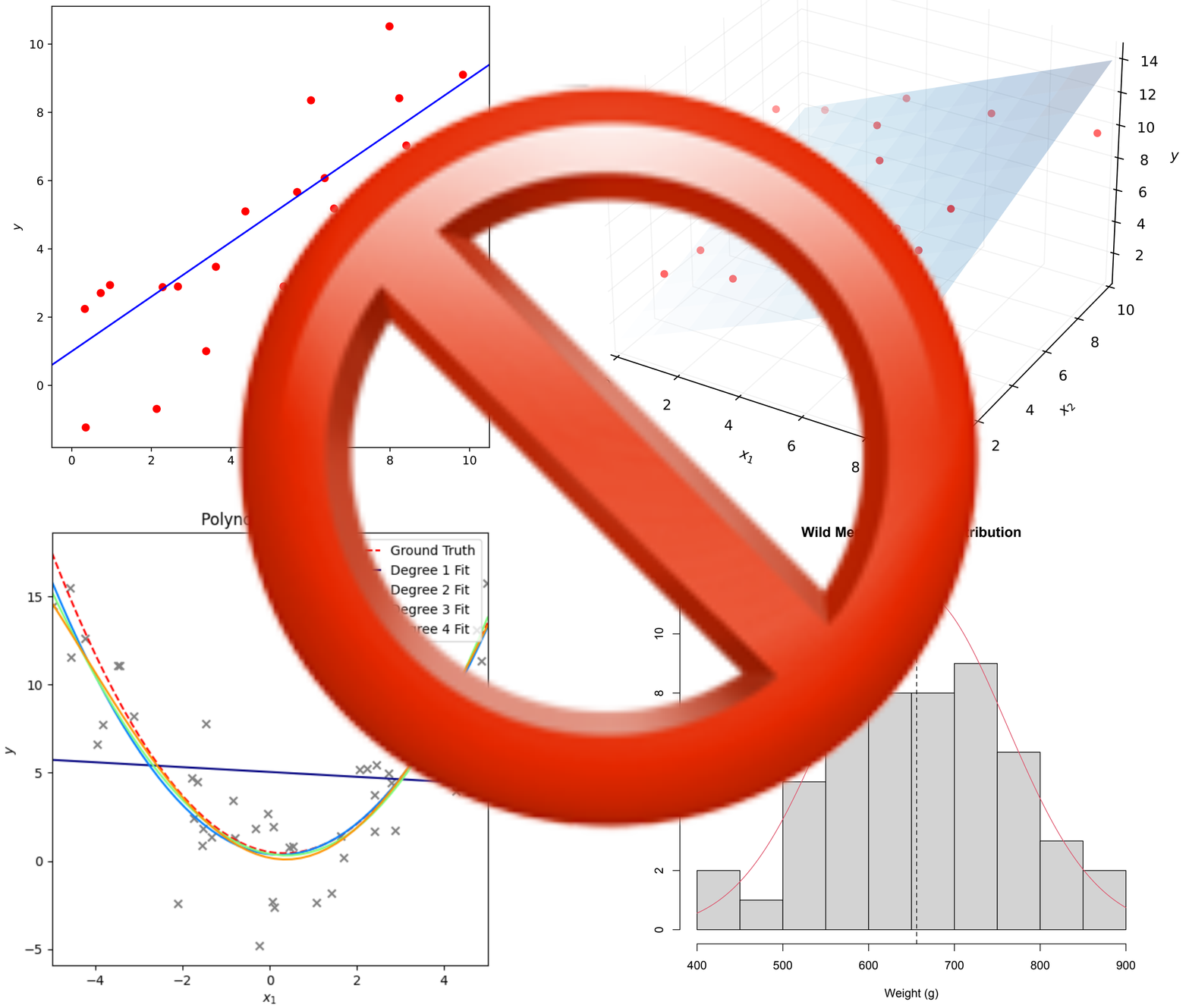


greed is good

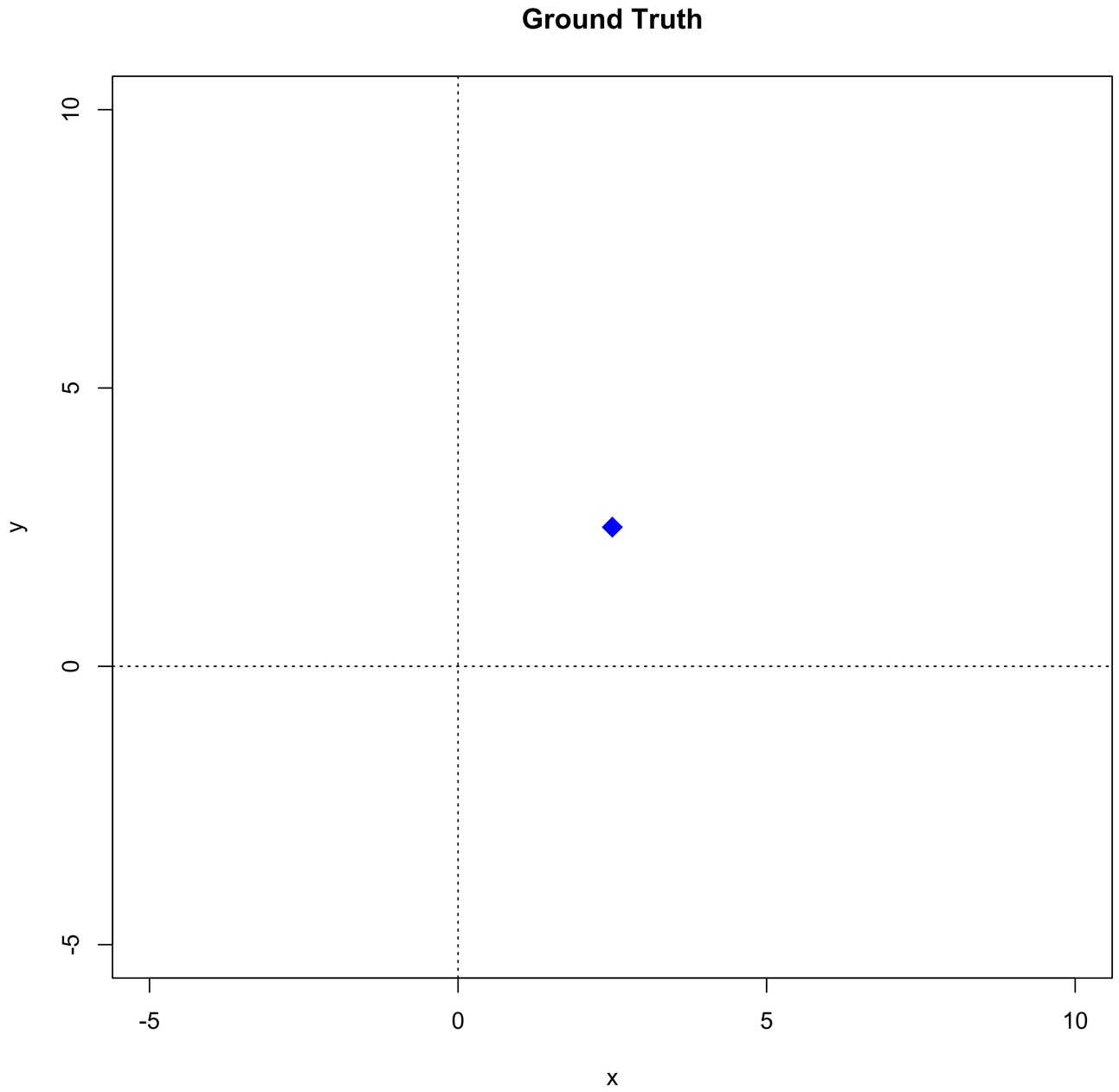

6
5
3
2
1
9
7
8
4
16
11
38
99

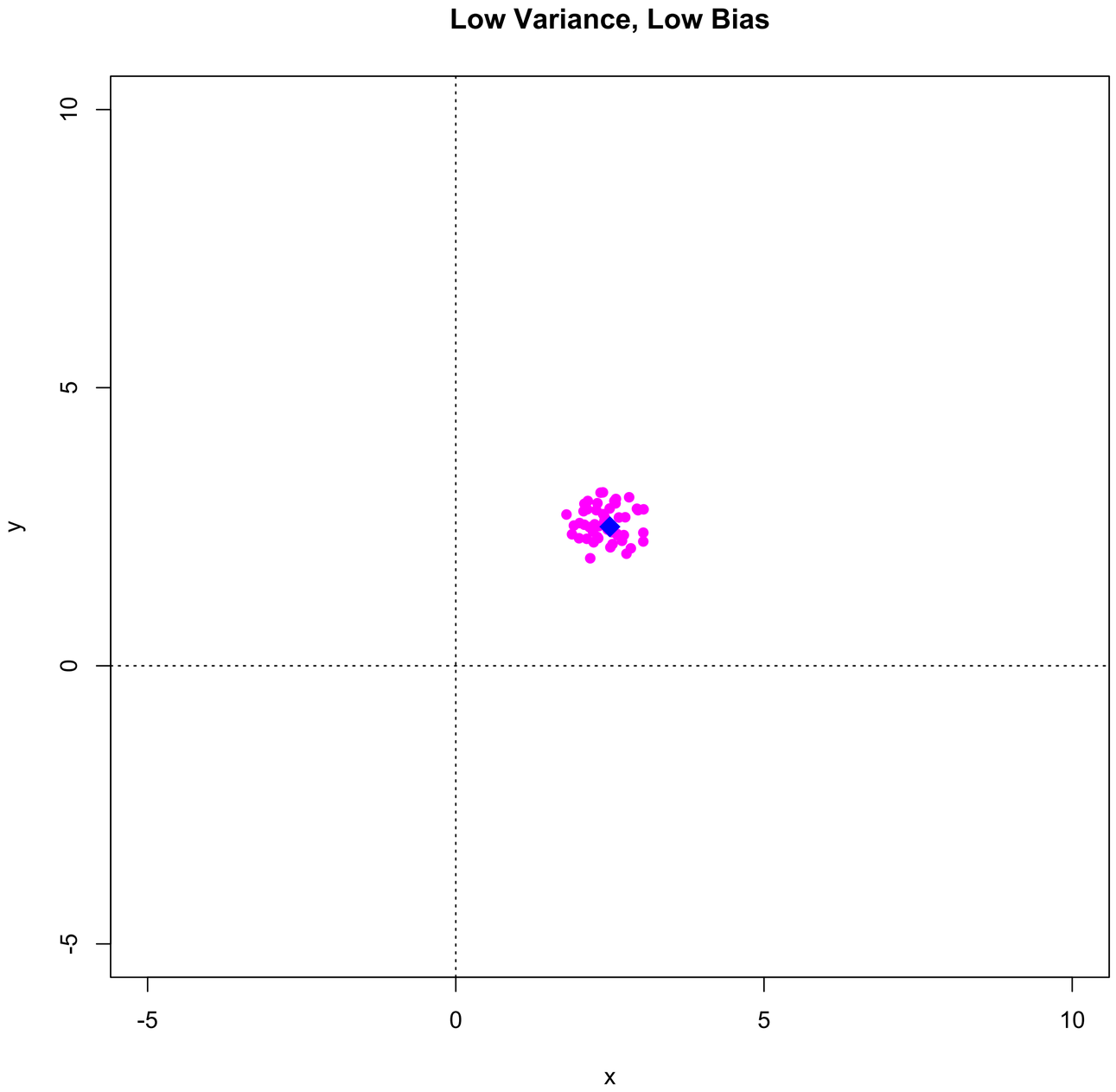

6
5
3
2
1
9
7
8
4
16
11
38
99

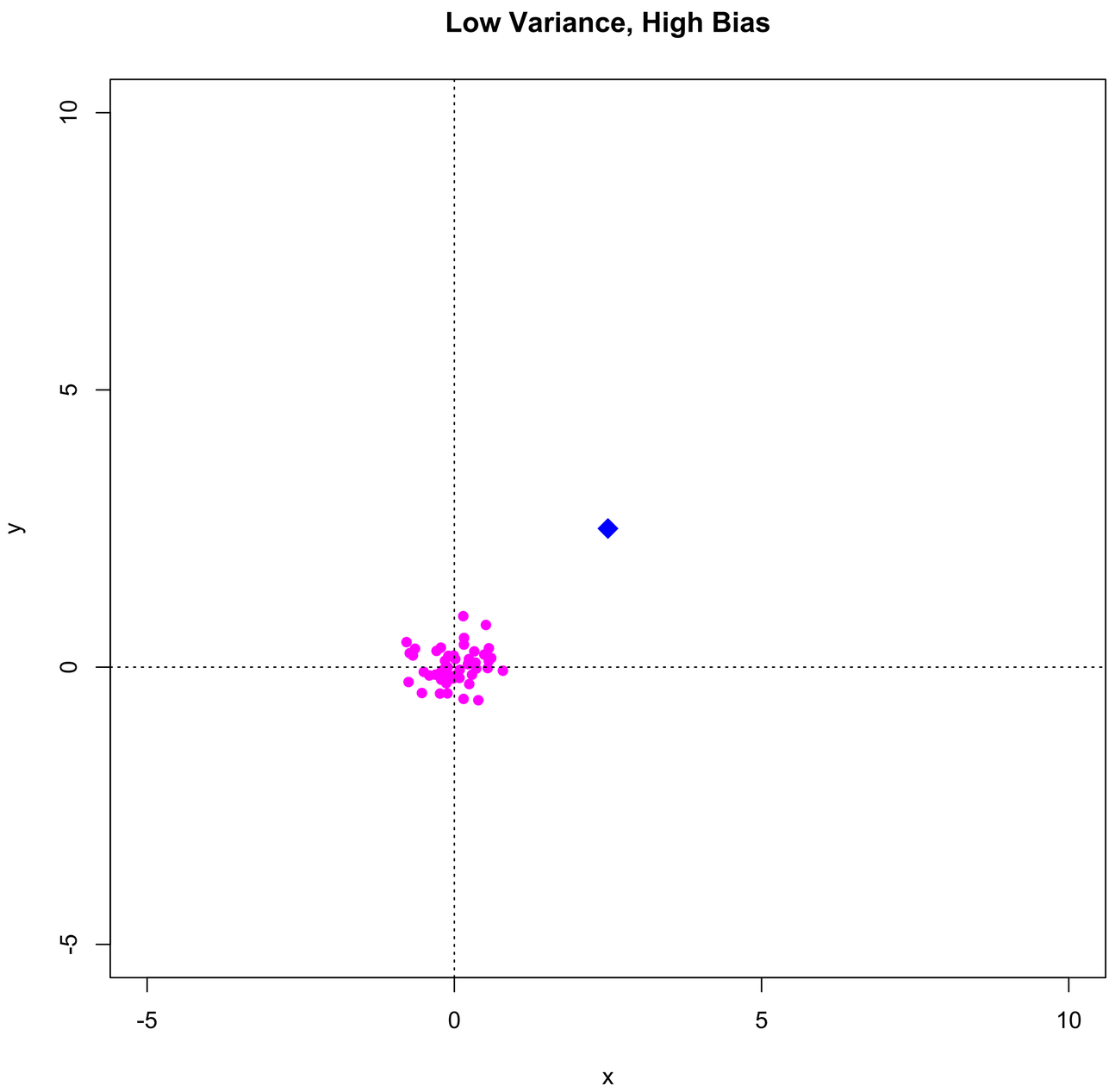

6
5
3
2
1
9
7
8
4
16
11
38
99

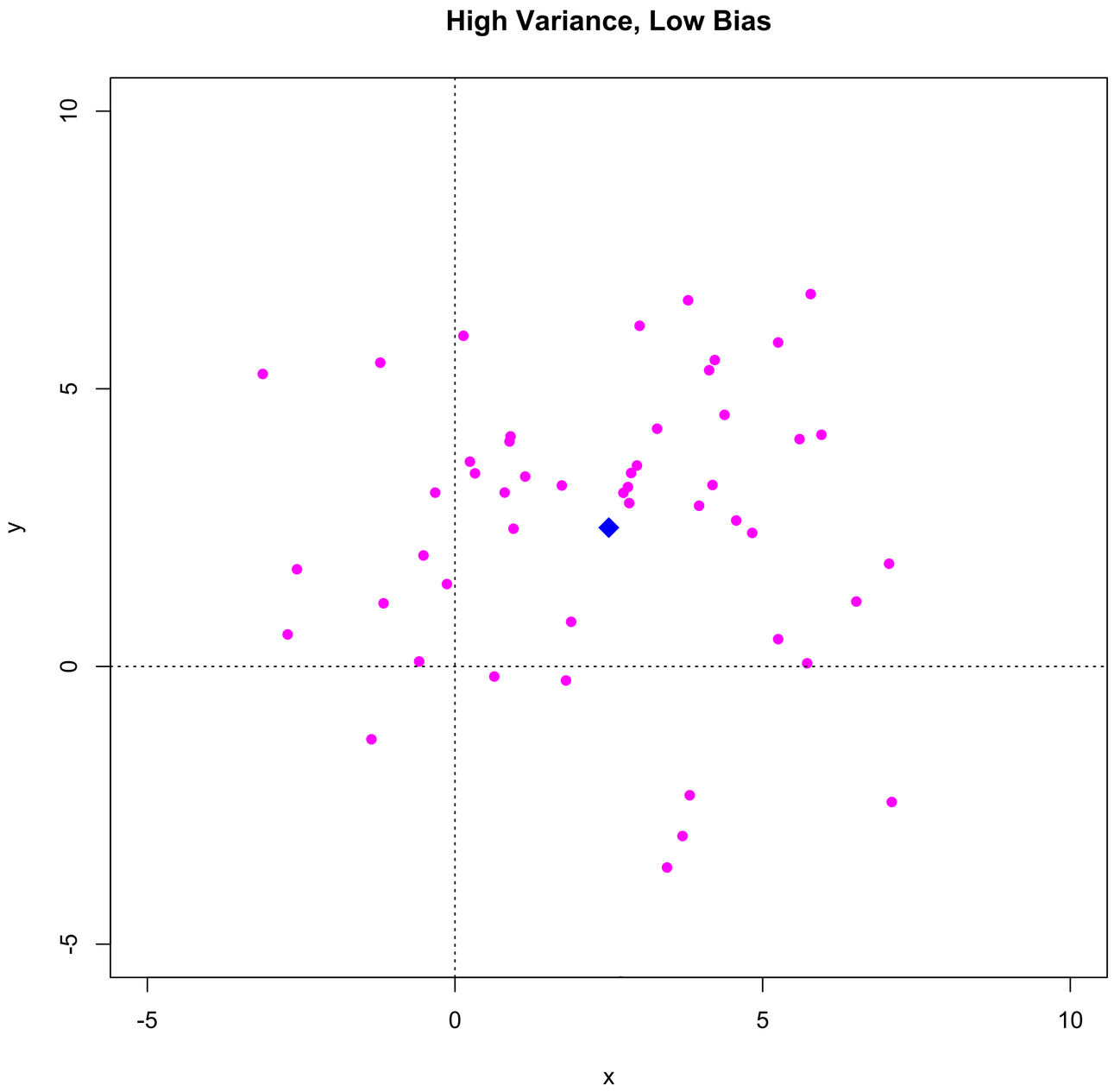

Ground Truth
Degree 1 Fit
Degree 2 Fit
Degree 3 Fit
Degree 4 Fit
Weight (g)

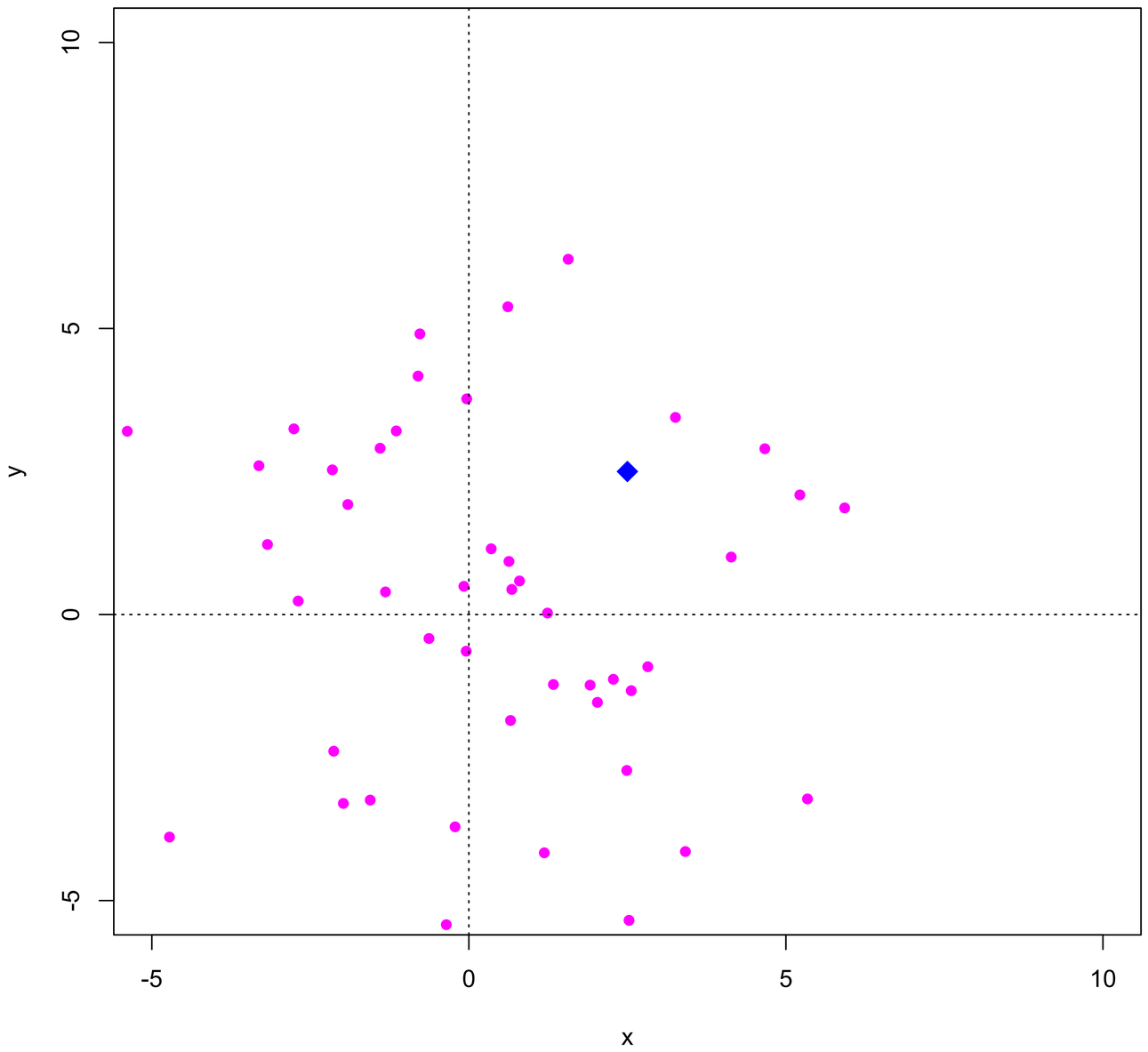
bias-variance tradeoff

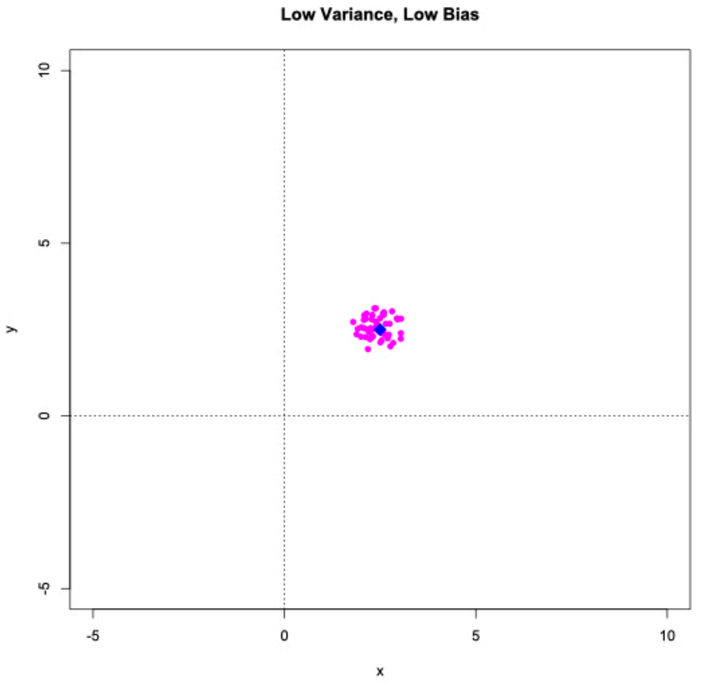

Ground Truth
10
5
0
-5
y
-5
0
5
10
x

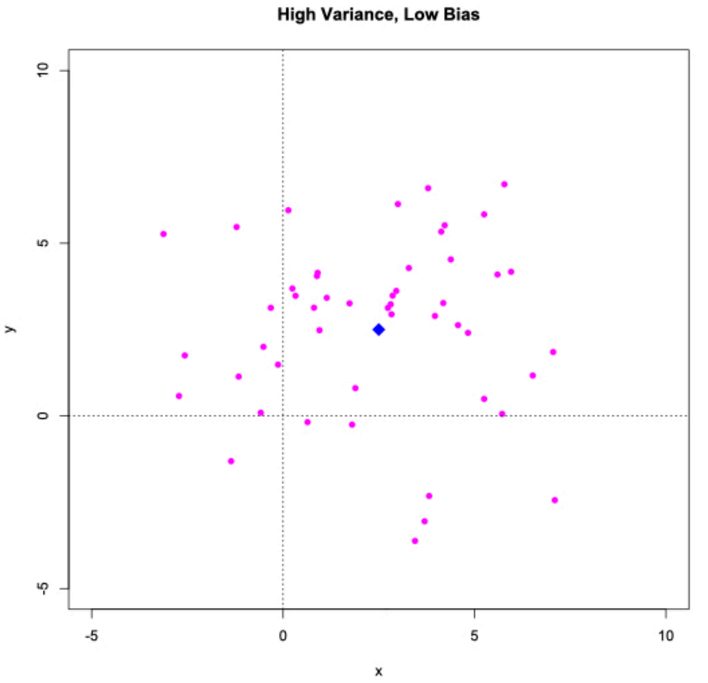

Low Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

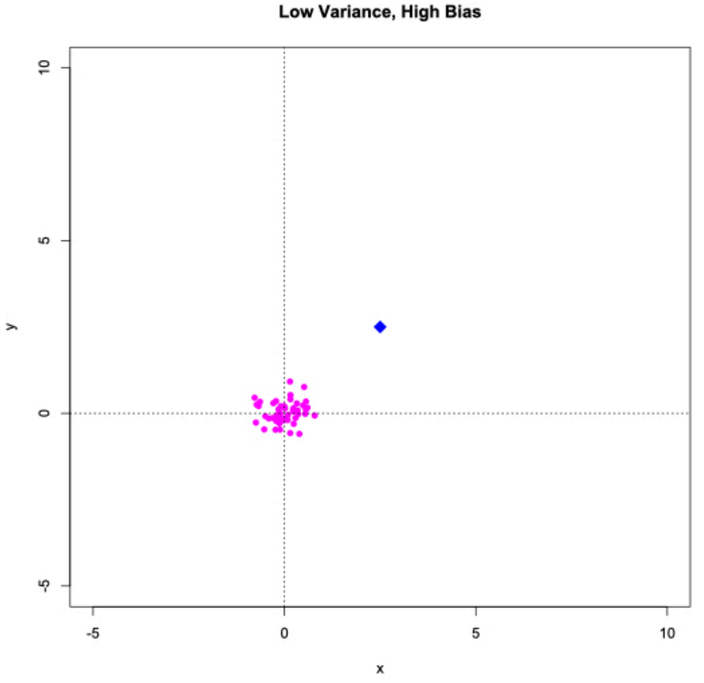

Low Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

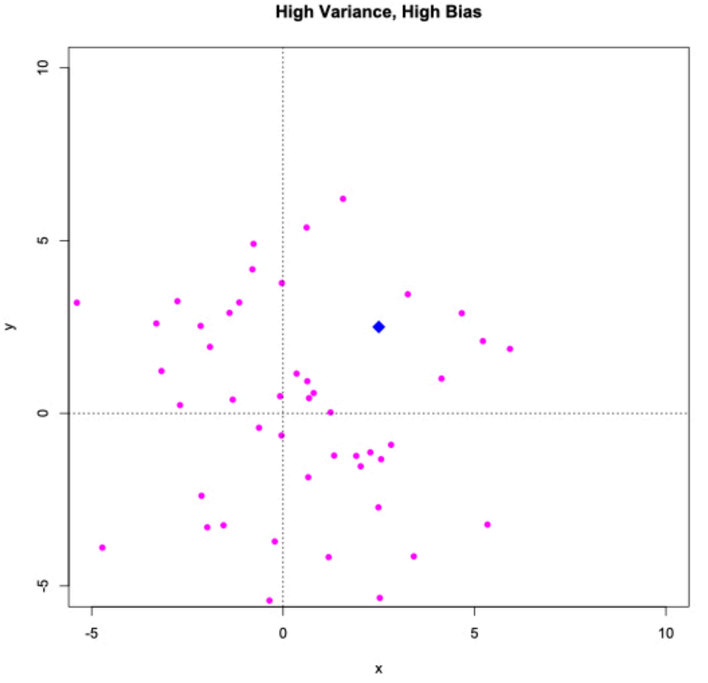

High Variance, Low Bias
y
x
10
5
0
-5
-5
0
5
10

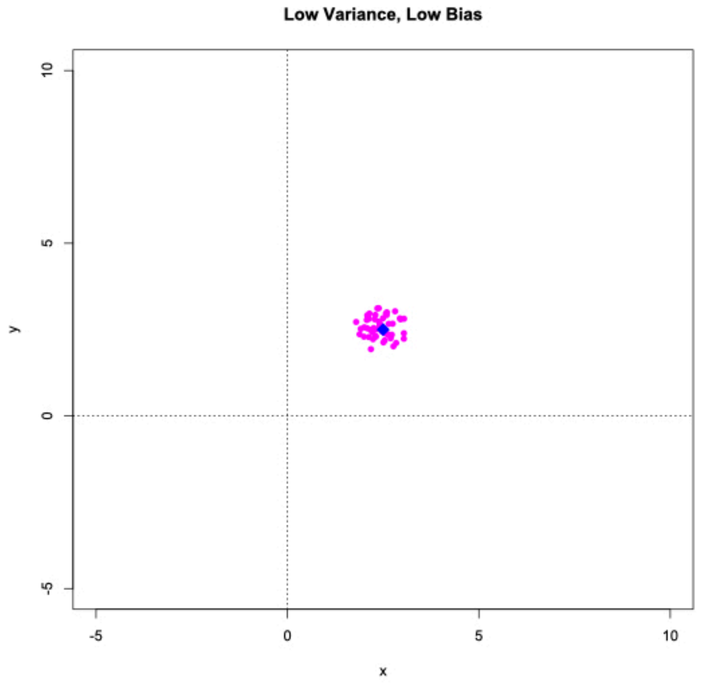

High Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

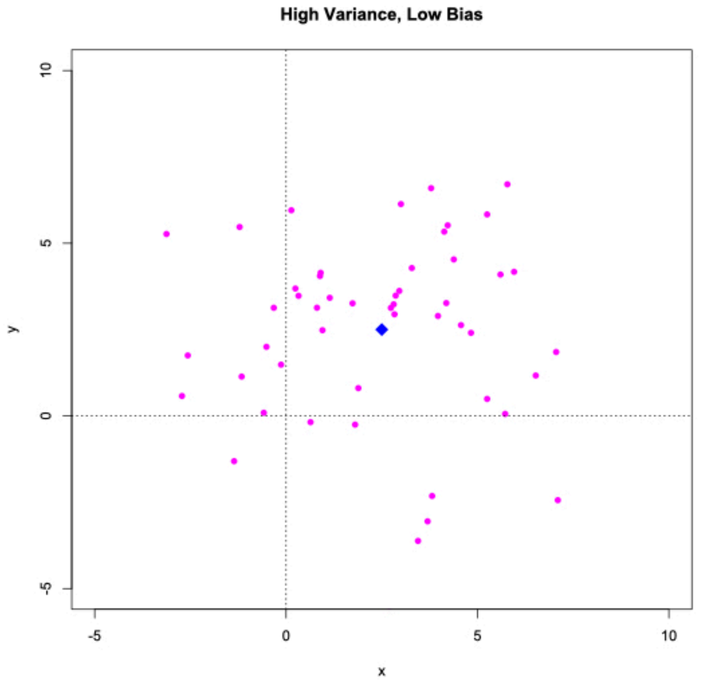

Low Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

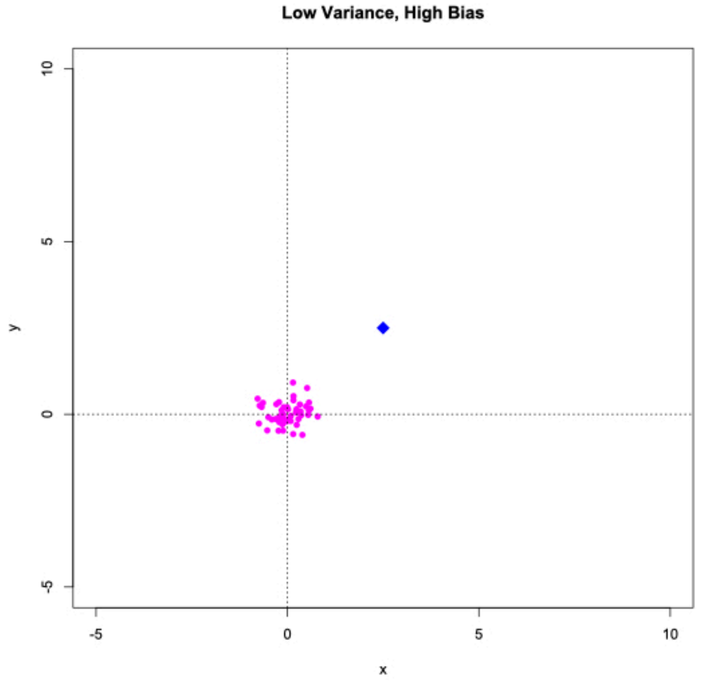

High Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

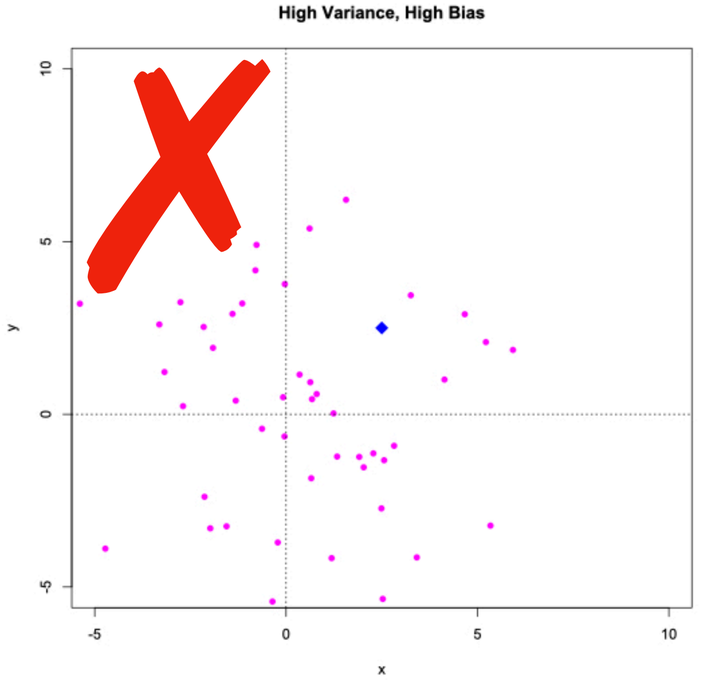

Low Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

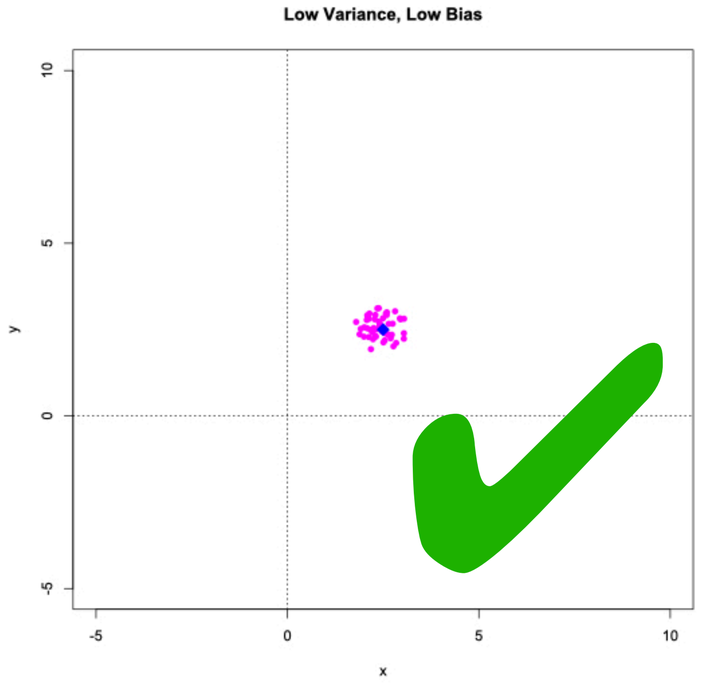

High Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

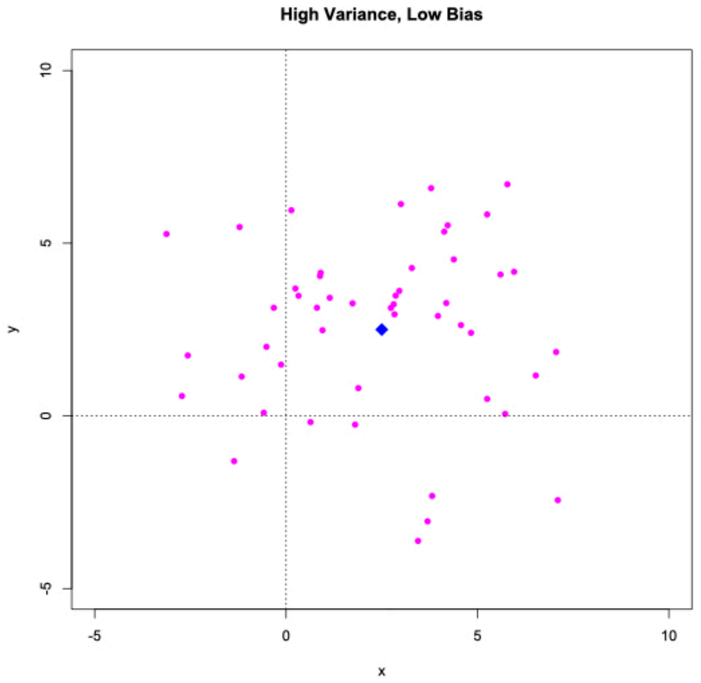

Low Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

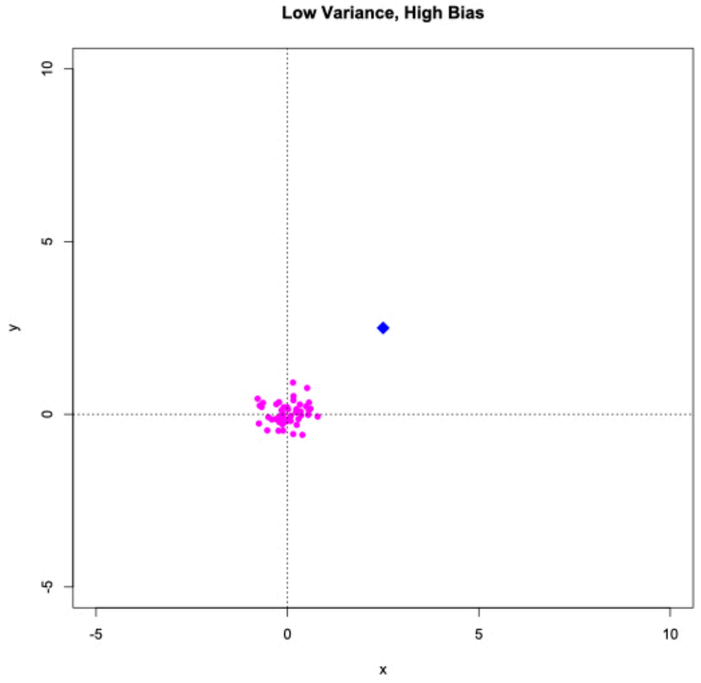

High Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

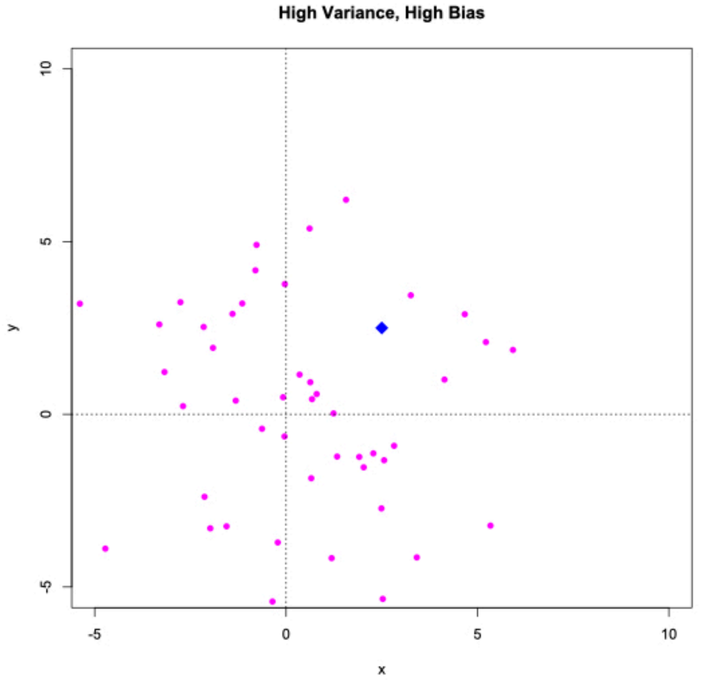

Low Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

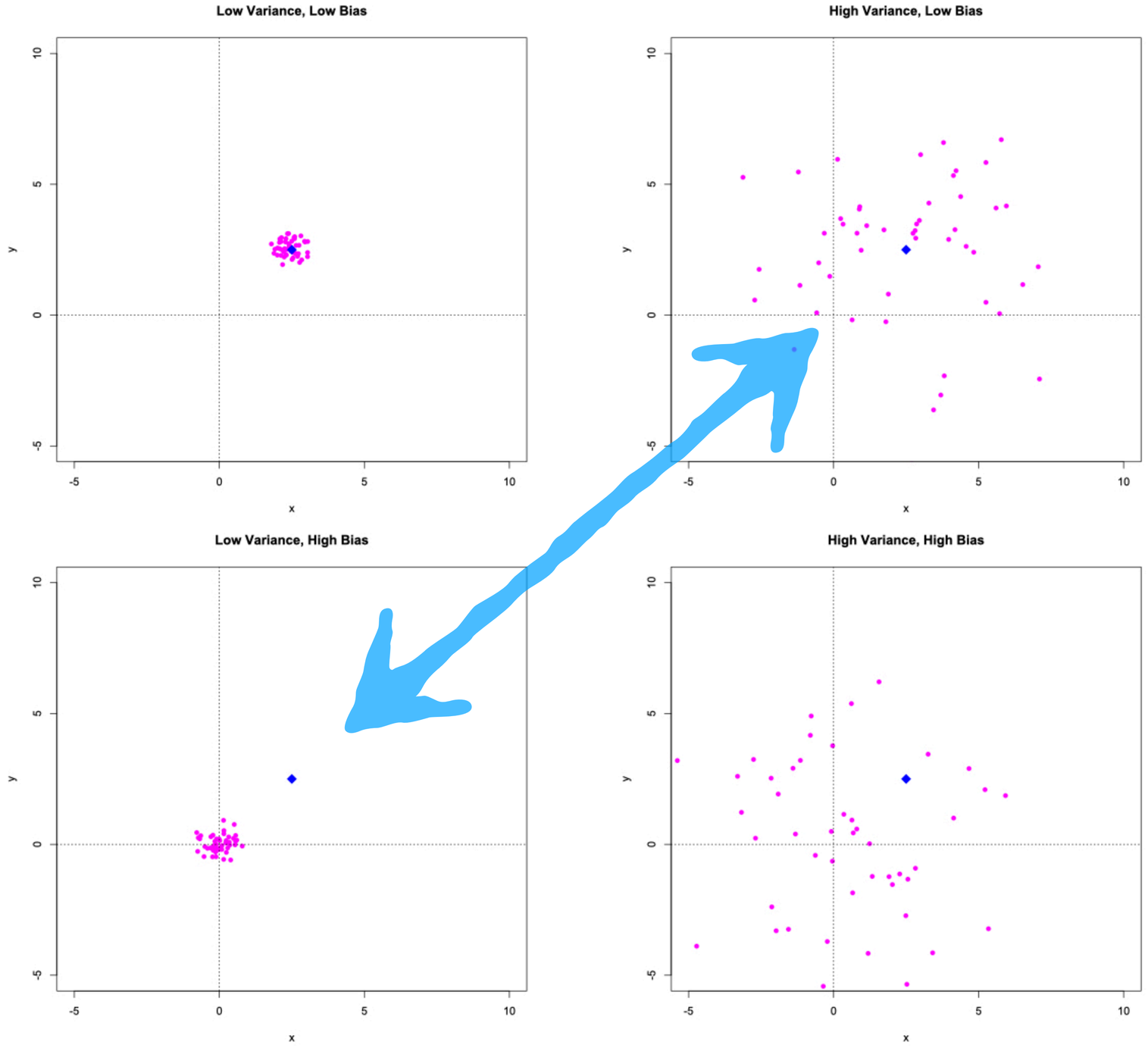

High Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

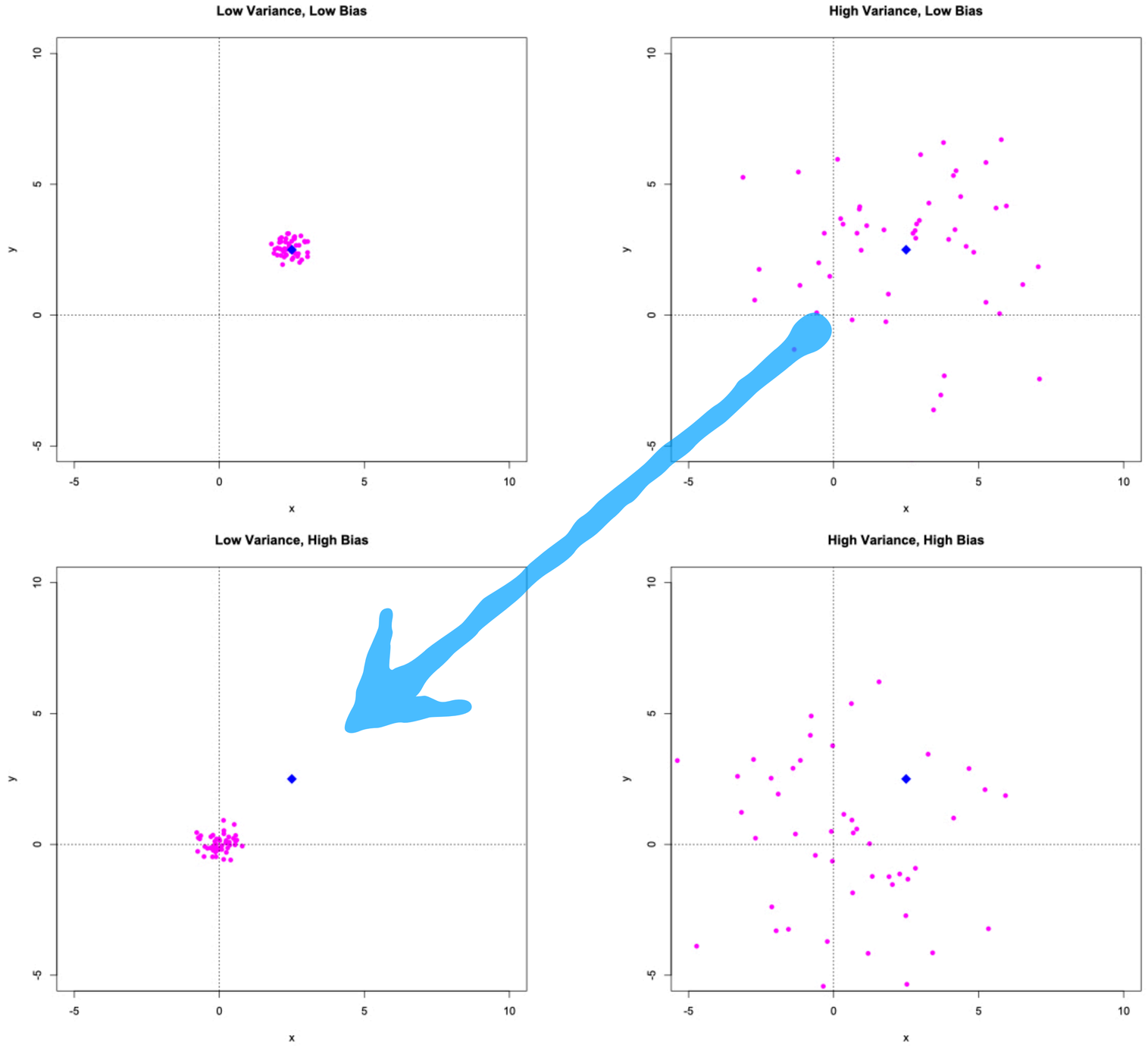

Low Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x


High Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

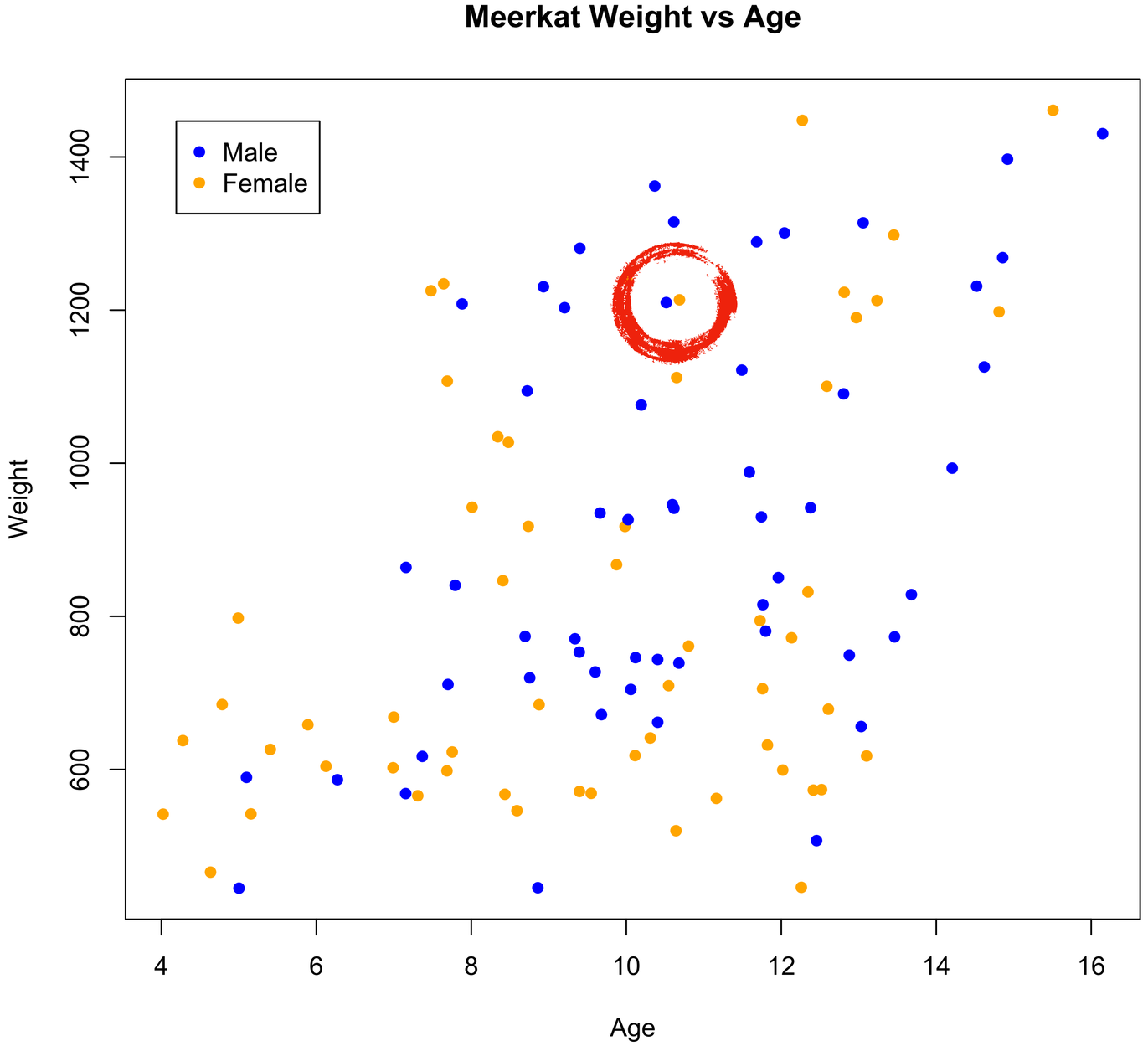

Low Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x

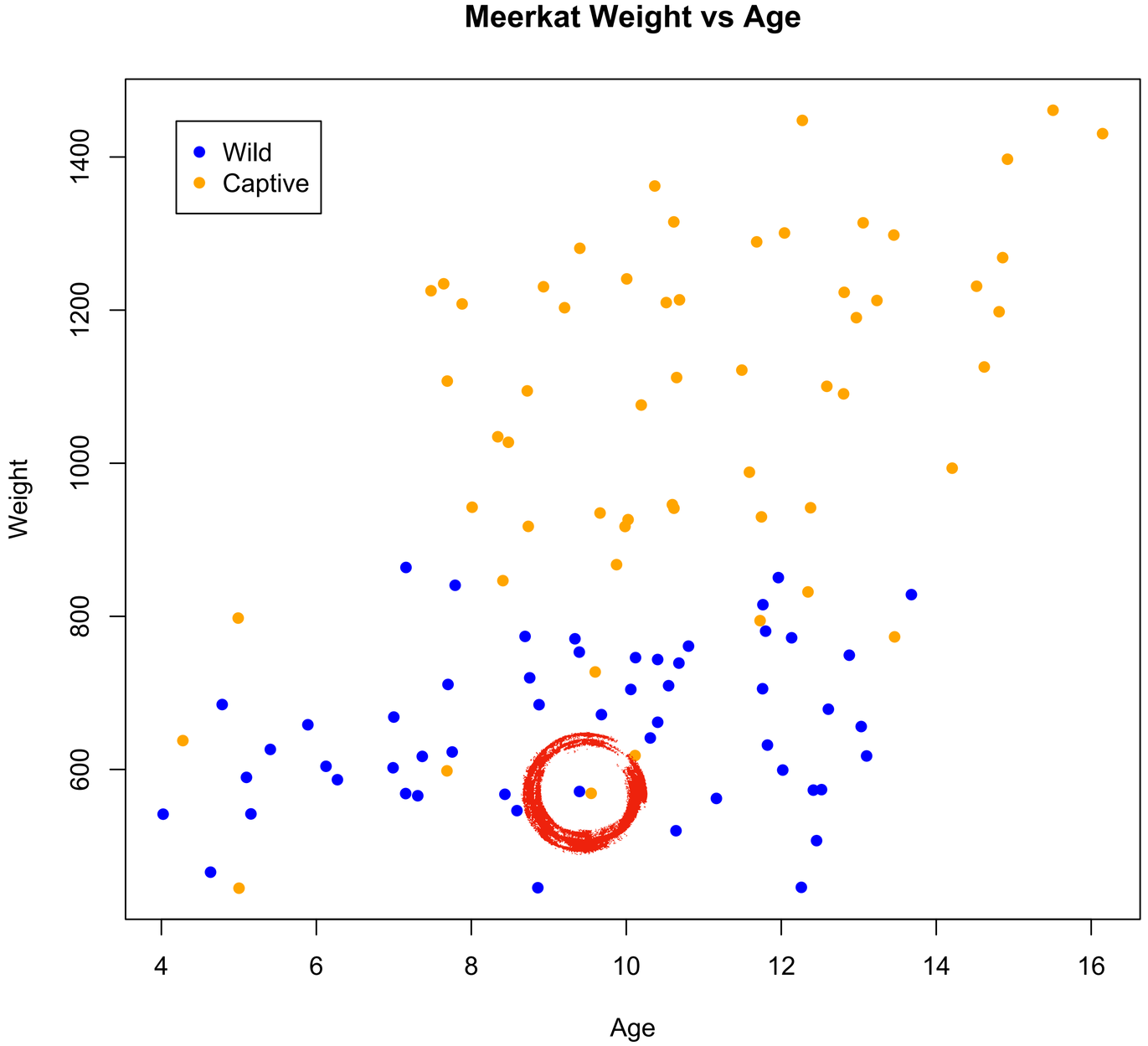

High Variance, High Bias
10
5
0
-5
y
-5
0
5
10
x


Low Variance, Low Bias
High Variance, Low Bias
Low Variance, High Bias
High Variance, High Bias
x
y
10
5
0
-5
-5
0
5
10


Low Variance, Low Bias
High Variance, Low Bias
Low Variance, High Bias
High Variance, High Bias
x
y
-5
0
5
10


Neighbors


Meerkat Weight vs Age
Male
Female
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age


Meerkat Weight vs Age
Wild
Captive
Weight
600
800
1000
1200
1400
4
6
8
10
12
14
16
Age

Meerkats

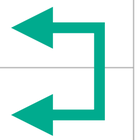
#	Age	Weight	Sex	Captive	...
1	11.8	815	1	0	
2	9.7	672	1	0	
3	8.9	446	1	0	
4	10.8	761	0	0	
5	8.3	1035	0	1	
6	11.7	930	1	1	
7	8.5	1027	0	1	
8	7.6	1234	0	1	
9	15.5	1461	0	1	
10	8.8	720	1	0	
11	12.8	1223	0	1	
12	7.7	711	1	0	
13	7.2	526			

Meerkats

#	Age	Weight	Sex	Captive	...
1	11.8	815	1	0	
2	9.7	672	1	0	
3	8.9	446	1	0	
4	10.8	761	0	0	
5	8.3	1035	0	1	
6	11.7	930	1	1	
7	8.5	1027	0	1	
8	7.6	1234	0	1	
9	15.5	1461	0	1	
10	8.8	720	1	0	
11	12.8	1223	0	1	
12	7.7	711	1	0	
13	7.2	526			

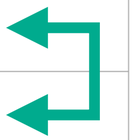
Meerkats

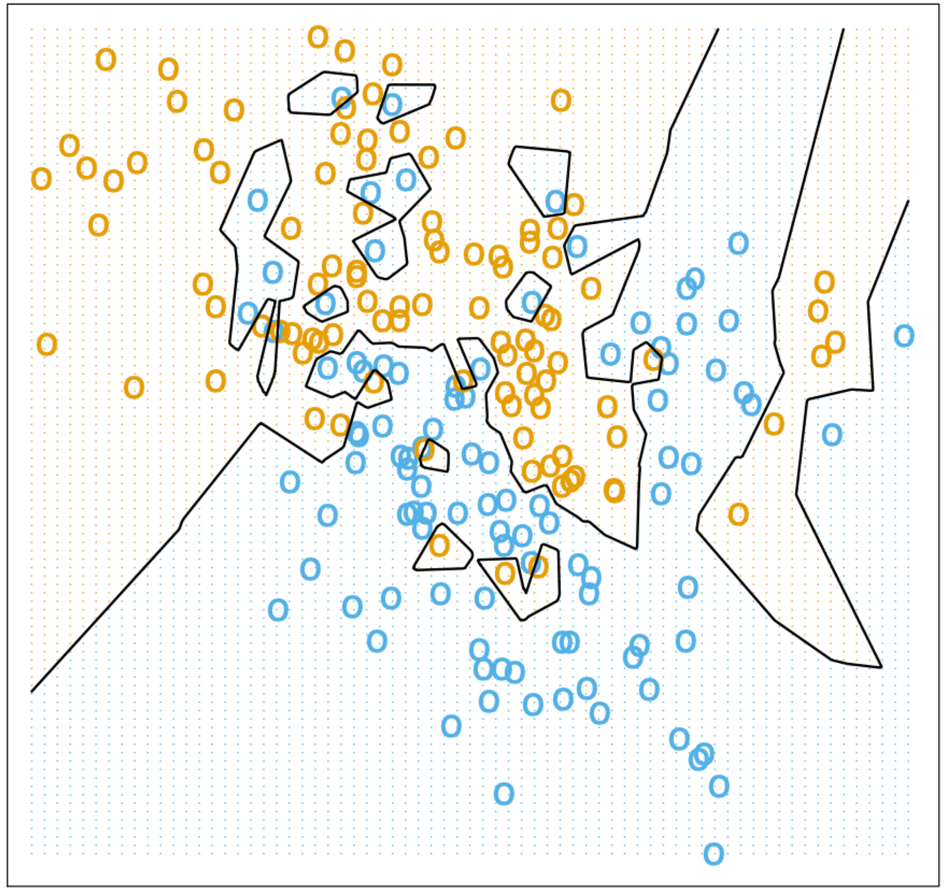
#	Age	Weight	Sex	Captive	...
1	11.8	815	1	0	
2	9.7	672	1	0	
3	8.9	446	1	0	
4	10.8	761	0	0	
5	8.3	1035	0	1	
6	11.7	930	1	1	
7	8.5	1027	0	1	
8	7.6	1234	0	1	
9	15.5	1461	0	1	
10	8.8	720	1	0	
11	12.8	1223	0	1	
12	7.7	711	1	0	



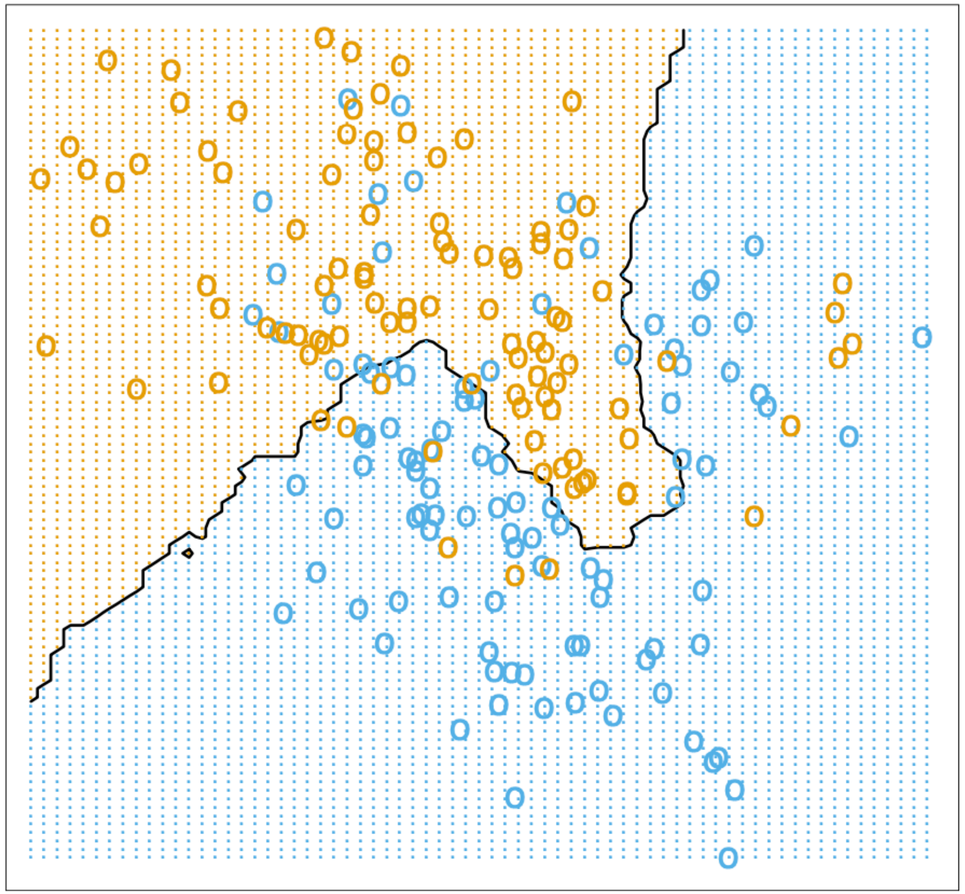
Meerkats

#	Age	Weight	Sex	Captive	...
1	11.8	815	1	0	
2	9.7	672	1	0	
3	8.9	446	1	0	
4	10.8	761	0	0	
5	8.3	1035	0	1	
6	11.7	930	1	1	
7	8.5	1027	0	1	
8	7.6	1234	0	1	
9	15.5	1461	0	1	
10	8.8	720	1	0	
11	12.8	1223	0	1	
12	7.7	711	1	0	

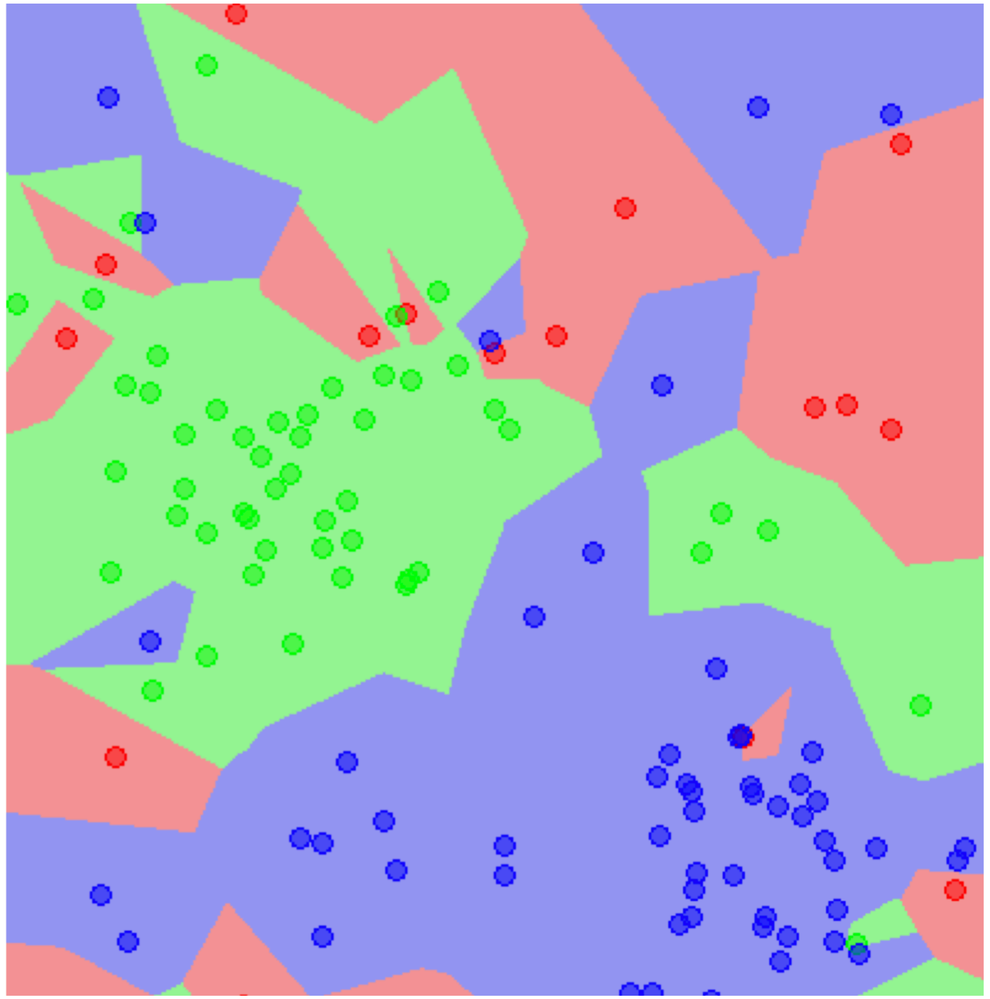




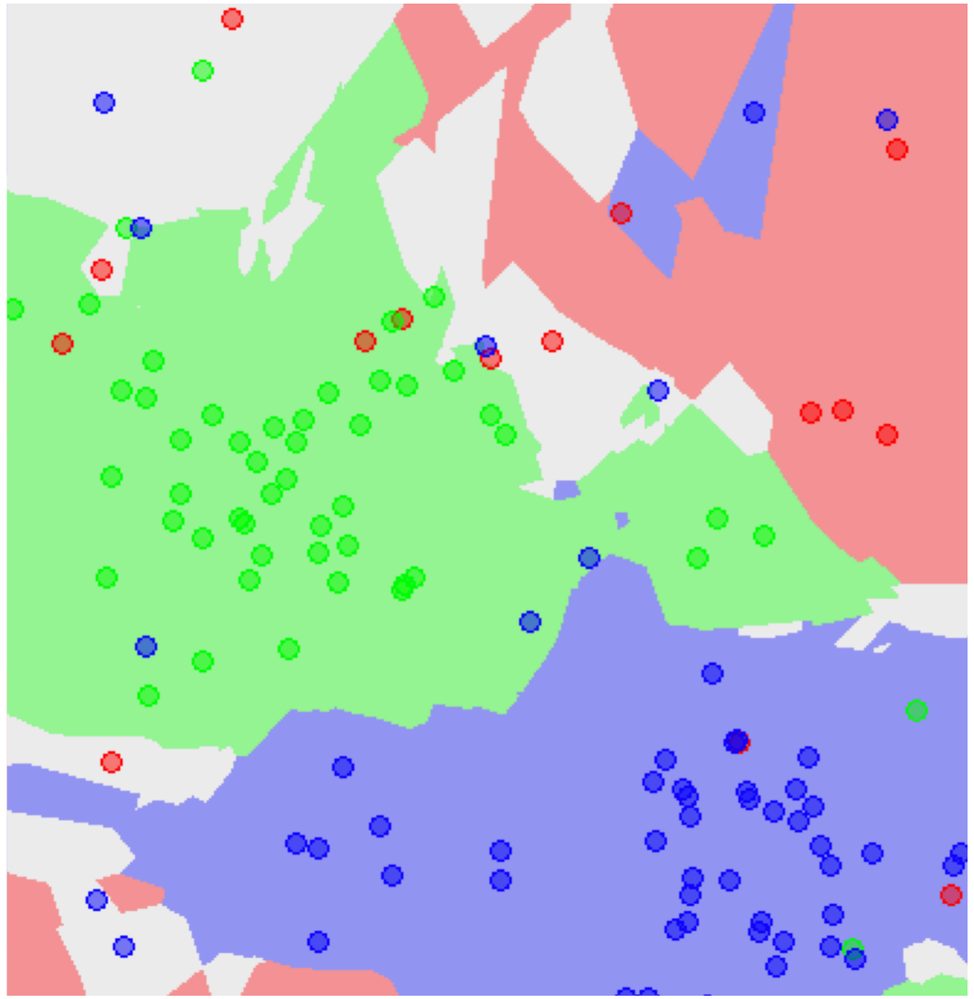
k=1



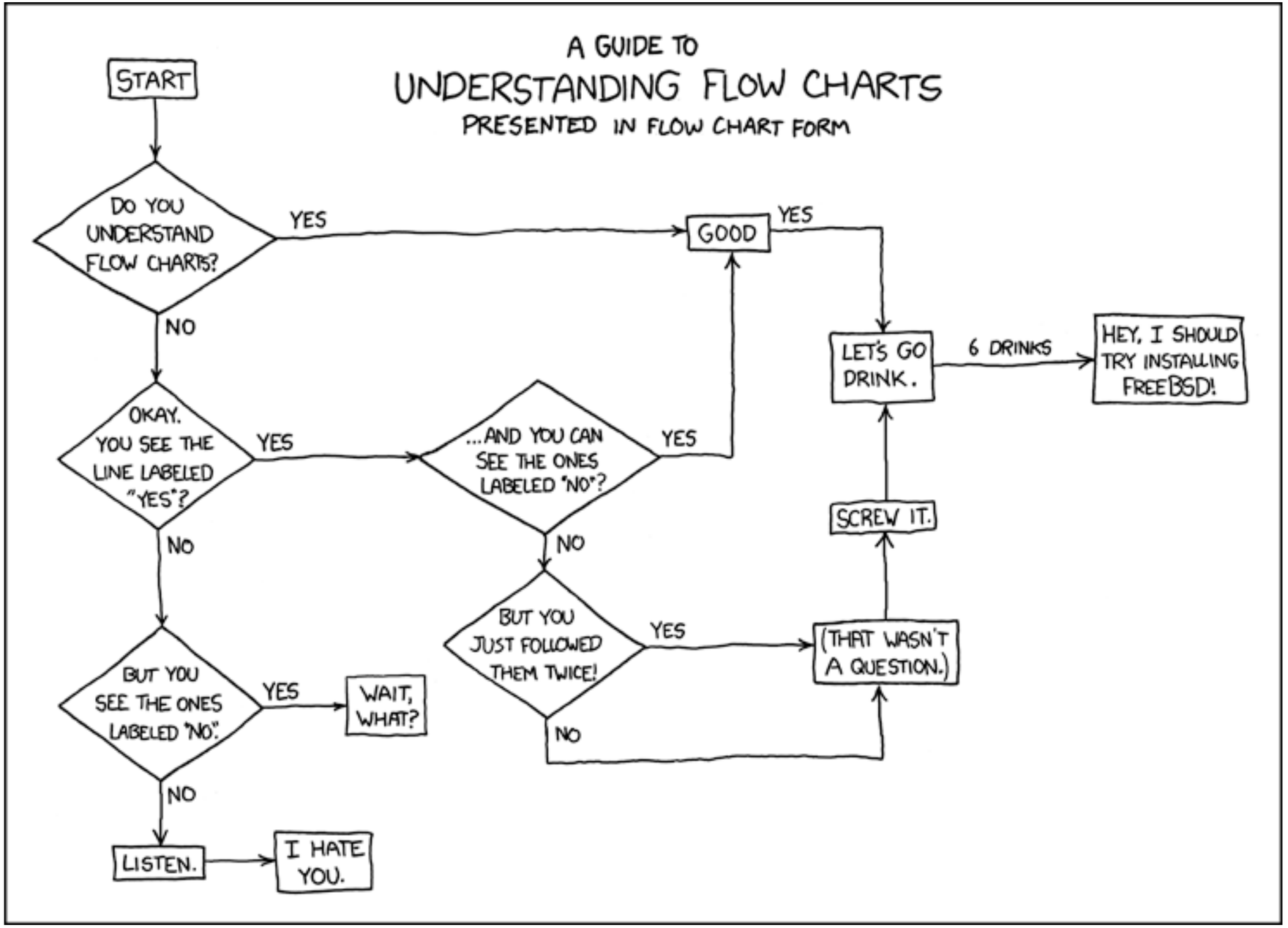
k=15

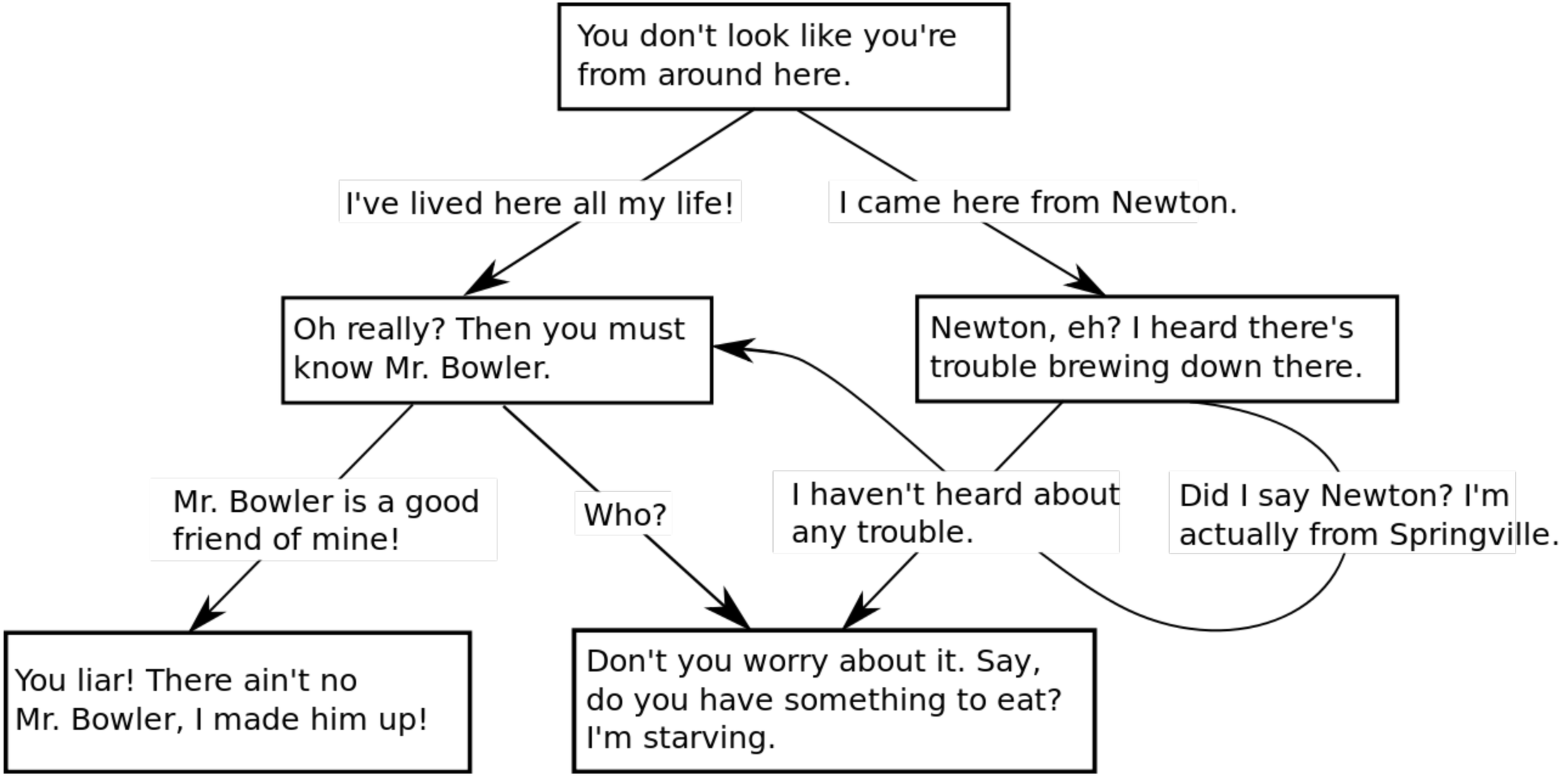


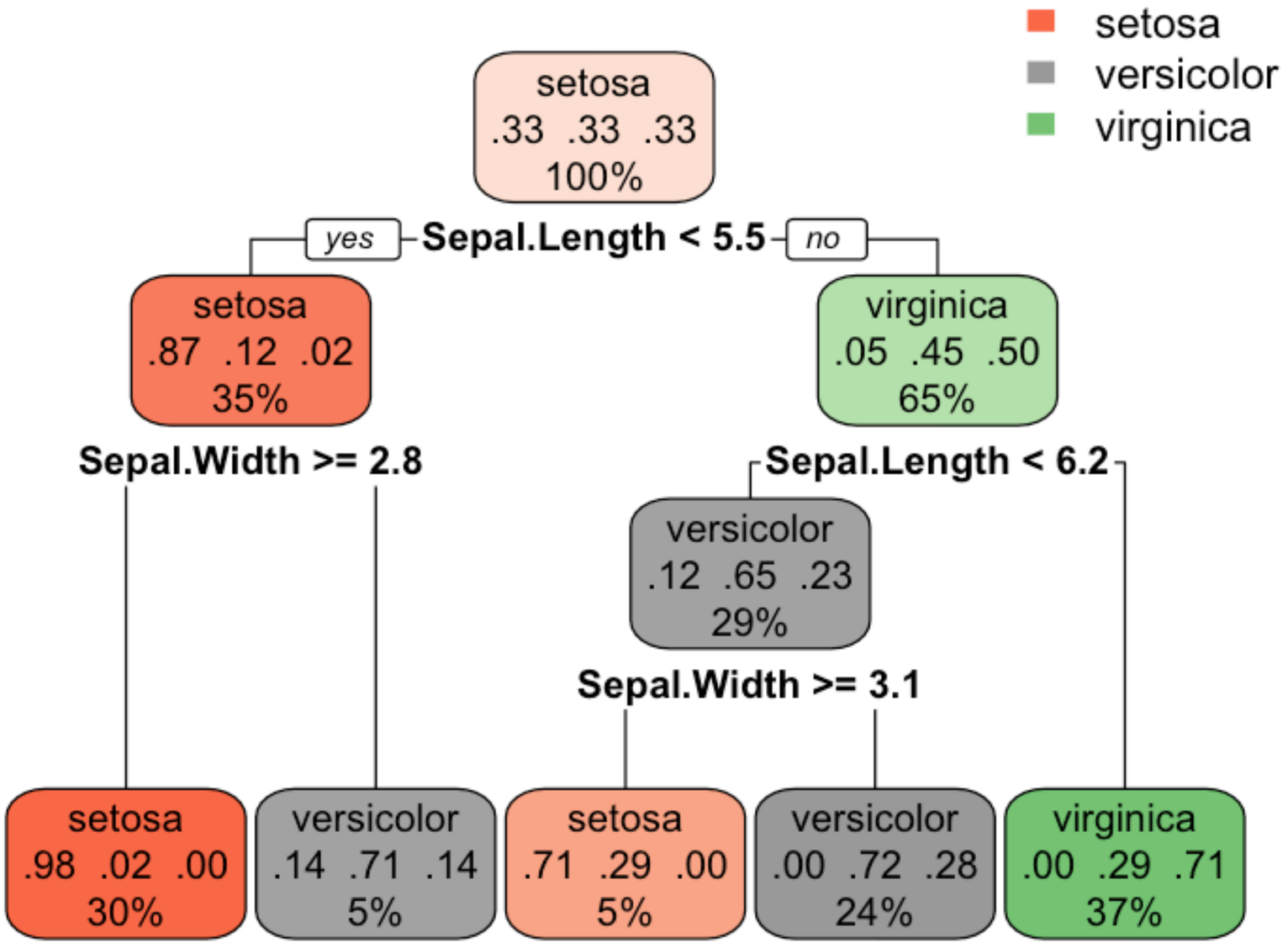
k=1

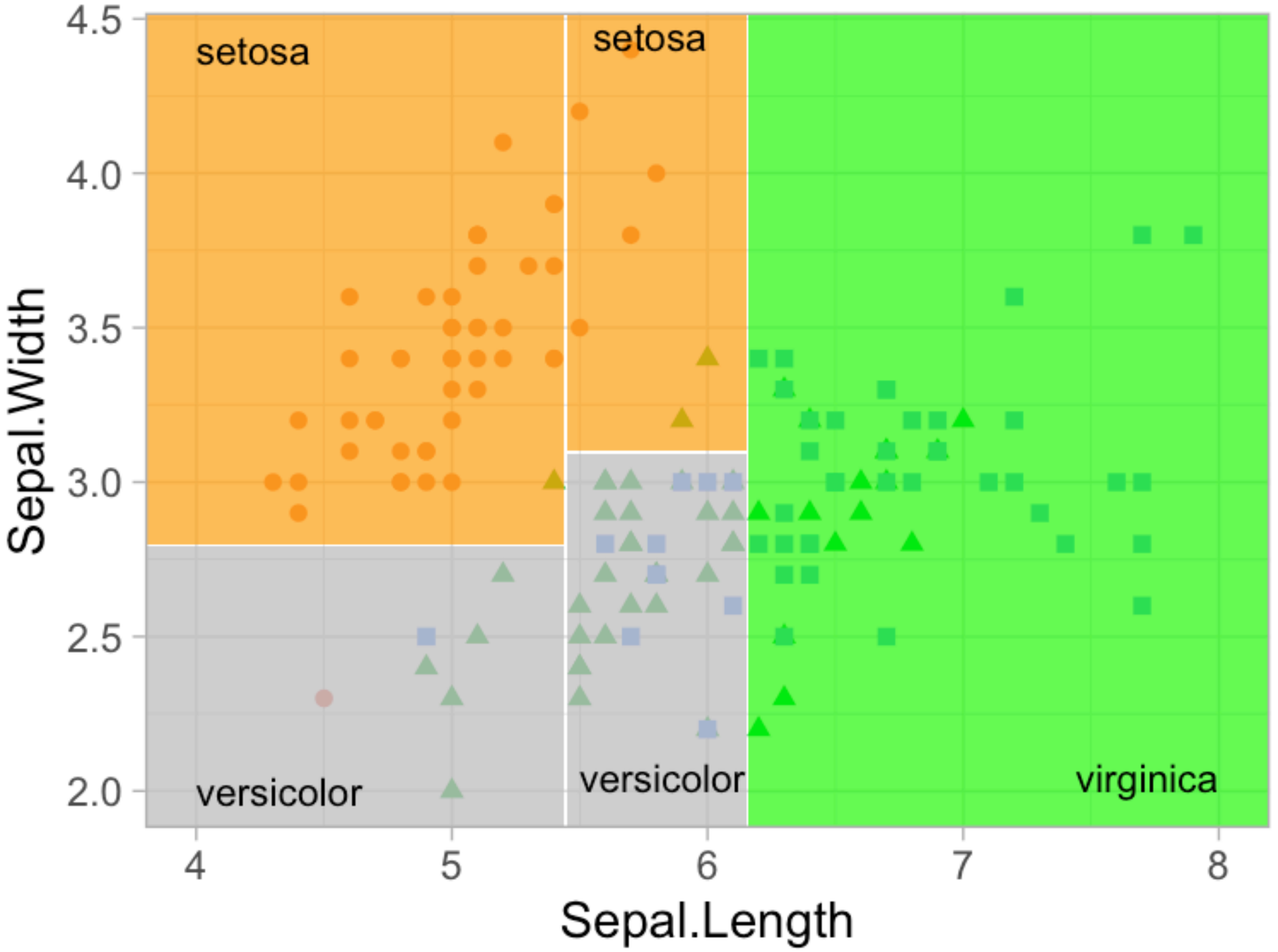


k=3


A GUIDE TO
UNDERSTANDING FLOW CHARTS
PRESENTED IN FLOW CHART FORM
START
DO YOU UNDERSTAND FLOW CHARTS?
YES
GOOD
YES
NO
LET'S GO DRINK.
6 DRINKS
HEY, I SHOULD TRY INSTALLING FREEBSD!
OKAY. YOU SEE THE LINE LABELED "YES"?
YES
...AND YOU CAN SEE THE ONES LABELED "NO"?
YES
NO
SCREW IT.
NO
BUT YOU JUST FOLLOWED THEM TWICE!
YES
(THAT WASN'T A QUESTION.)
BUT YOU SEE THE ONES LABELED "NO".
YES
WAIT, WHAT?
NO
NO
LISTEN.
I HATE YOU.


You don't look like you're from around here.
I've lived here all my life!
I came here from Newton.
Oh really? Then you must know Mr. Bowler.
Newton, eh? I heard there's trouble brewing down there.
Mr. Bowler is a good friend of mine!
Who?
I haven't heard about any trouble.
Did I say Newton? I'm actually from Springville.
You liar! There ain't no Mr. Bowler, I made him up!
Don't you worry about it. Say, do you have something to eat? I'm starving.


setosa
versicolor
virginica
setosa
.33 .33 .33
100%
yes
Sepal.Length < 5.5
no
setosa
.87 .12 .02
35%
virginica
.05 .45 .50
65%
Sepal.Width >= 2.8
Sepal.Length < 6.2
versicolor
.12 .65 .23
29%
Sepal.Width >= 3.1
setosa
.98 .02 .00
30%
versicolor
.14 .71 .14
5%
setosa
.71 .29 .00
5%
versicolor
.00 .72 .28
24%
virginica
.00 .29 .71
37%


setosa
setosa
versicolor
versicolor
virginica
Sepal.Width
Sepal.Length
4.5
4.0
3.5
3.0
2.5
2.0
4
5
6
7
8

$$\frac{1}{n_R} \sum_{i \in R} \mathbf{1}(y_i \neq \hat{y}_R)$$

Misclassification Error

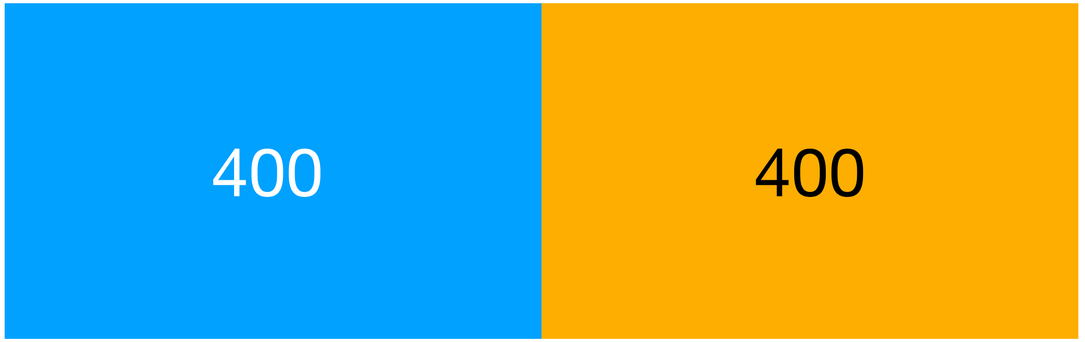
$$-\sum_{k \in R} p_k \log p_k$$

(Cross) Entropy

$$\sum_{k \in R} p_k (1 - p_k)$$

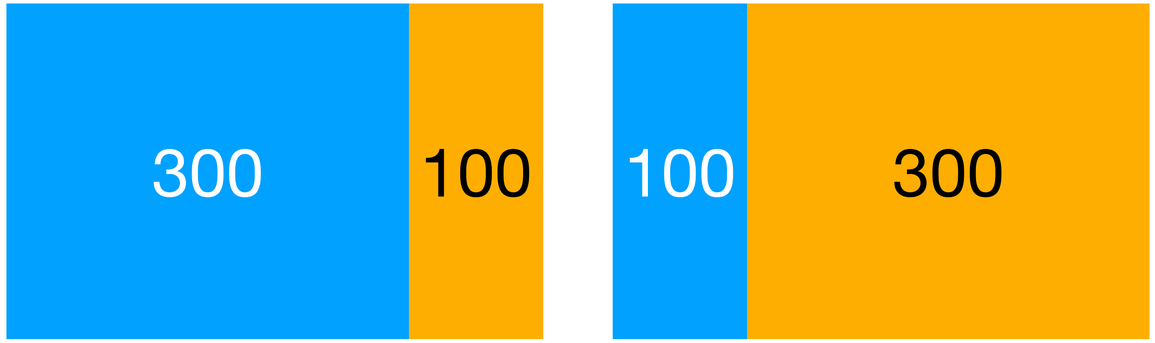
Gini Impurity

Parent



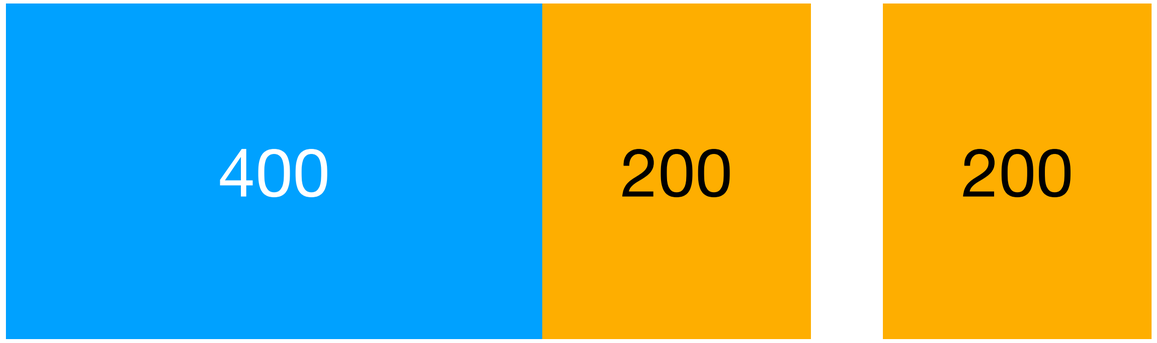
Misclassification: 0.5
Gini: 0.5
Entropy: 0.693

Split Option 1

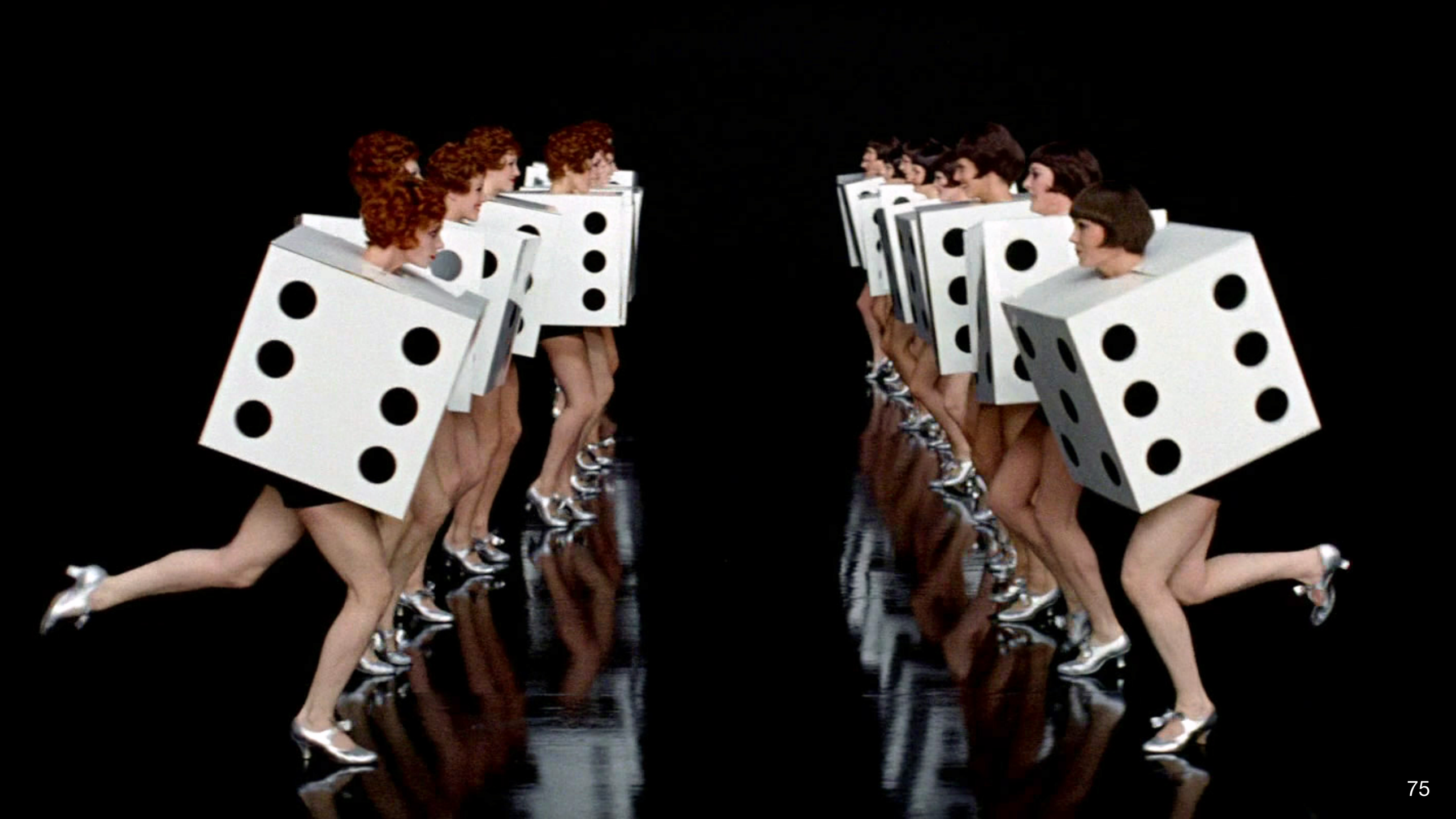


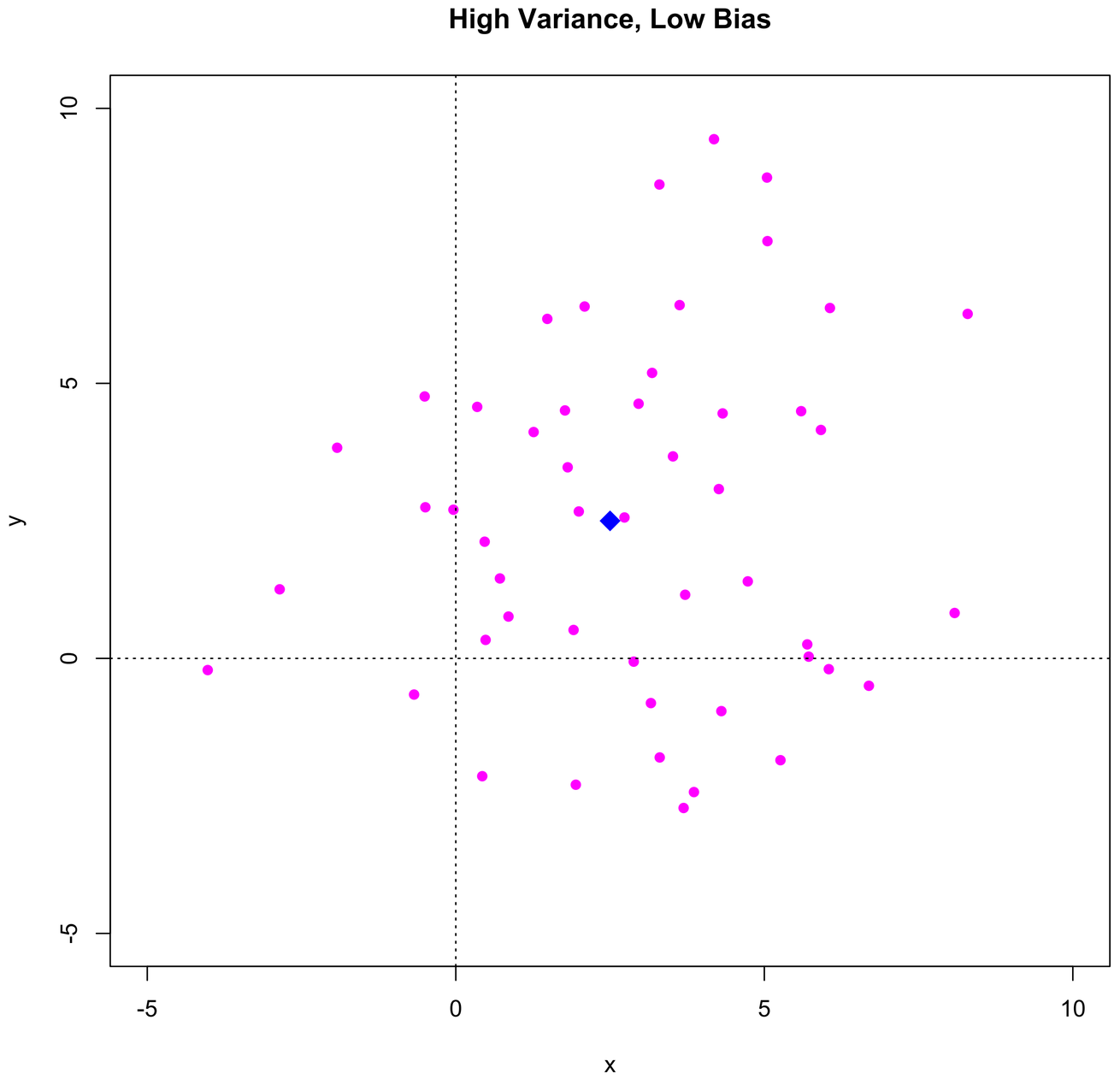
Misclassification: 0.25
Gini: 0.375
Entropy: 0.56

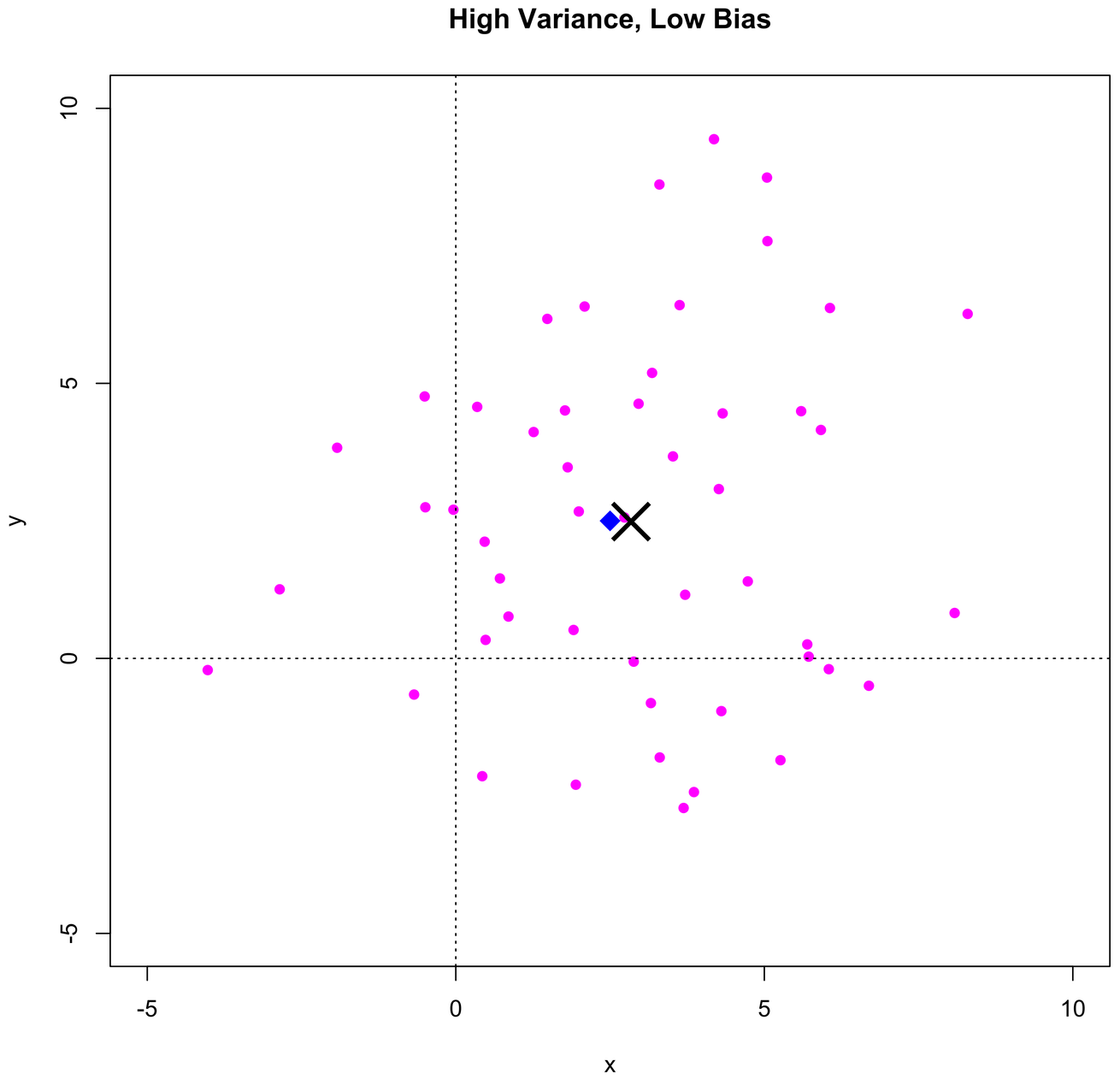
Split Option 2



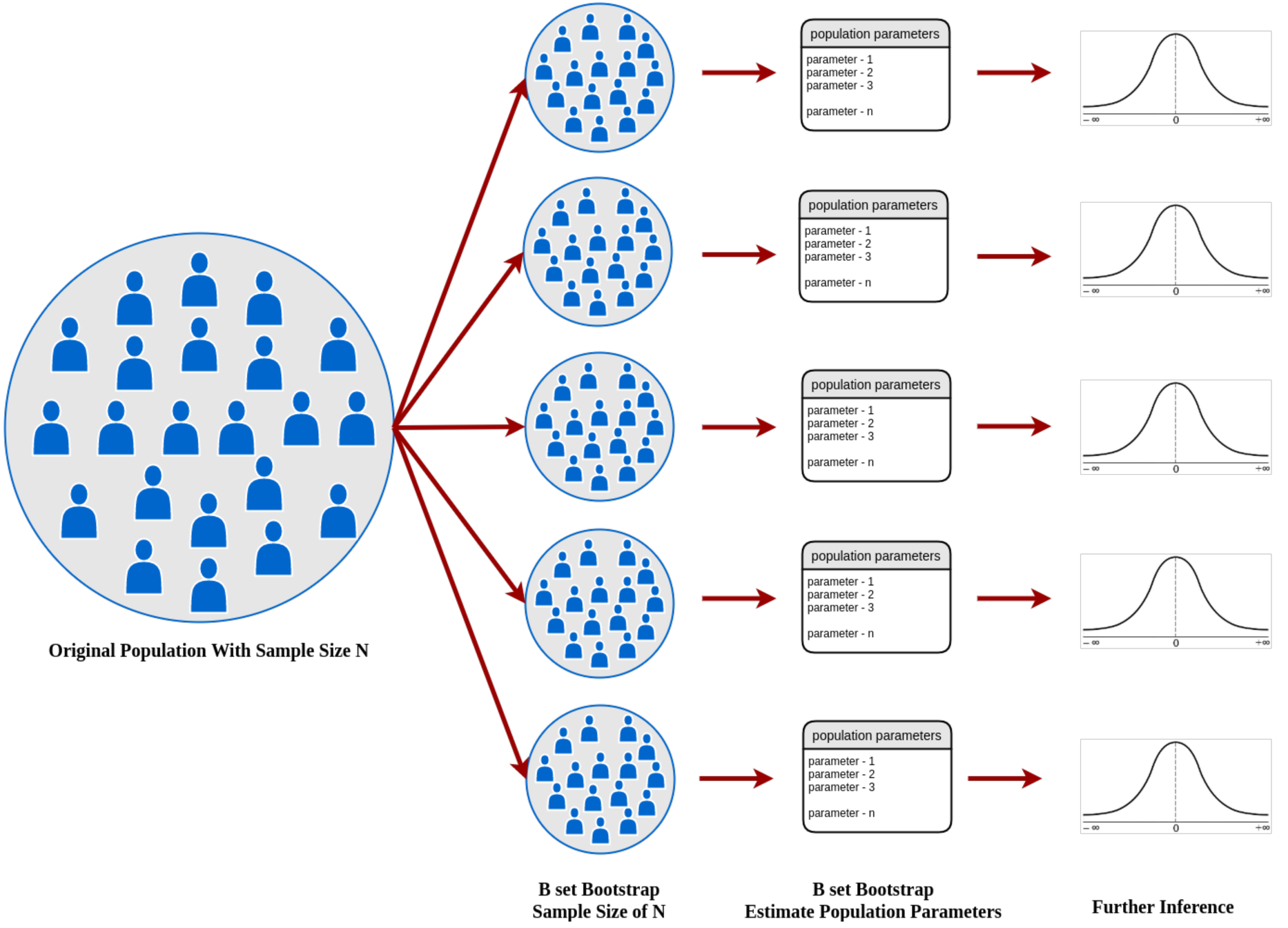
Misclassification: 0.25
Gini: 0.333
Entropy: 0.477



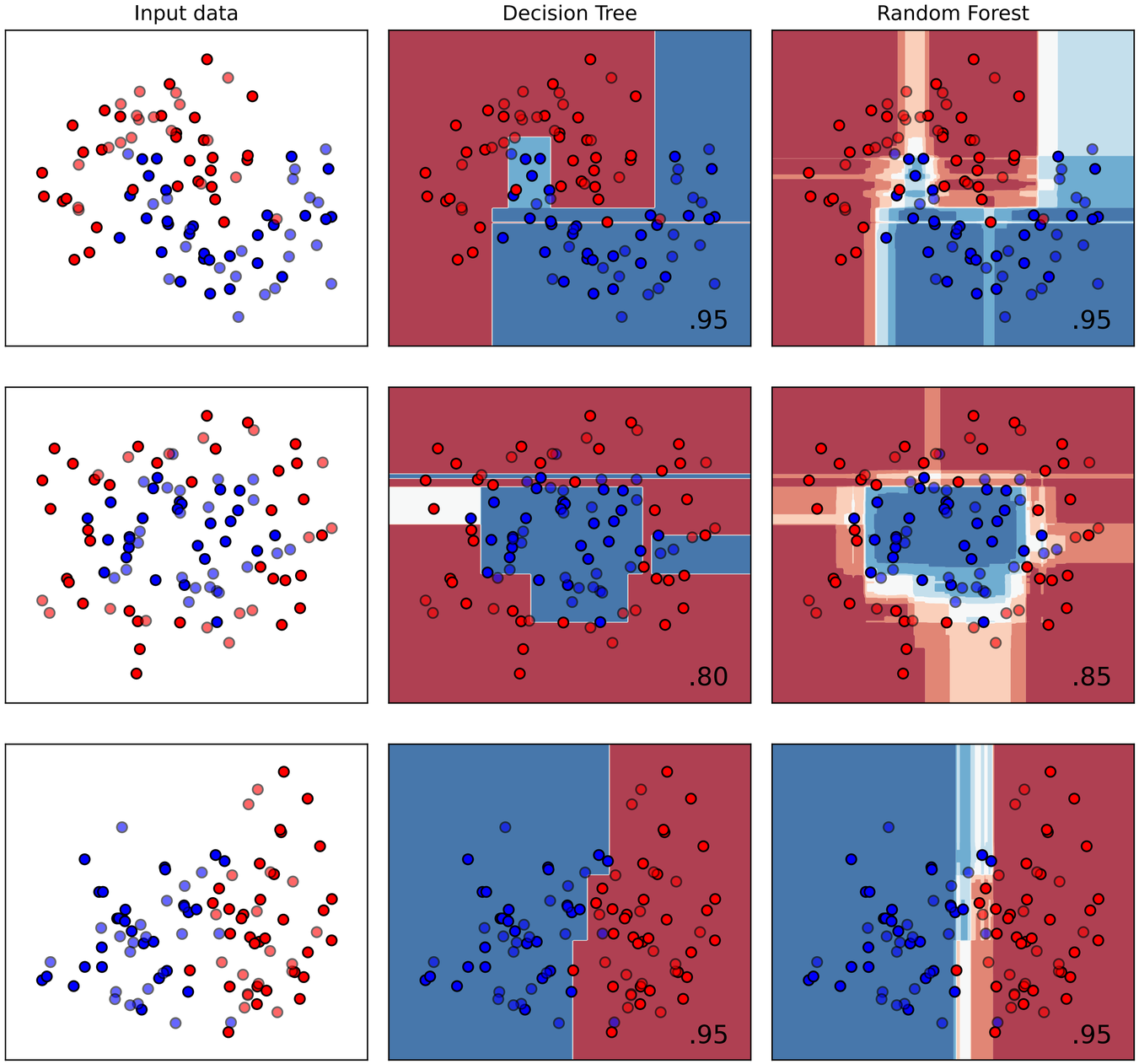

High Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

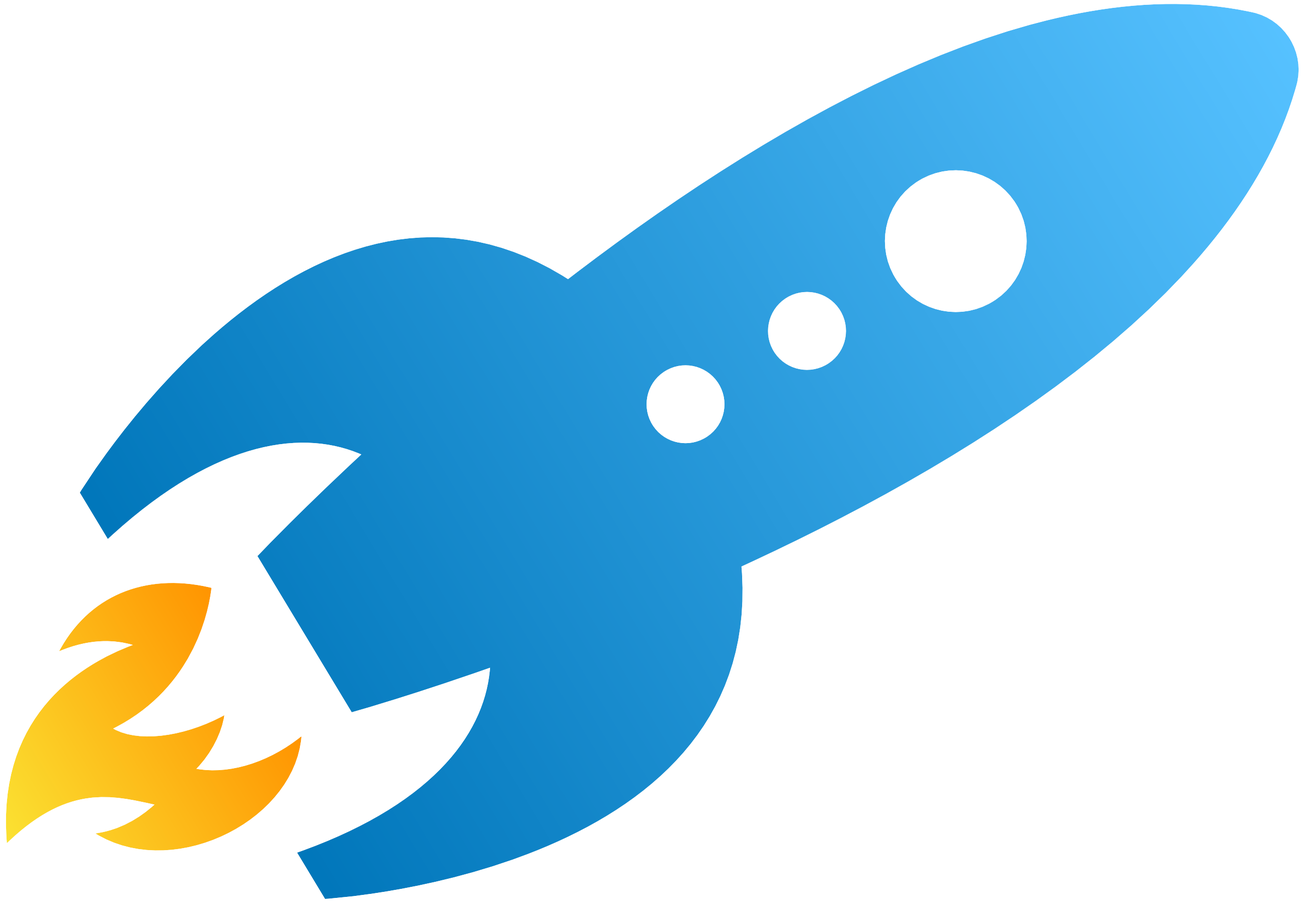

High Variance, Low Bias
10
5
0
-5
y
-5
0
5
10
x

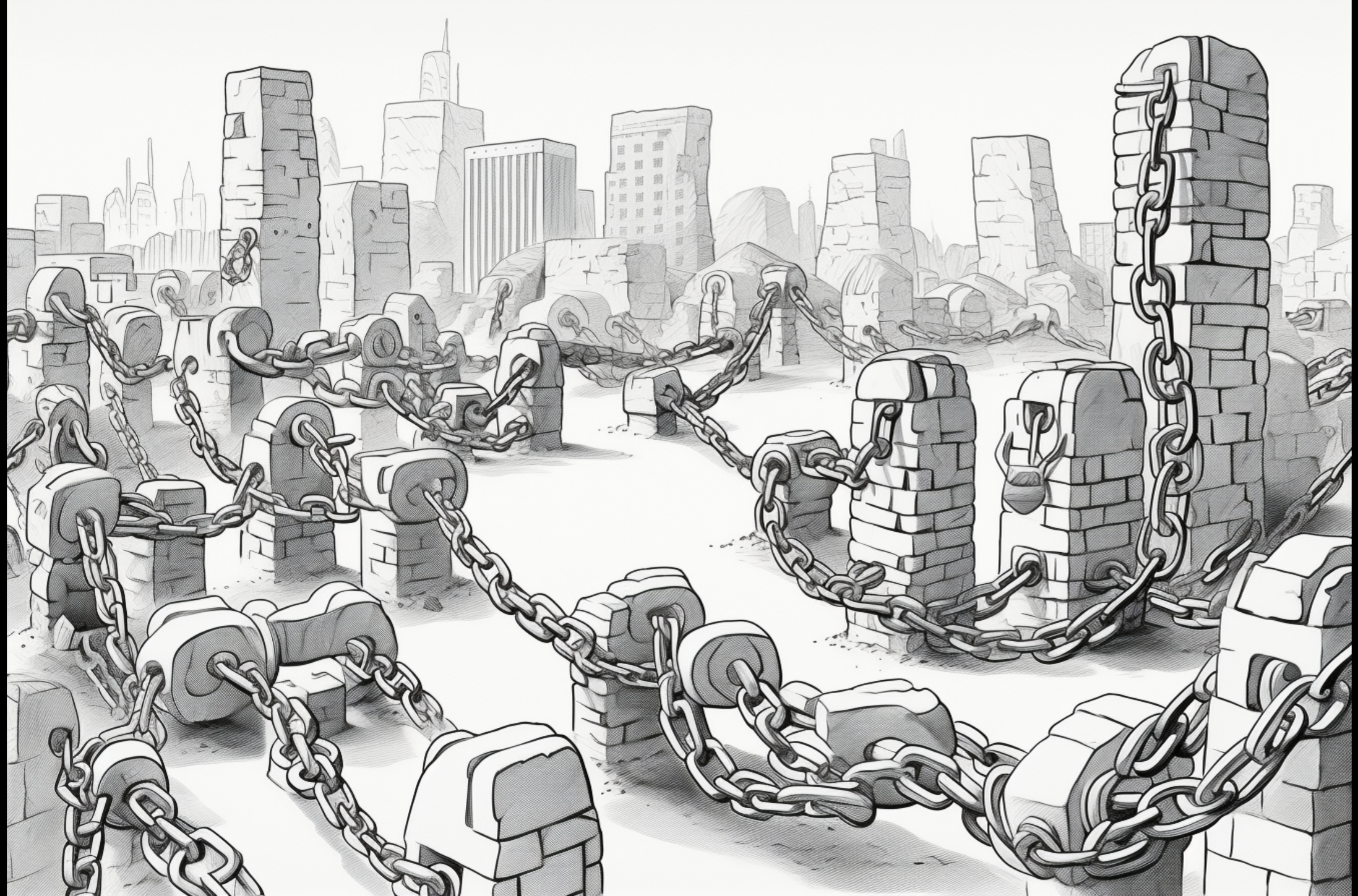


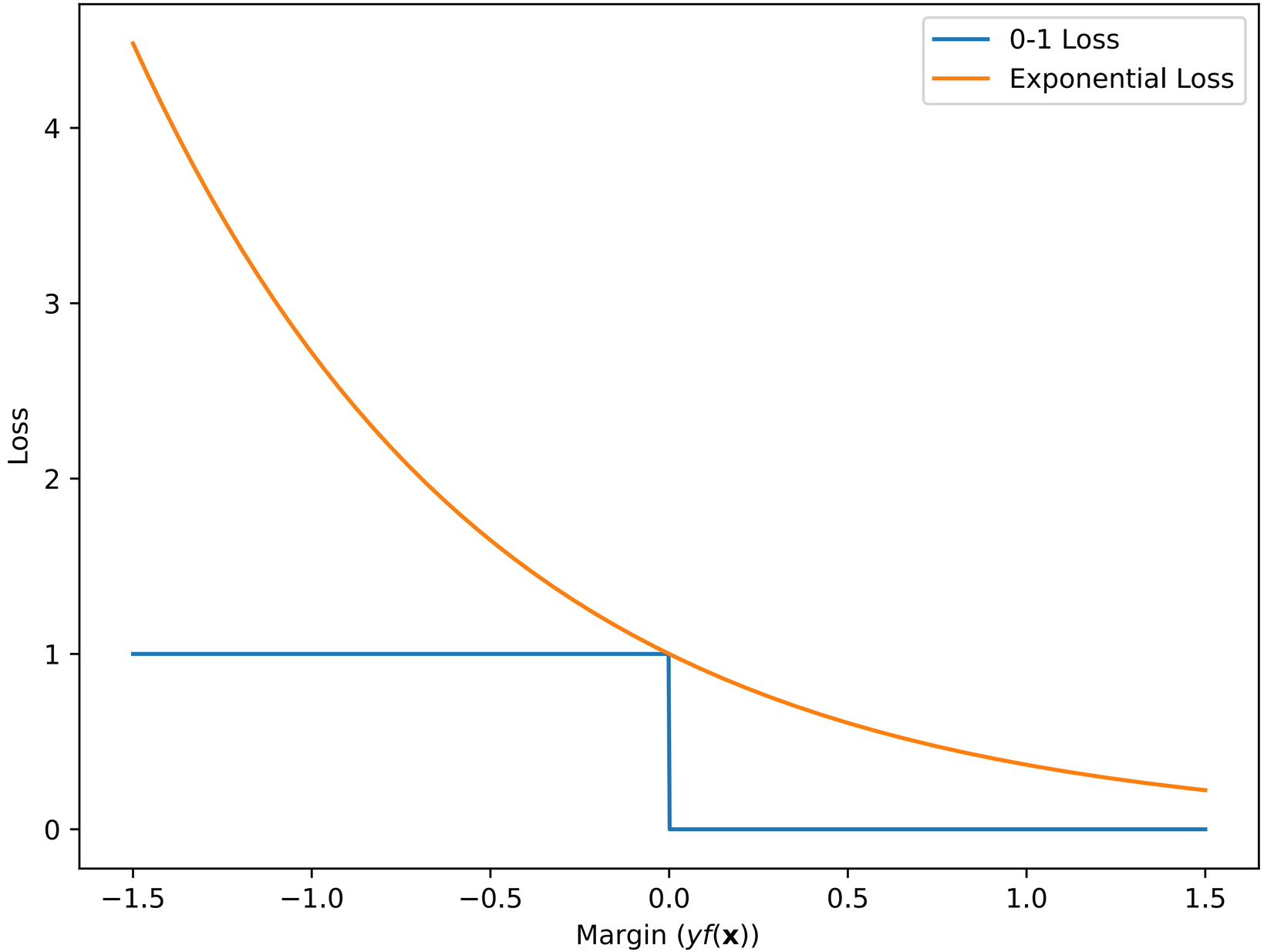

population parameters
parameter - 1
parameter - 2
parameter - 3
parameter - n
population parameters
parameter - 1
parameter - 2
parameter - 3
parameter - n
population parameters
parameter - 1
parameter - 2
parameter - 3
parameter - n
population parameters
parameter - 1
parameter - 2
parameter - 3
parameter - n
population parameters
parameter - 1
parameter - 2
parameter - 3
parameter - n
Original Population With Sample Size N
B set Bootstrap
Sample Size of N
B set Bootstrap
Estimate Population Parameters
Further Inference

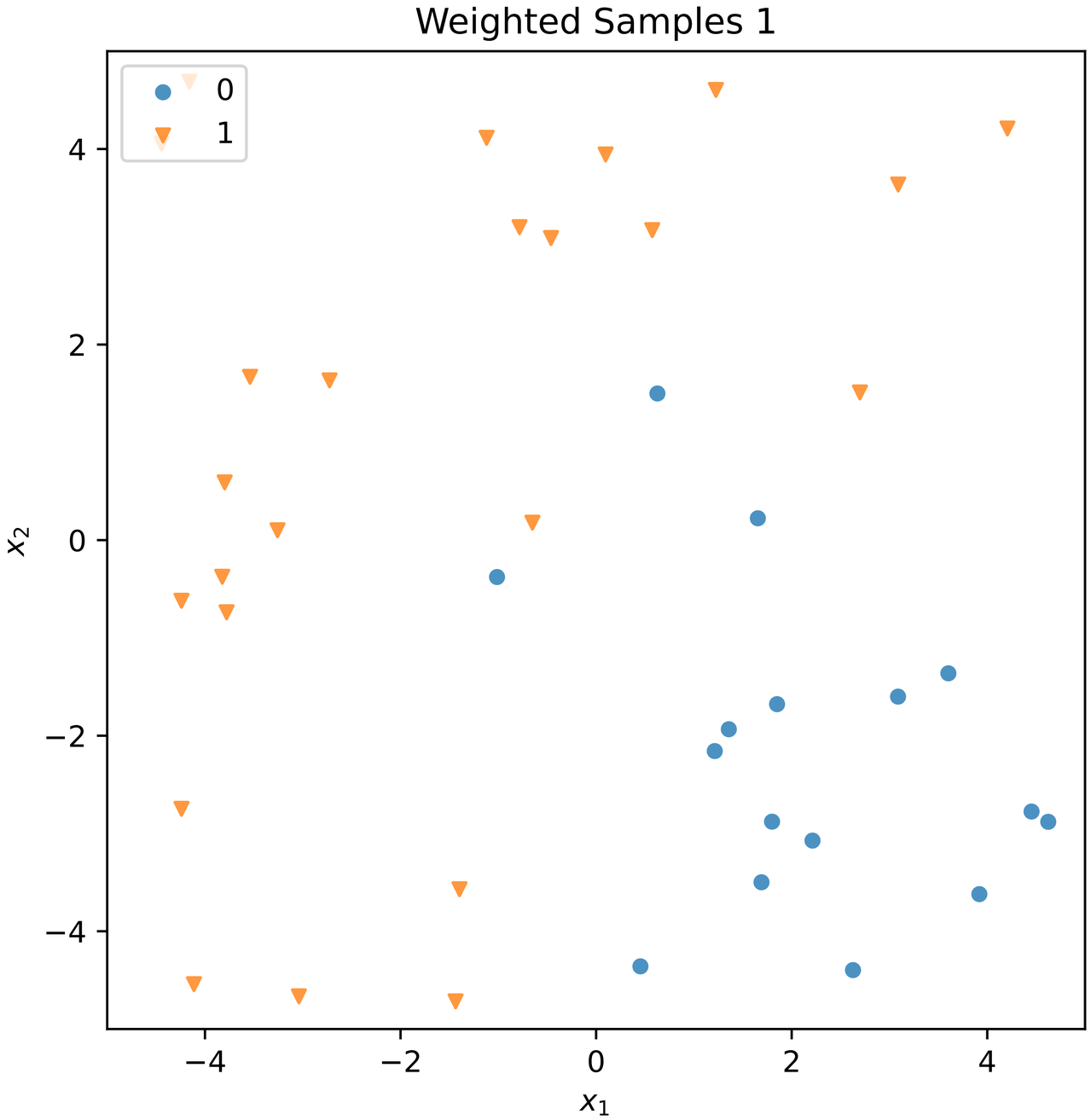


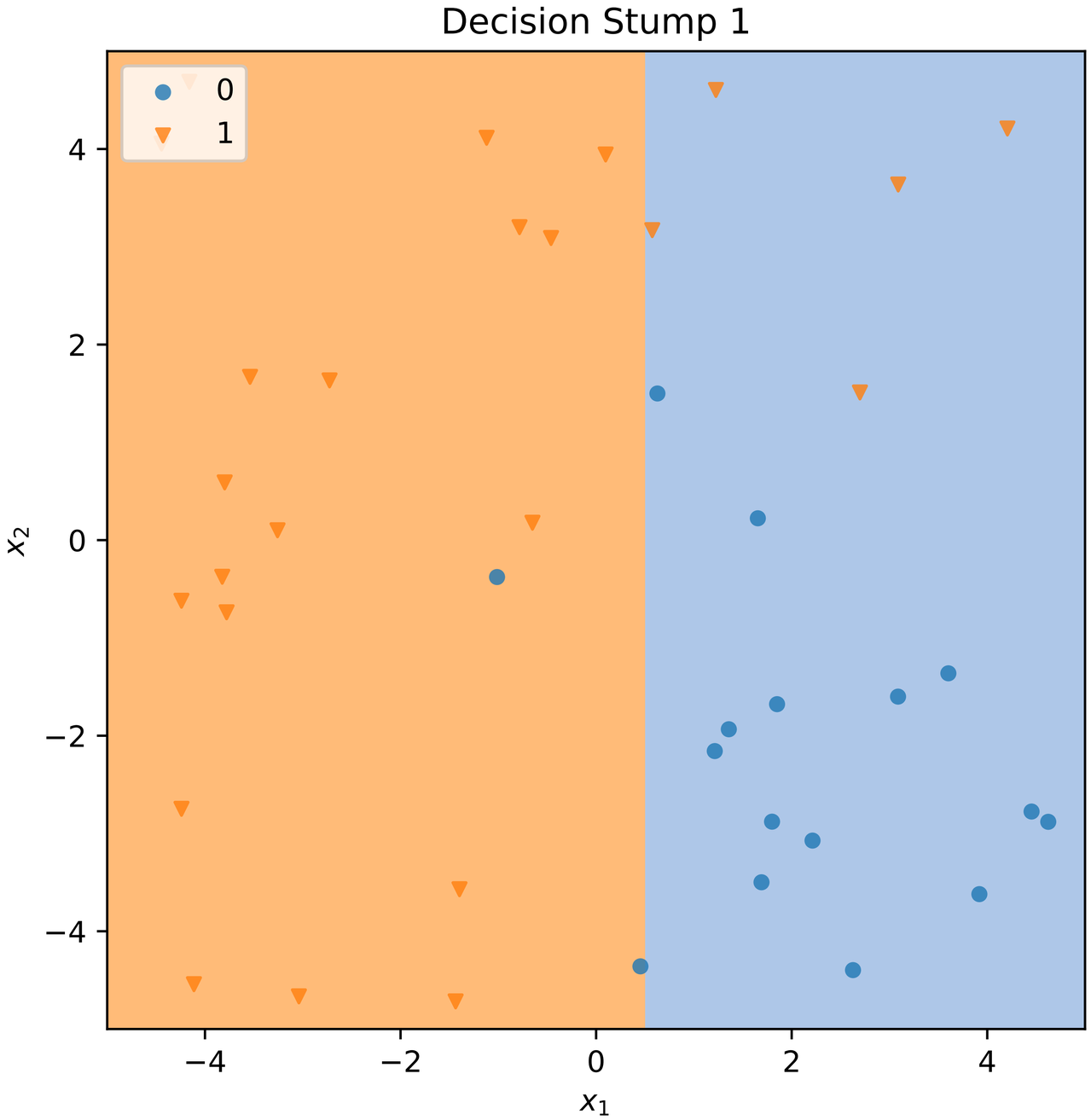

Input data
Decision Tree
Random Forest
.95
.95
.80
.85
.95
.95

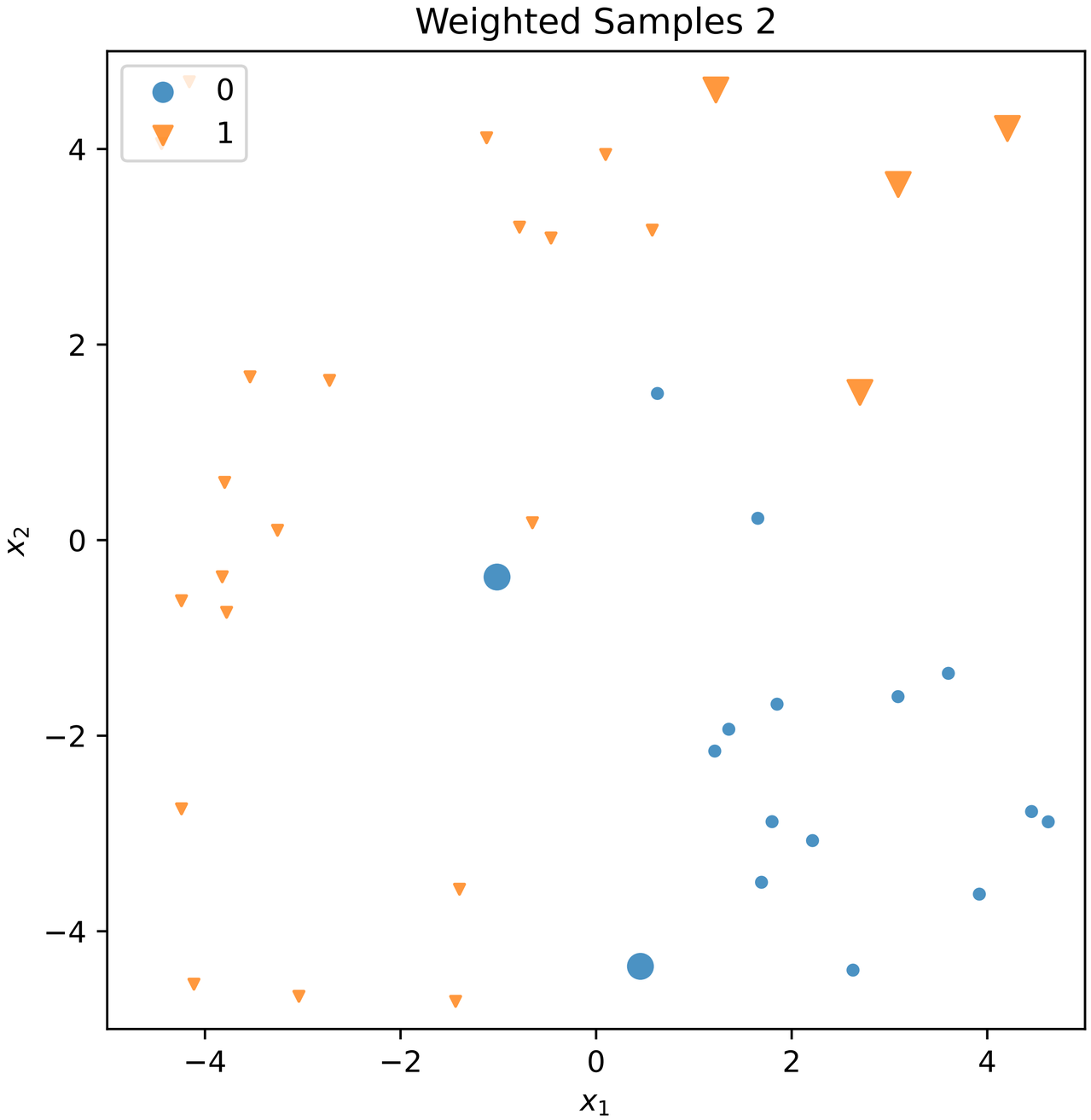


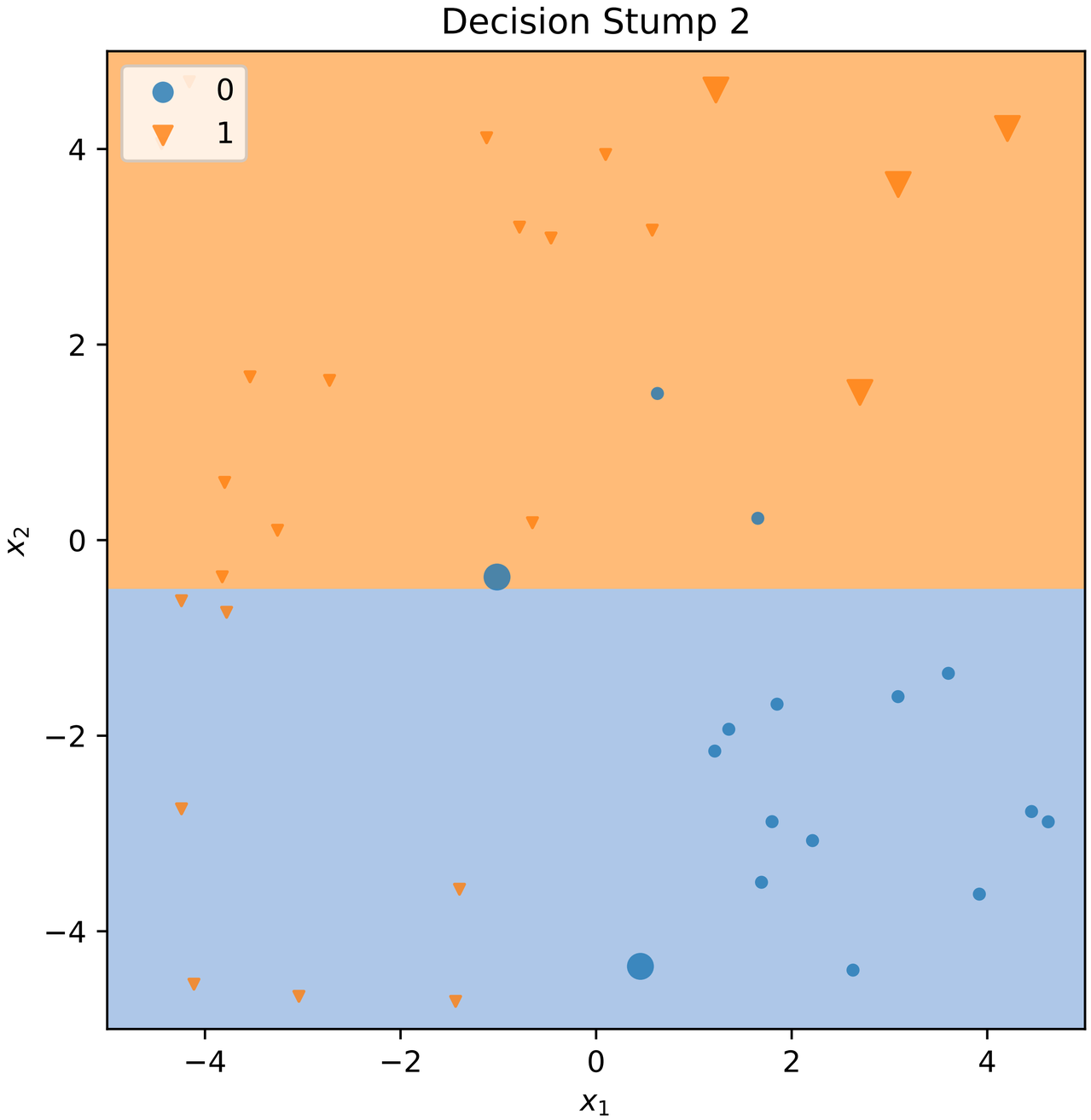


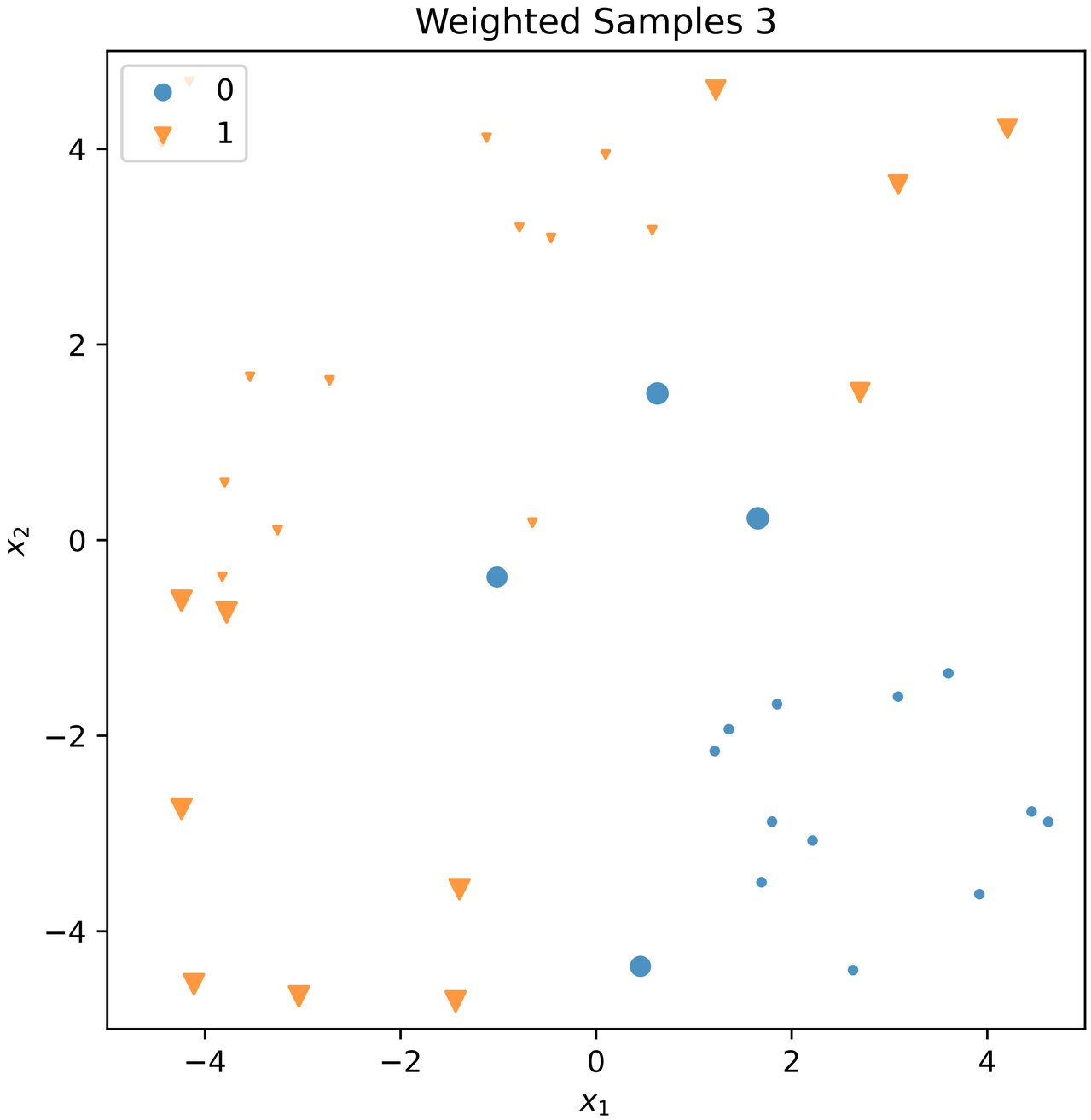

0-1 Loss
Exponential Loss
Loss
4
3
2
1
0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Margin ($yf(\mathbf{x})$)

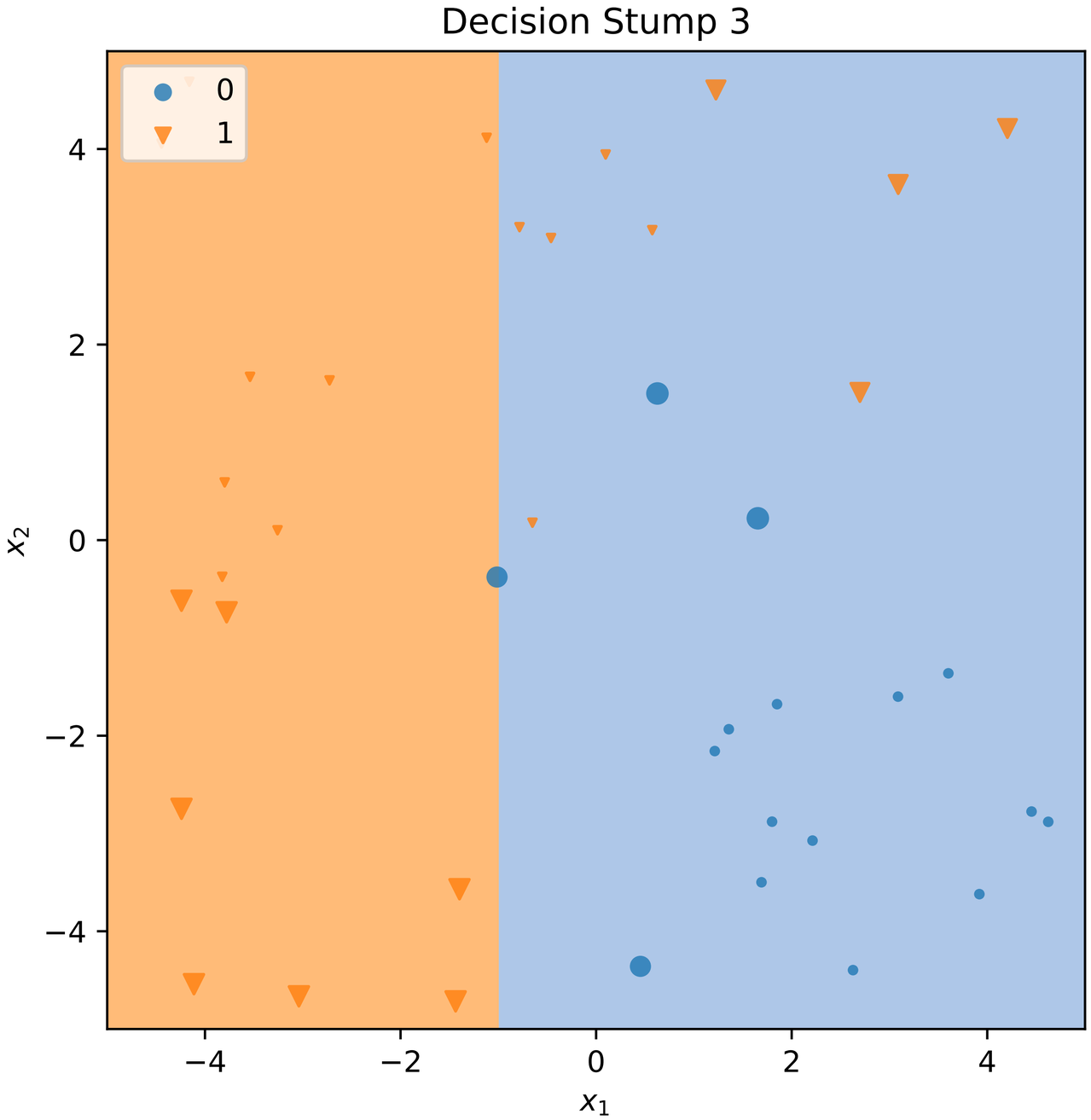

Weighted Samples 1
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

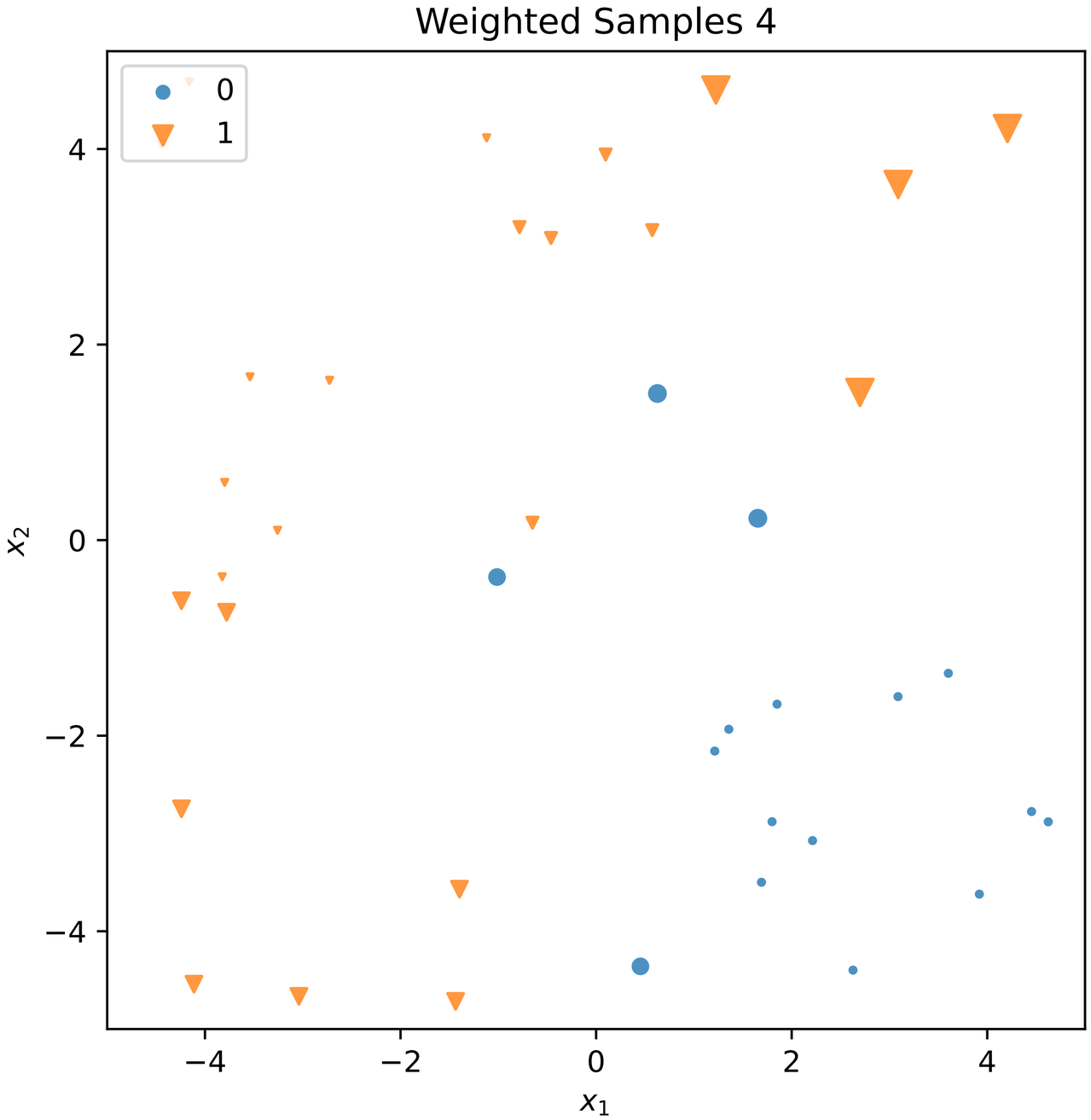

Decision Stump 1
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

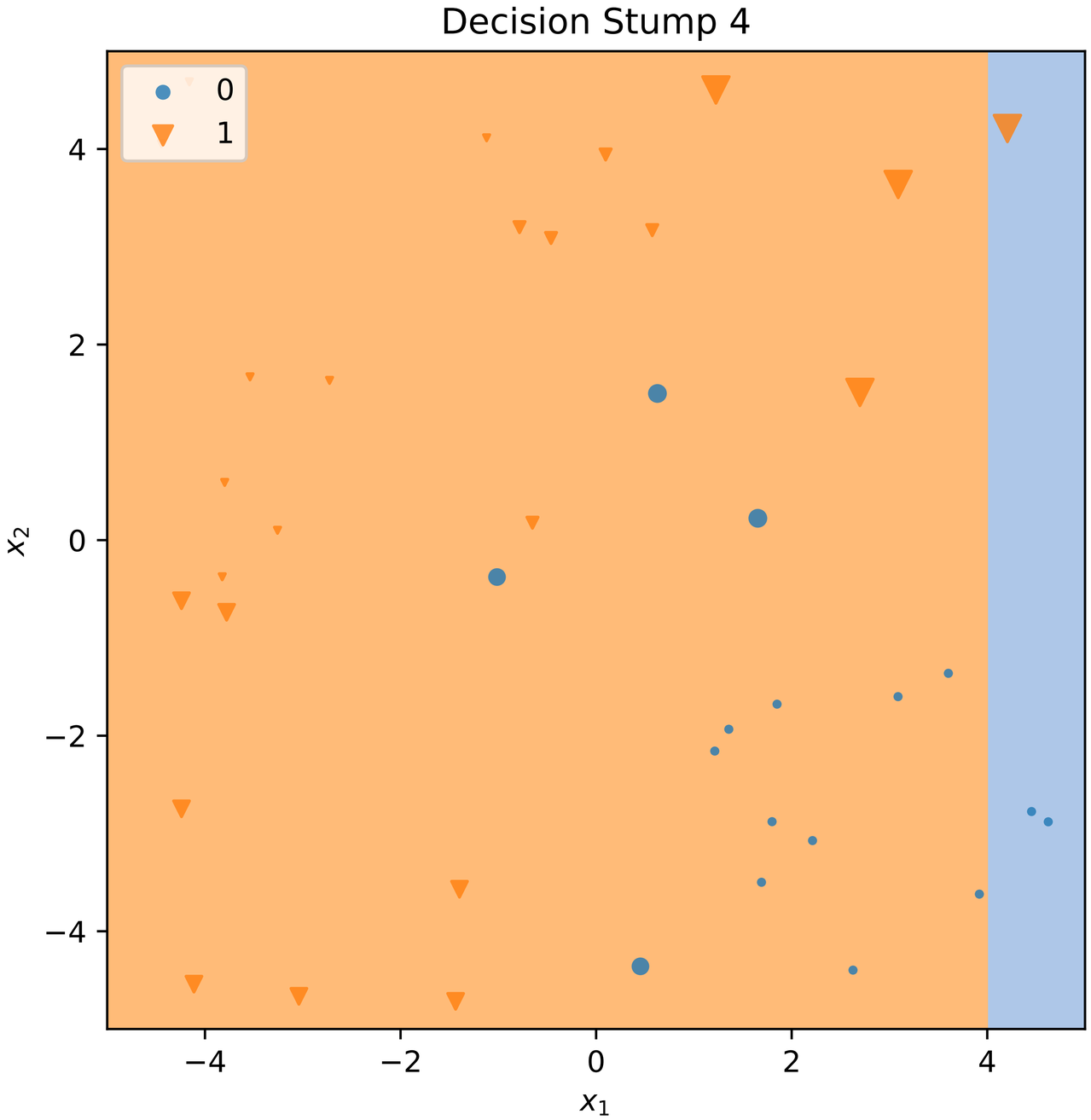

Weighted Samples 2
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

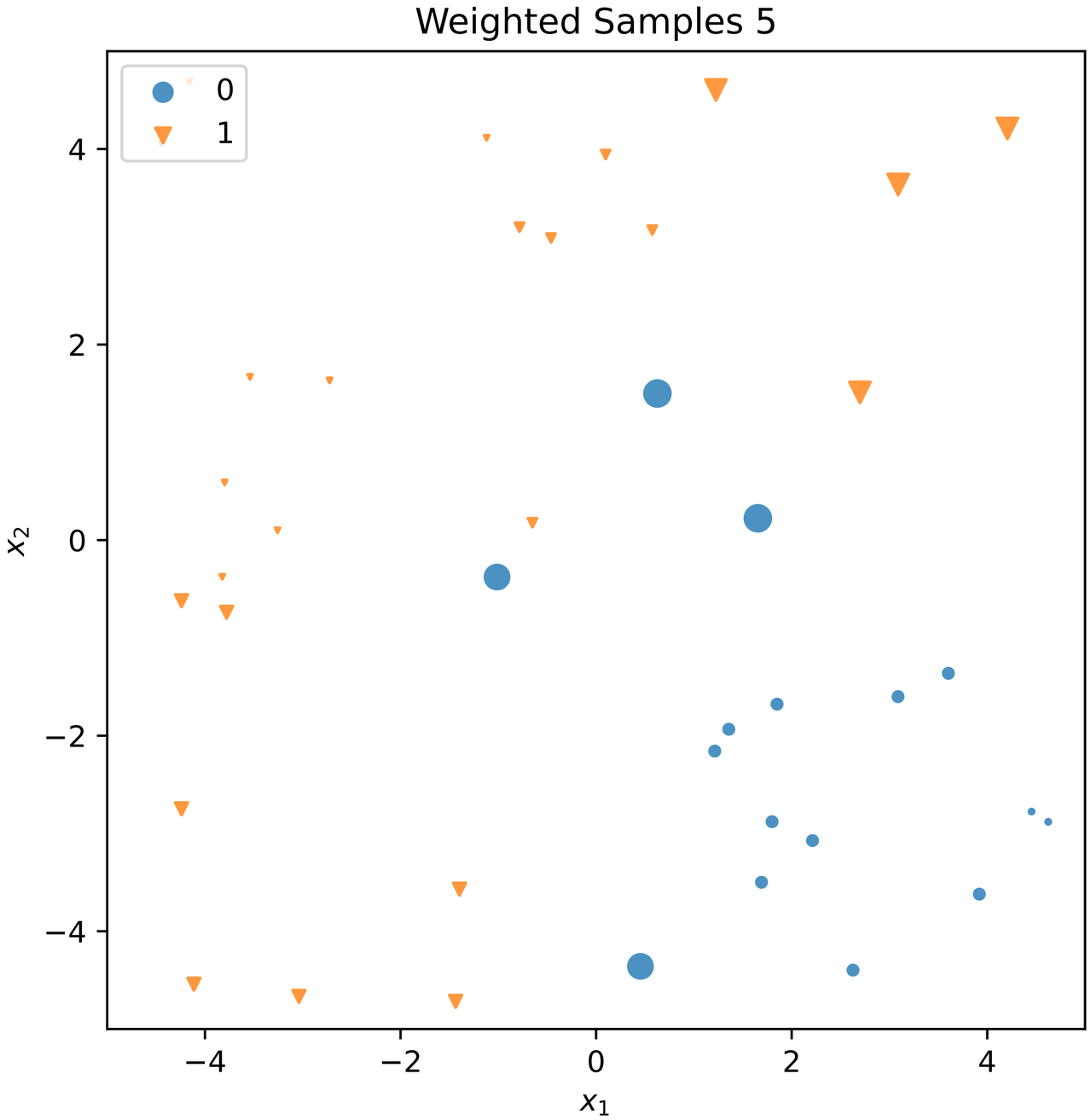

Decision Stump 2
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

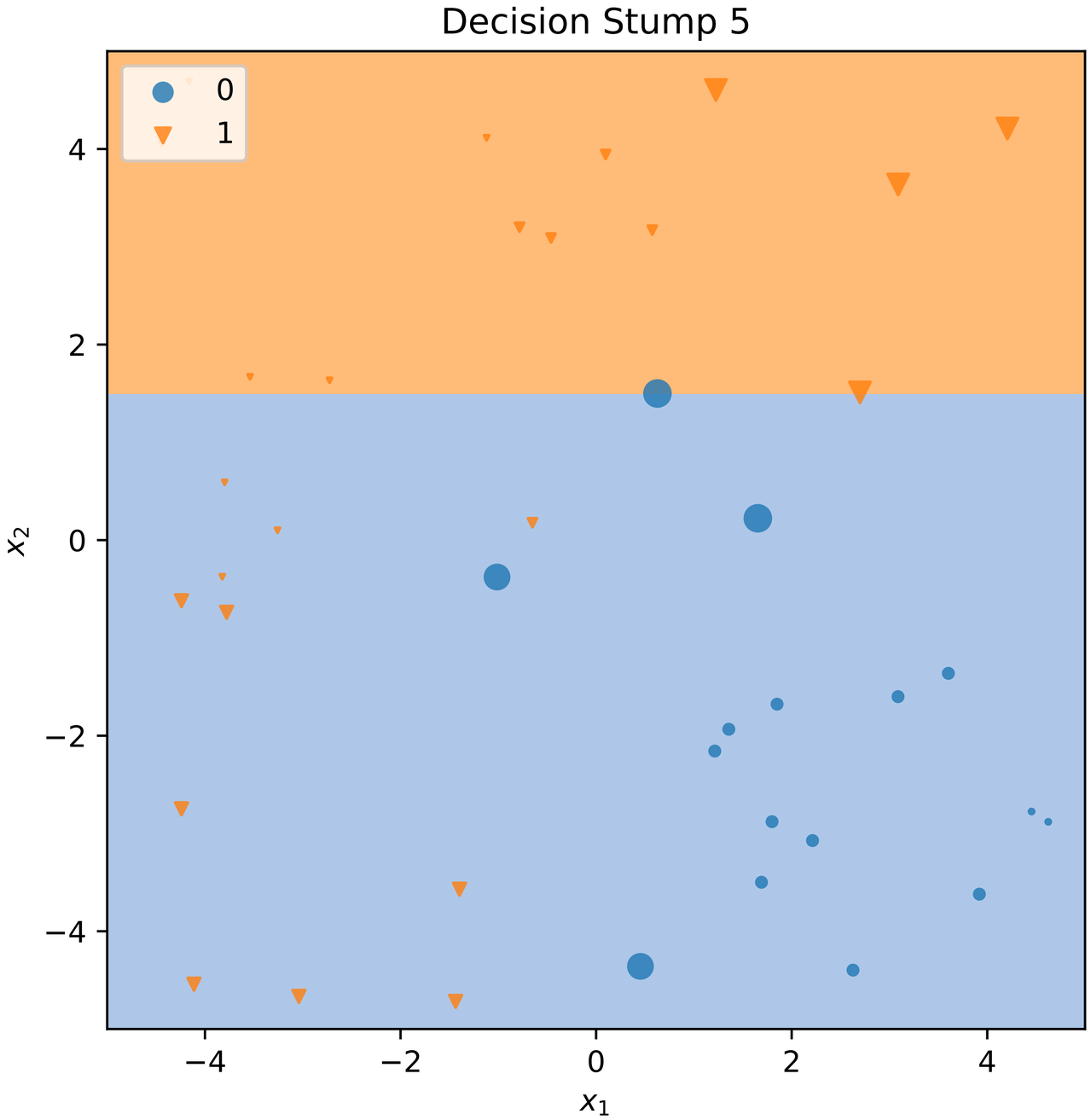

Weighted Samples 3
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

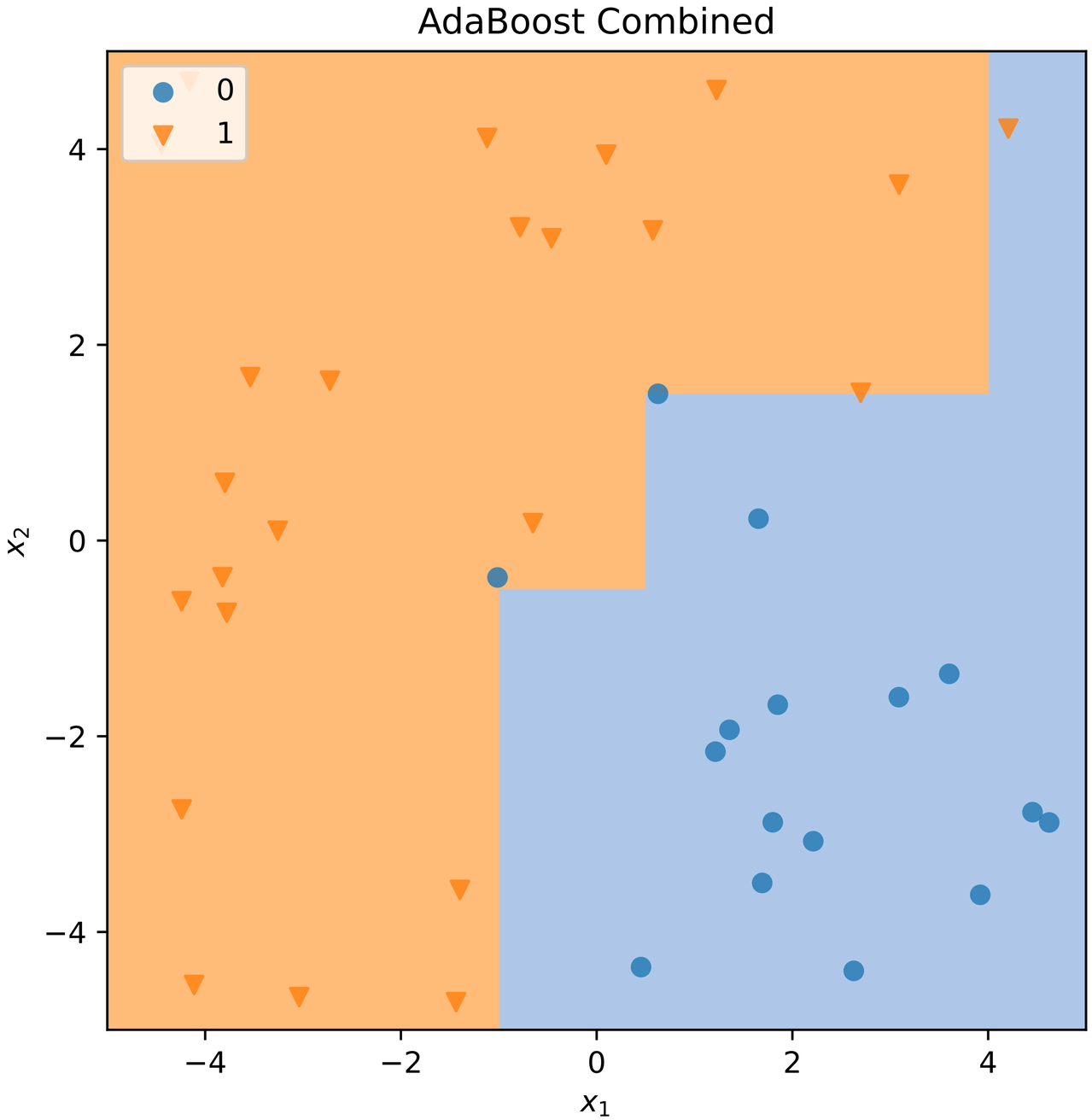

Decision Stump 3
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1


Weighted Samples 4
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1

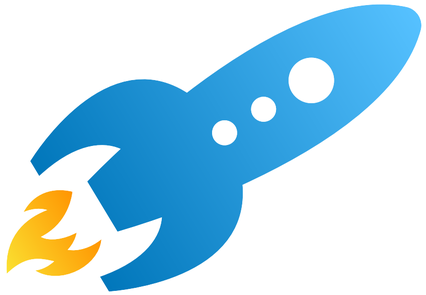

Decision Stump 4
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1


Weighted Samples 5
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1


Decision Stump 5
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1


AdaBoost Combined
0
1
4
2
0
−2
−4
x_2
−4
−2
0
2
4
x_1





strong learners

parallel

reduce variance

weak learners

sequential

increase accuracy

Questions?

Next: Linear Models Revisited



